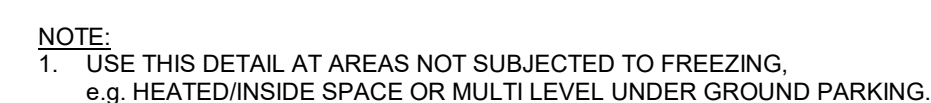
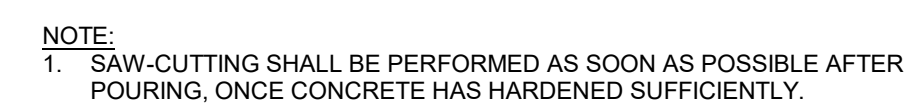


		TABLE 1 - TENSION DEVELOPMENT AND LAP SPLICE LENGTHS FOR UNCOATED BARS (ACI 318, INCH-POUND VALUES)																											
		LENGTHS (IN.) PER CONCRETE STRENGTH (PSI)																											
BAR SIZE	LAP CLASS	3000 PSI				4000 PSI				5000 PSI				6000 PSI				7000 PSI				8000 PSI				10000 PSI & GREATER			
		TOP BARS	OTHER BARS	TOP BARS	OTHER BARS	TOP BARS	OTHER BARS	TOP BARS	OTHER BARS	TOP BARS	OTHER BARS	TOP BARS	OTHER BARS	TOP BARS	OTHER BARS	TOP BARS	OTHER BARS	TOP BARS	OTHER BARS	TOP BARS	OTHER BARS	TOP BARS	OTHER BARS	TOP BARS	OTHER BARS	TOP BARS	OTHER BARS	TOP BARS	OTHER BARS
		CASE1	CASE2	CASE1	CASE2	CASE1	CASE2	CASE1	CASE2	CASE1	CASE2	CASE1	CASE2	CASE1	CASE2	CASE1	CASE2	CASE1	CASE2	CASE1	CASE2	CASE1	CASE2	CASE1	CASE2	CASE1	CASE2	CASE1	CASE2
#3	A	22	33	17	25	19	28	15	22	17	25	13	20	16	23	12	18	14	21	12	17	14	20	12	16	12	15	12	14
	B	28	42	22	32	25	37	19	28	22	33	17	25	20	30	16	23	19	28	14	21	18	26	14	20	16	23	12	18
#4	A	29	43	22	33	25	37	19	29	23	34	17	26	21	31	16	24	19	28	15	22	18	27	14	21	16	24	12	18
	B	38	56	29	43	33	49	25	37	29	44	23	34	27	40	21	31	25	37	19	28	23	35	18	27	21	31	16	24
#5	A	36	54	28	42	31	47	24	36	28	42	22	32	26	38	20	30	24	35	18	27	22	33	17	26	20	30	15	23
	B	47	70	36	54	41	61	31	47	36	54	28	42	33	50	26	38	31	46	24	35	29	43	22	33	26	39	20	30
#6	A	43	65	33	50	37	56	29	43	34	50	26	39	31	46	24	35	28	42	22	33	27	40	21	31	24	36	18	27
	B	56	84	43	65	49	73	37	56	44	65	34	50	40	59	31	46	37	55	28	42	35	52	27	40	31	46	24	35
#7	A	63	94	48	72	54	81	42	63	49	73	38	56	45	67	34	51	41	62	32	48	38	58	30	45	34	52	27	40
	B	81	122	63	94	71	106	54	81	63	95	49	73	58	86	45	67	54	80	41	62	50	75	39	58	45	67	34	52
#8	A	72	107	55	83	62	93	48	72	56	83	43	64	51	76	39	59	47	70	36	54	44	66	34	51	39	59	30	45
	B	93	139	72	107	81	121	62	93	72	108	56	83	68	99	51	76	61	91	47	70	57	86	44	66	51	77	39	59
#9	A	81	121	62	93	70	105	54	81	63	94	48	72	57	86	44	66	53	79	41	61	50	74	38	57	44	66	34	51
	B	105	157	81	121	91	136	70	105	81	122	63	94	74	111	57	86	69	103	53	79	64	96	50	74	58	86	44	66
#10	A	91	136	70	105	79	118	61	91	71	106	54	81	64	96	50	74	60	89	46	69	56	84	43	64	50	75	39	58
	B	118	177	91	136	102	153	79	118	92	137	71	106	84	125	64	96	77	116	60	89	72	108	56	84	65	97	50	75
#11	A	101	151	78	116	87	131	67	101	78	117	60	90	71	107	55	82	68	99	51	76	62	93	48	71	55	83	43	64
	B	131	198	101	151	114	170	87	131	102	152	78	117	93	139	71	107	86	128	66	99	80	120	62	93	72	108	55	83
#14	N/A	121	181	93	140	105	157	81	121	94	141	72	108	86	128	66	99	79	119	61	92	74	111	57	86	67	100	51	77
#18	N/A	161	242	124	186	140	209	108	161	125	187	96	144	114	171	88	131	106	158	81	122	99	148	76	114	89	133	68	102
NOTES:		1. TABULATED VALUES BASED ON GRADE 60 REINF. AND NORMAL WEIGHT CONCRETE. FOR LIGHTWEIGHT AGGREGATE CONCRETE, MULTIPLY THE TABULATED VALUES BY 1.33 FOR GRADE 80 REINF. BARS, MULTIPLY TABULATED VALUES BY 1.533.																											
		2. TENSION DEVELOPMENT LENGTHS AND LAP SPLICE LENGTHS ARE CALCULATED BASED ON ACI 318. TABULATED VALUES FOR BEAMS AND COLUMNS ARE BASED ON TRANSVERSE REINF. AND CONCRETE COVER MEETING MINIMUM CODE REQUIREMENTS.																											
		3. CASES 1 AND 2, WHICH DEPEND ON THE TYPE OF STRUCTURAL ELEMENT, CONCRETE COVER, AND CENTER-TO-CENTER SPACING, ARE DEFINED AS: BEAM OR COLUMNS: CASE 1: COVER AT LEAST 1.0 db AND c-c SPACING AT LEAST 2.0 db AND STRIRUPS OR TIES THROUGHOUT Ld NOT LESS THAN THE CODE MIN. CASE 2: COVER LESS THAN 1.0 db OR c-c SPACING LESS THAN 2.0 db ALL OTHERS: CASE 1: COVER AT LEAST 1.0 db AND c-c SPACING AT LEAST 3.0 db CASE 2: COVER LESS THAN 1.0 db OR c-c SPACING LESS THAN 3.0 db																											
		4. LAP SPLICE LENGTHS ARE MULTIPLES OF DEVELOPMENT LENGTHS: CLASS A = 1.0 ld CLASS B = 1.3 ld																											
		5. ACI 318 DOES NOT ALLOW TENSION LAP SPLICES OF #14 AND #18 BARS. TABULATED VALUES FOR THOSE BARS ARE TENSION DEVELOPMENT LENGTHS.																											
		6. TOP BARS ARE HORIZONTAL BARS WITH MORE THAN 12-INCHES OF CONCRETE CAST BELOW THE BARS.																											

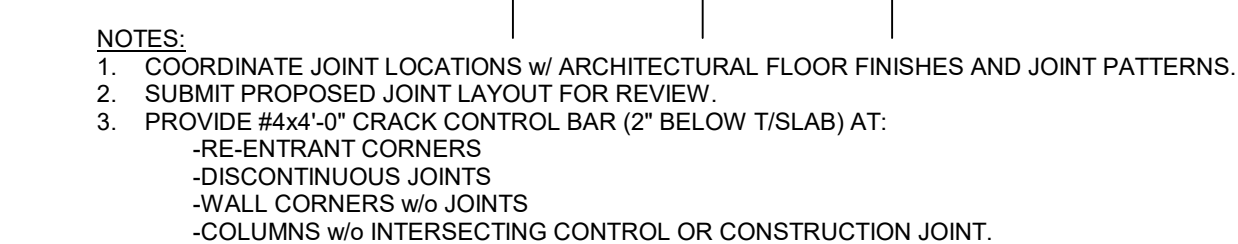
ACI 318-19



TYPICAL THICKENED
SLAB-ON-GRADE



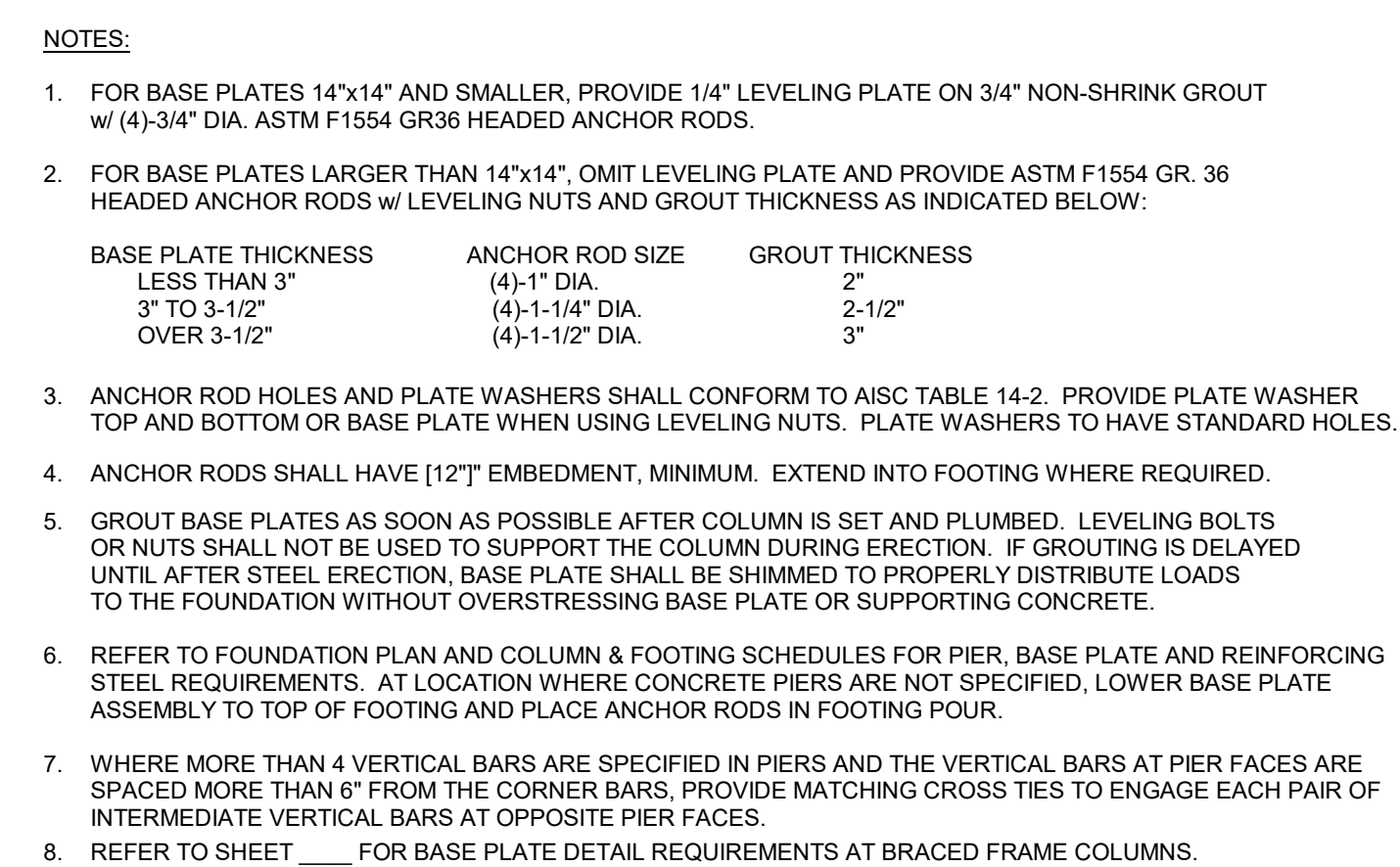
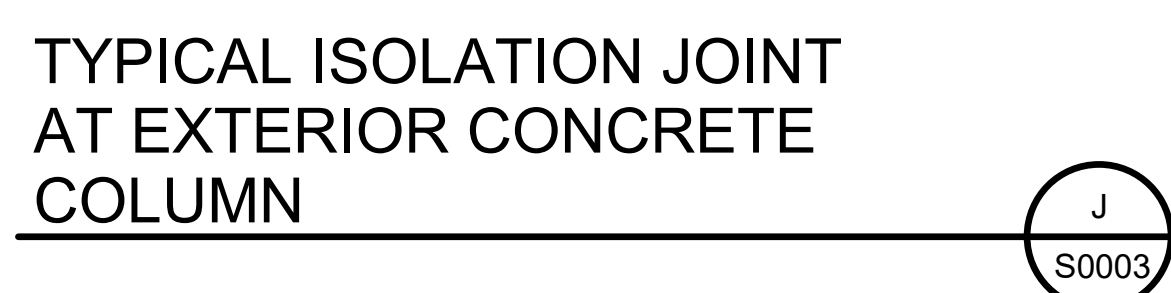
TYPICAL SAWED CONTROL JOINT



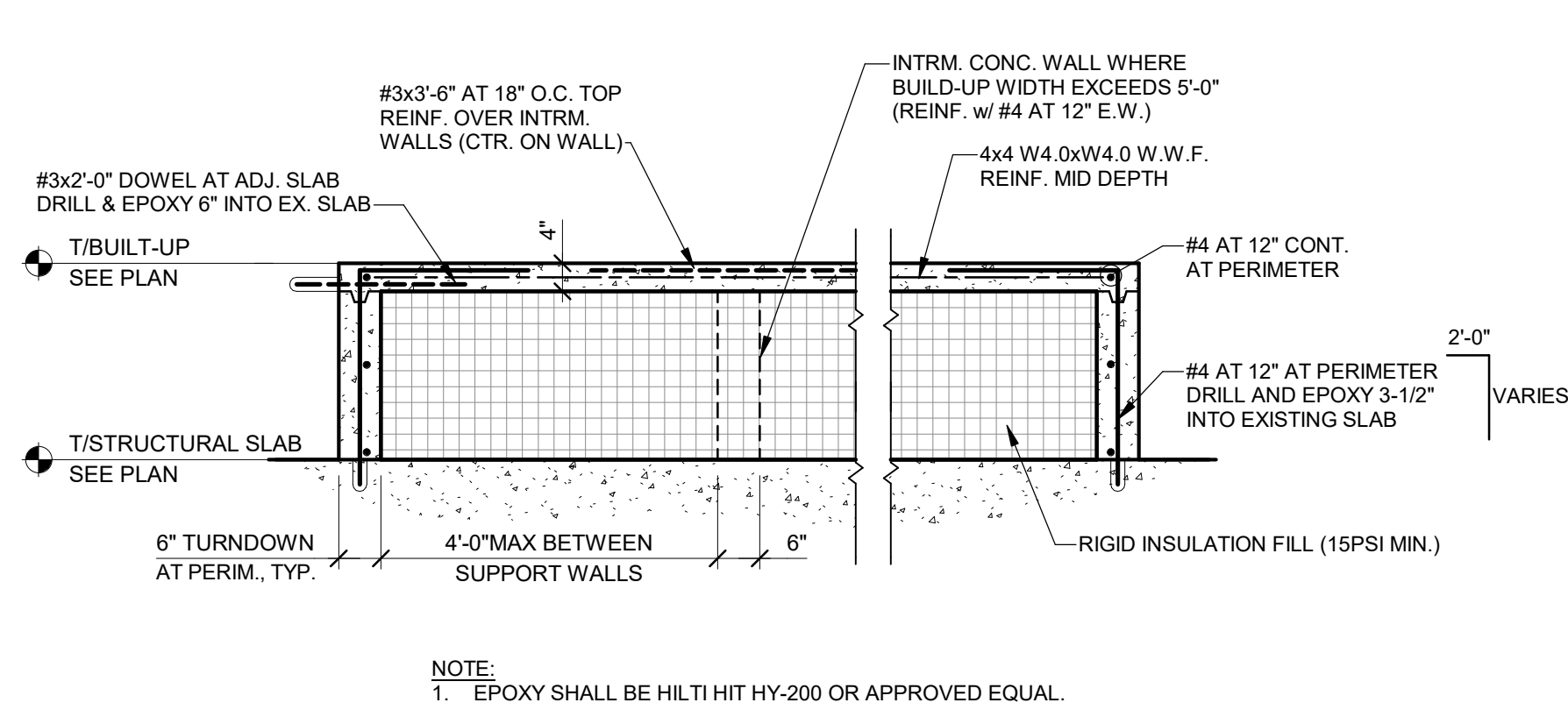
TYPICAL SLAB ON GRADE
JOINTING



TYPICAL ISOLATION JOINT AT
INTERIOR CONCRETE COLUMN

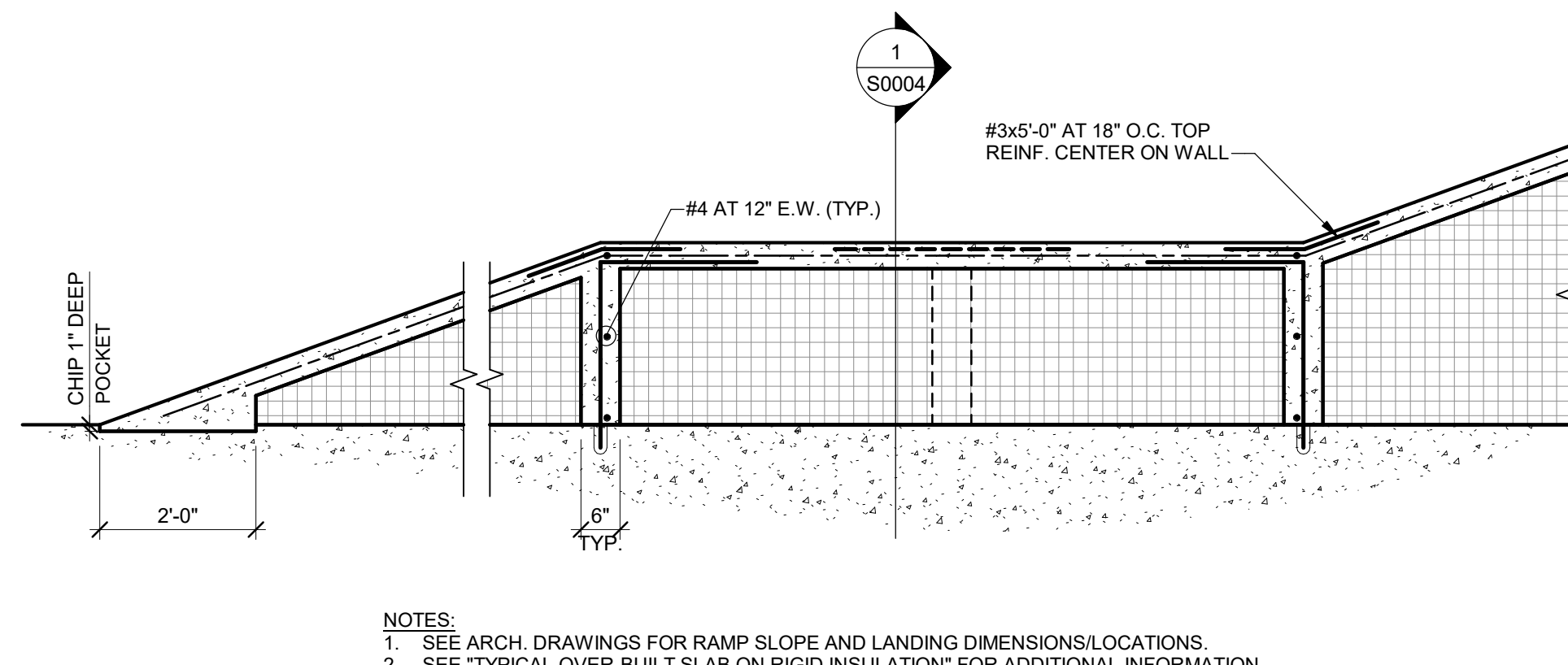


TYPICAL FOOTING AND
STEEL COLUMN BASE



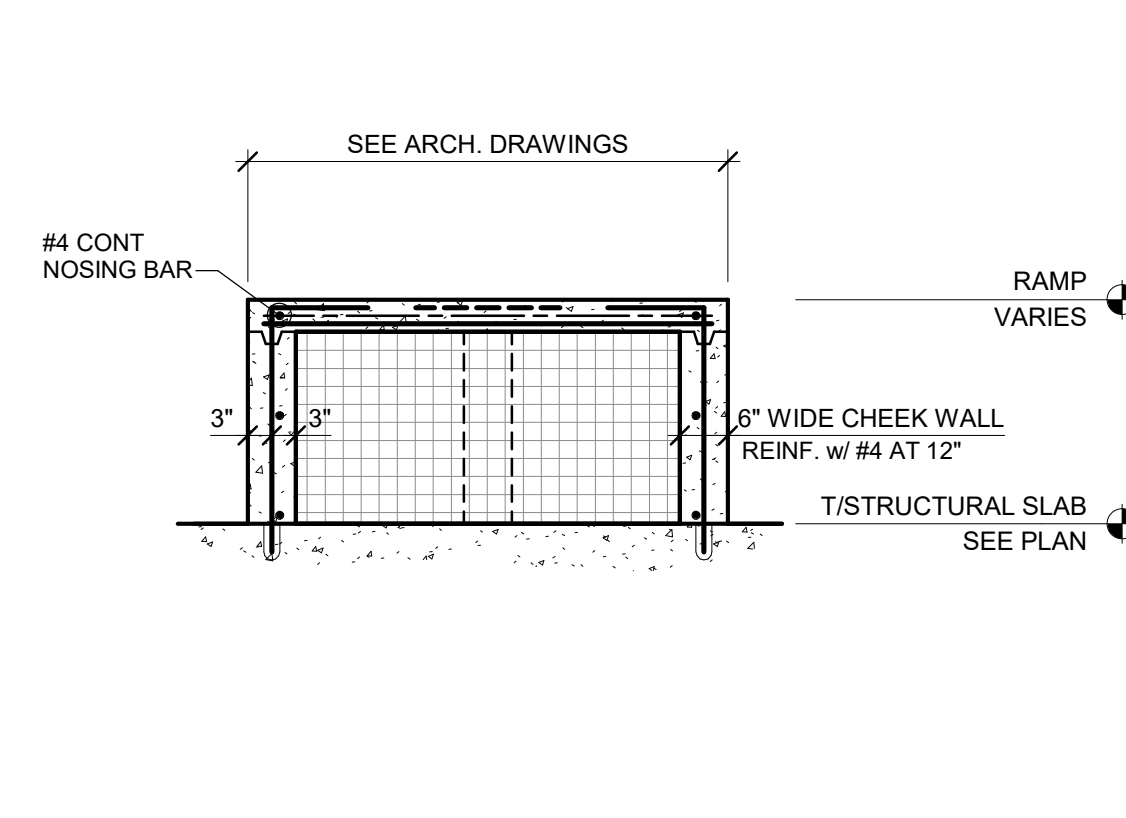
TYPICAL OVER-BUILT SLAB ON RIGID INSULATION

A
S0004



TYPICAL OVER-BUILT CONCRETE RAMP ON RIGID INSULATION

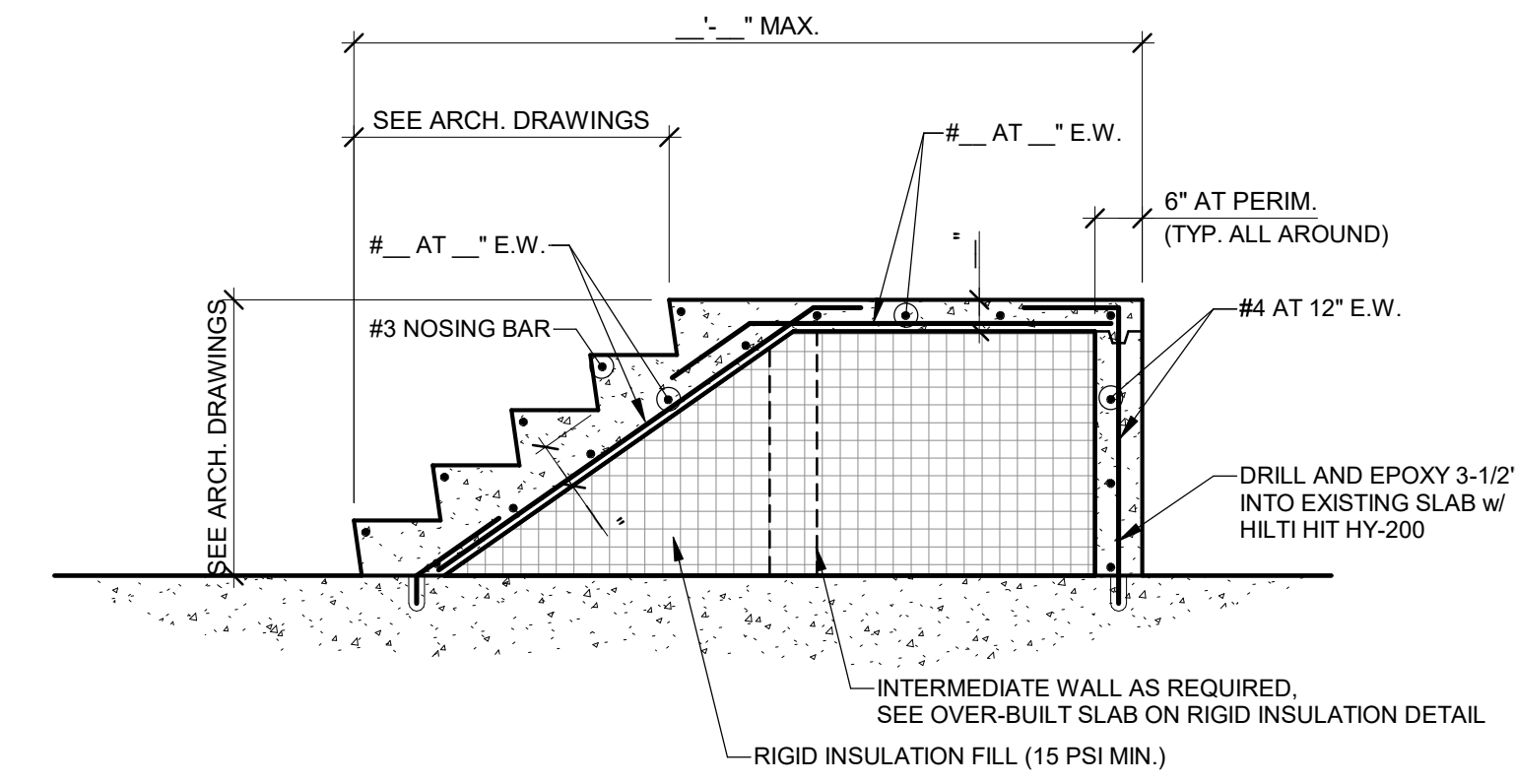
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SECTION 1

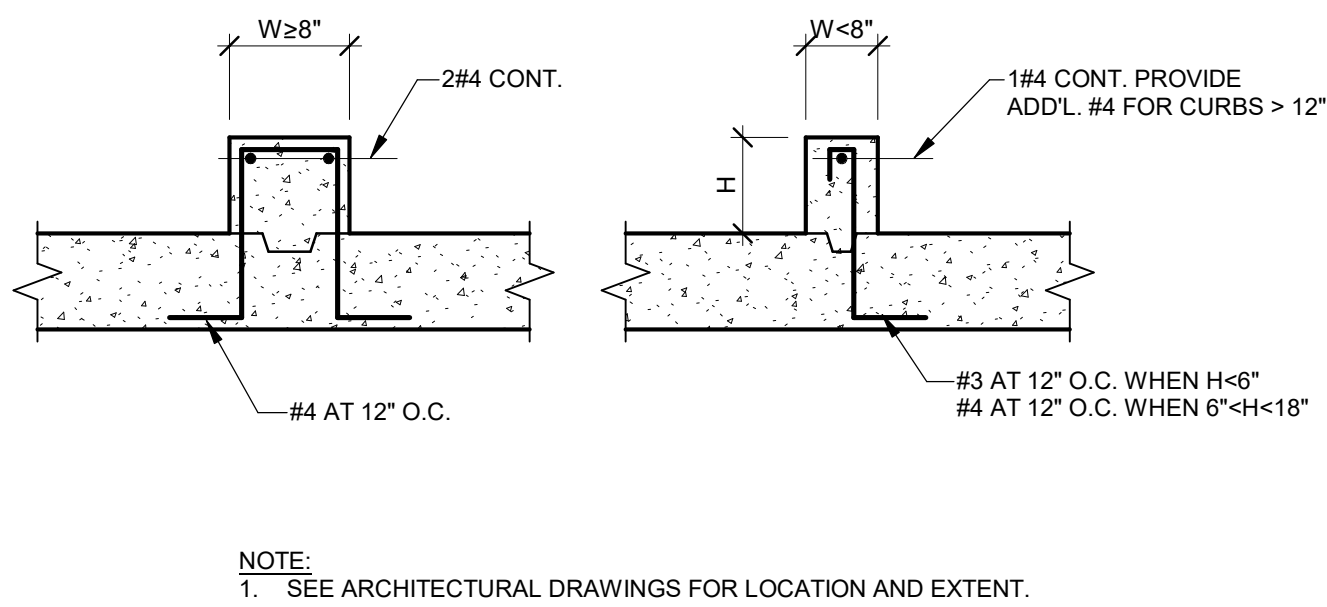
SCALE: 1/2" = 1'-0"

1
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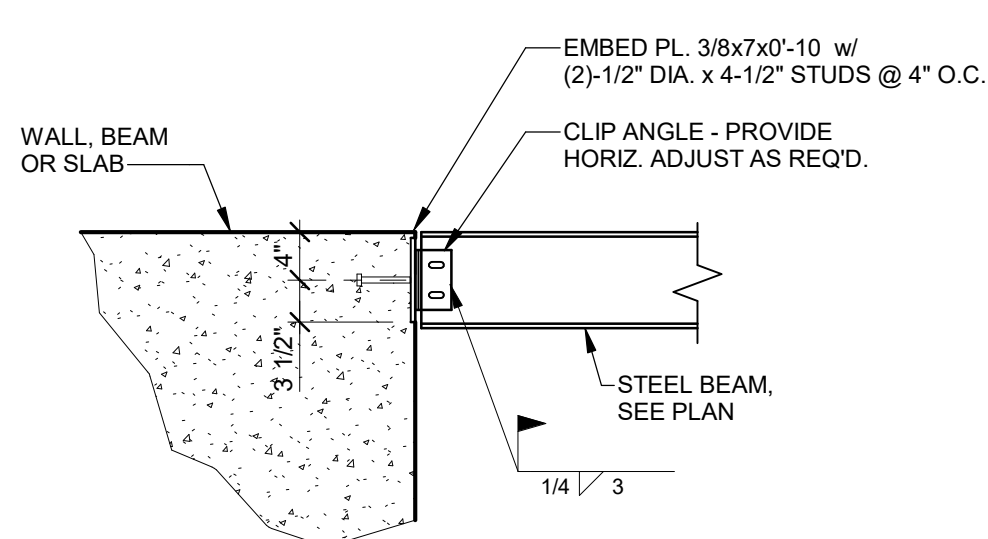
TYPICAL OVER-BUILT SLAB

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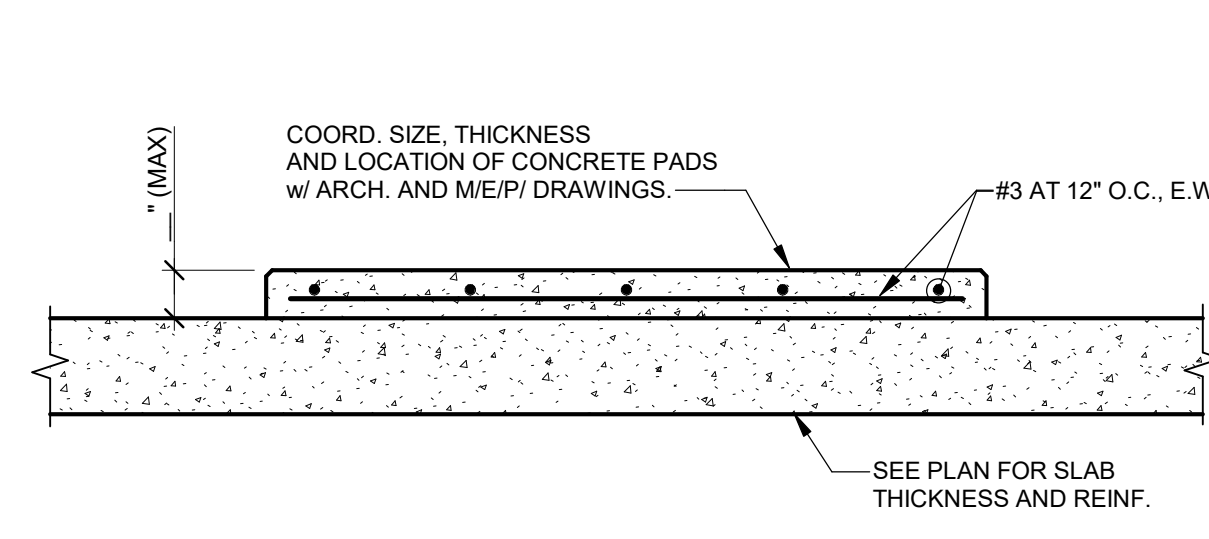
TYPICAL CONCRETE CURBS

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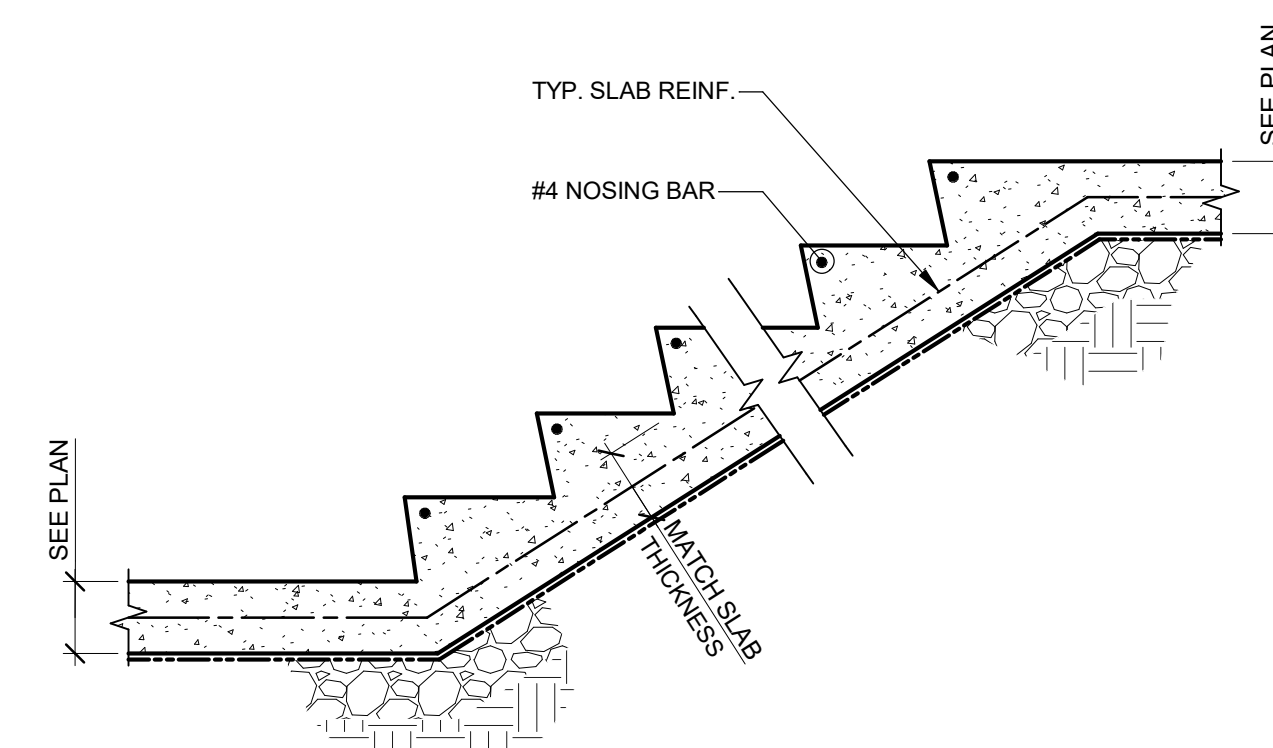
TYPICAL ELEVATOR DIVIDER BEAM CONNECTION

E
S0004



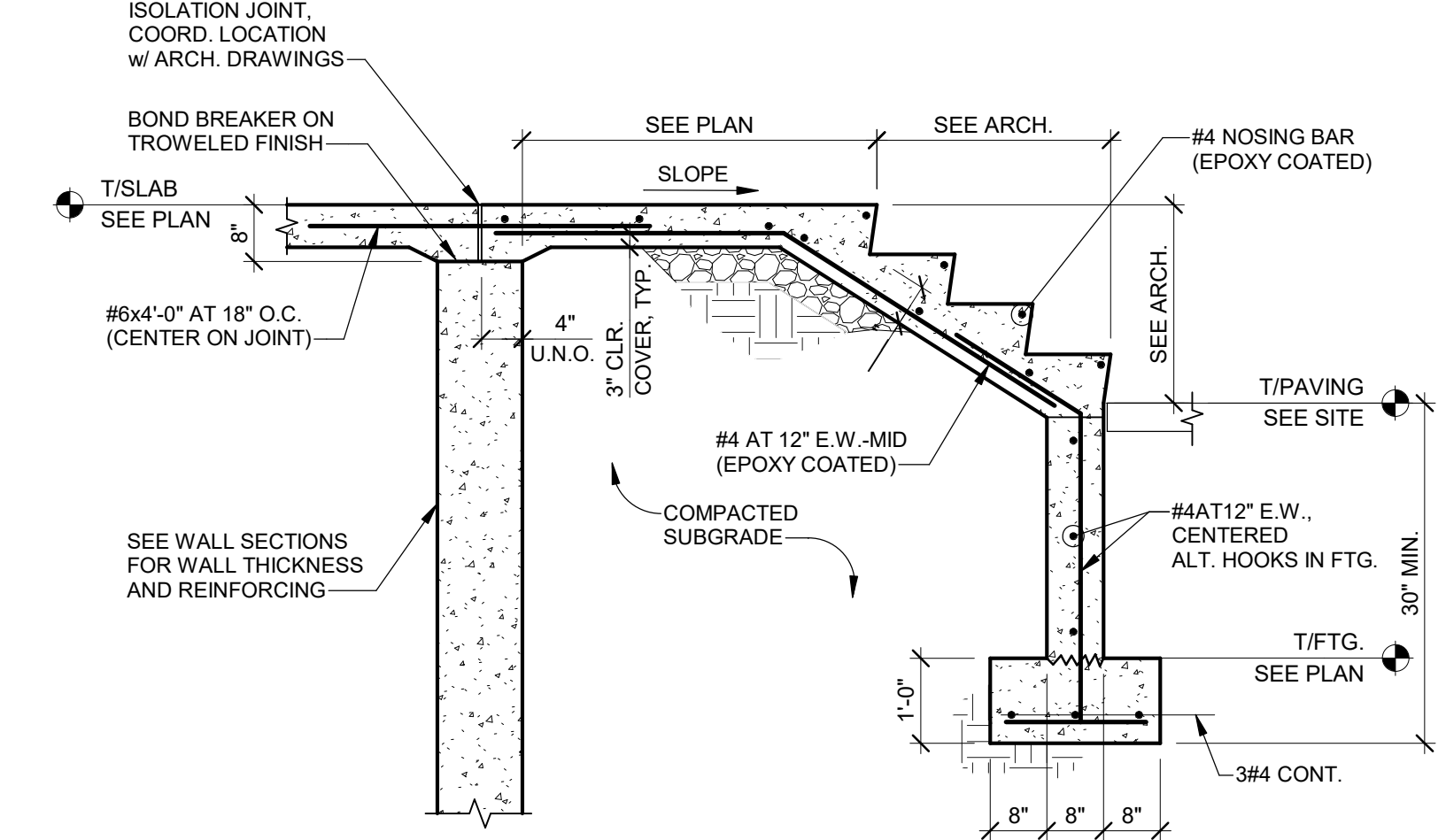
TYPICAL HOUSEKEEPING PAD (FOR M/E/P EQUIPMENT)

F
S0004



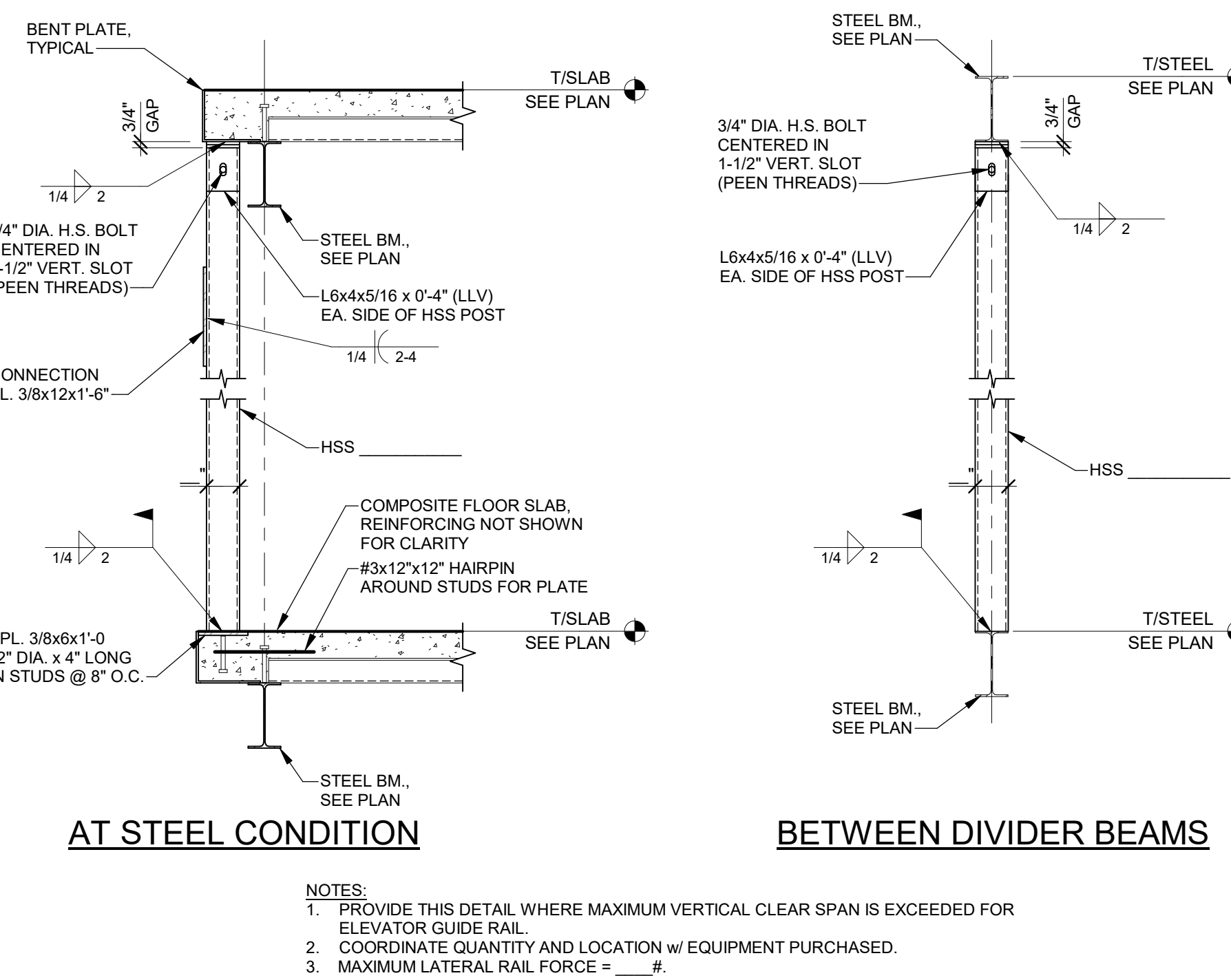
TYPICAL INTERIOR STAIR-ON-GRADE

4
S0004



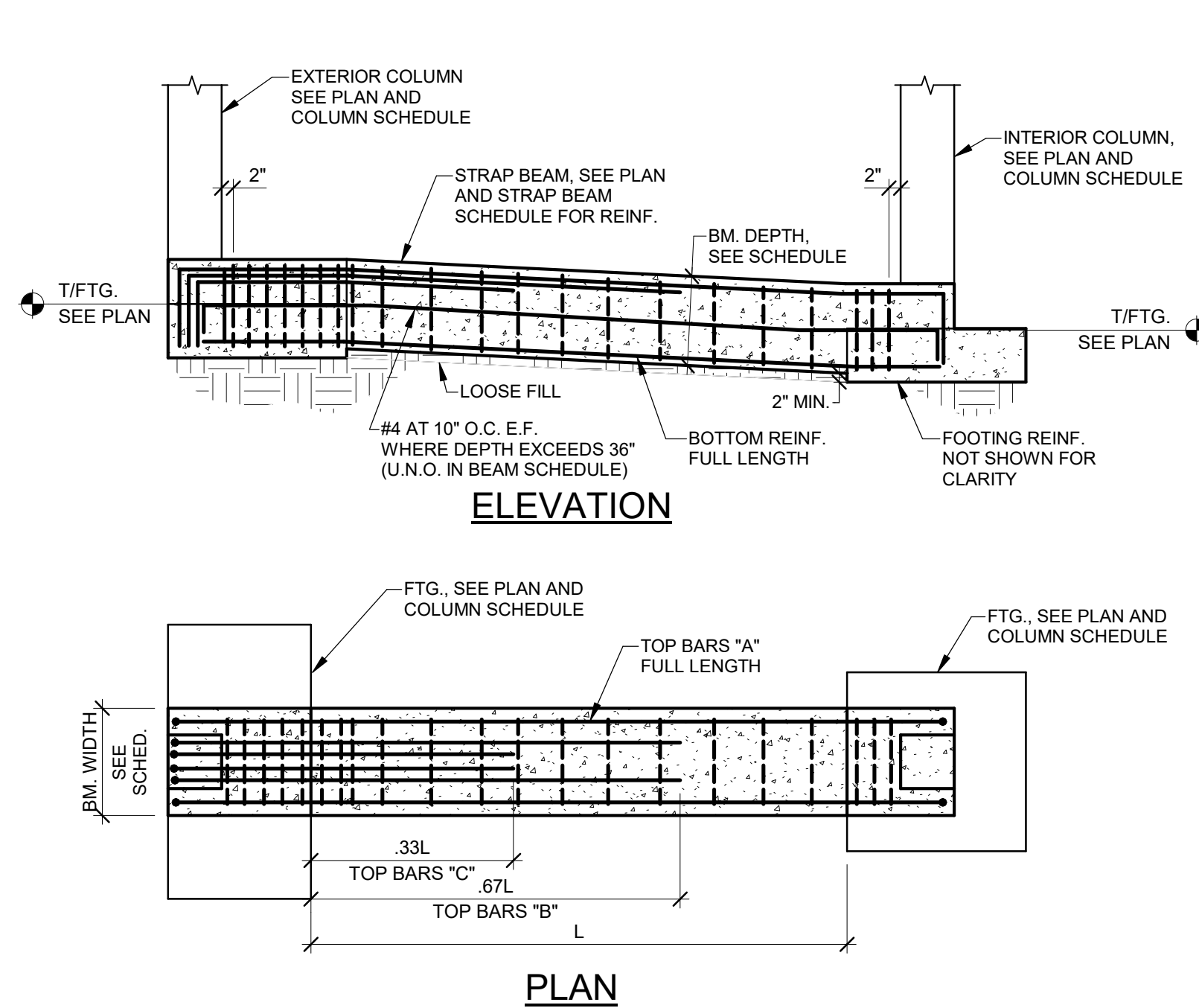
TYPICAL EXTERIOR STAIR

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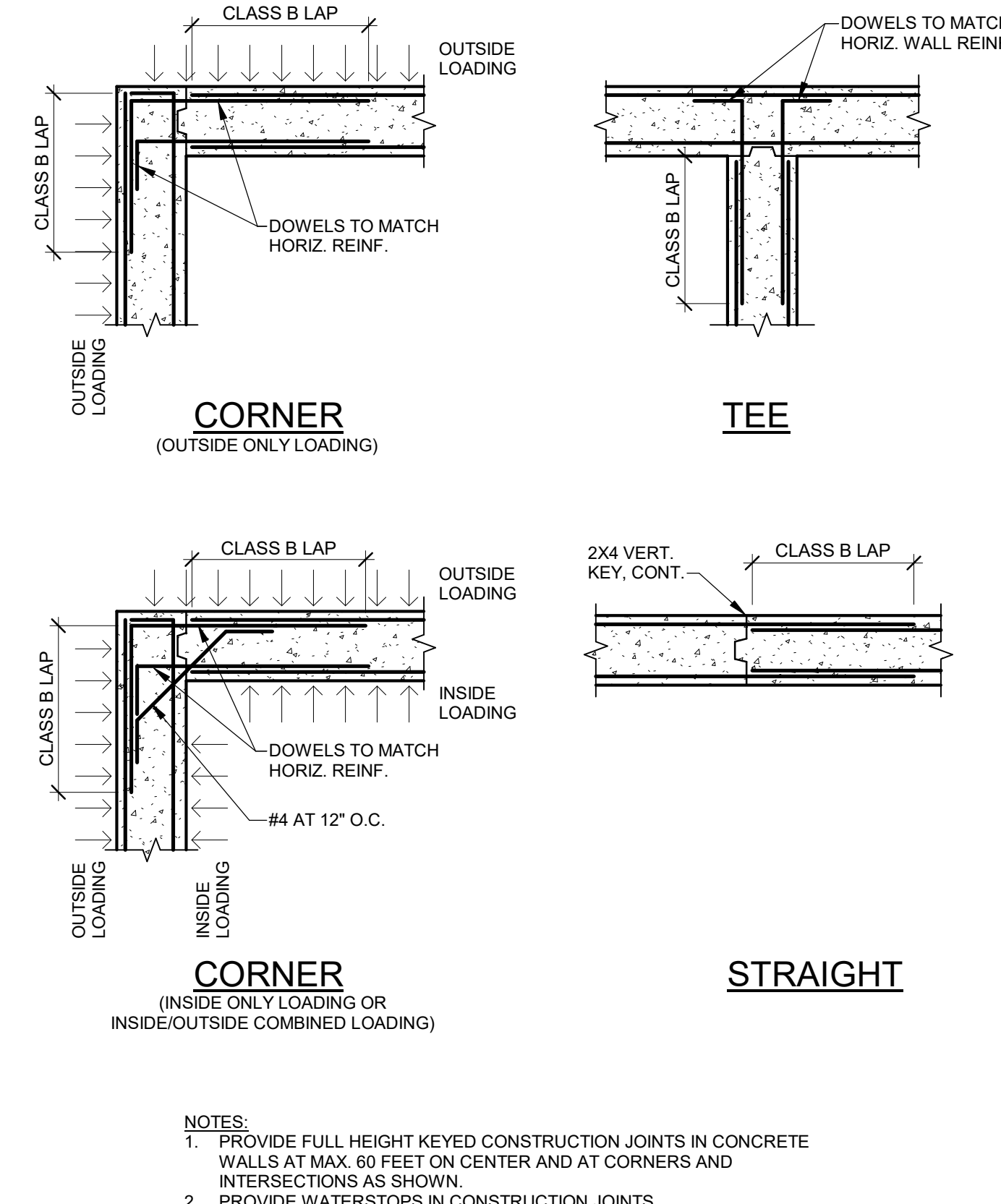
TYPICAL ELEVATOR RAIL SUPPORT

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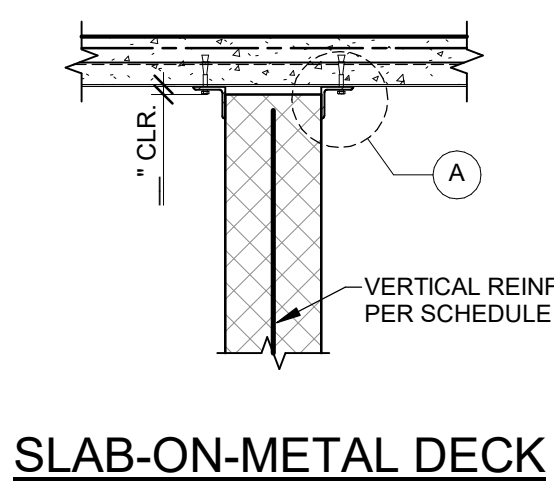
TYPICAL STRAP BEAM DETAIL

K
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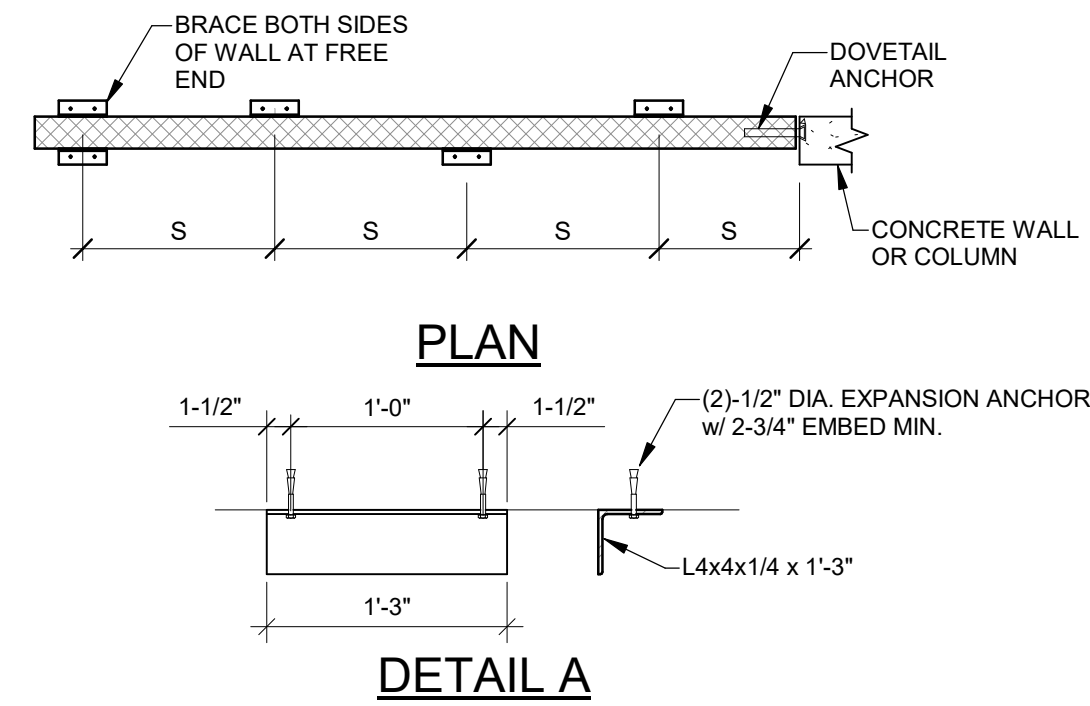


TYPICAL WALL CONSTRUCTION JOINT PLAN DETAILS

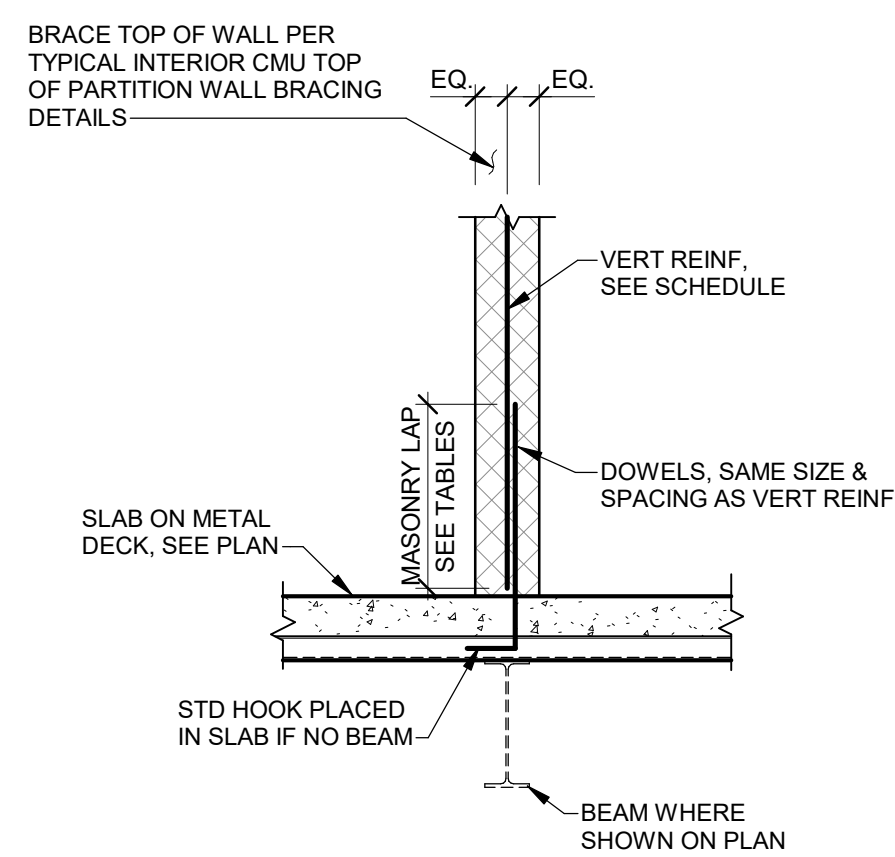
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SLAB-ON-METAL DECK

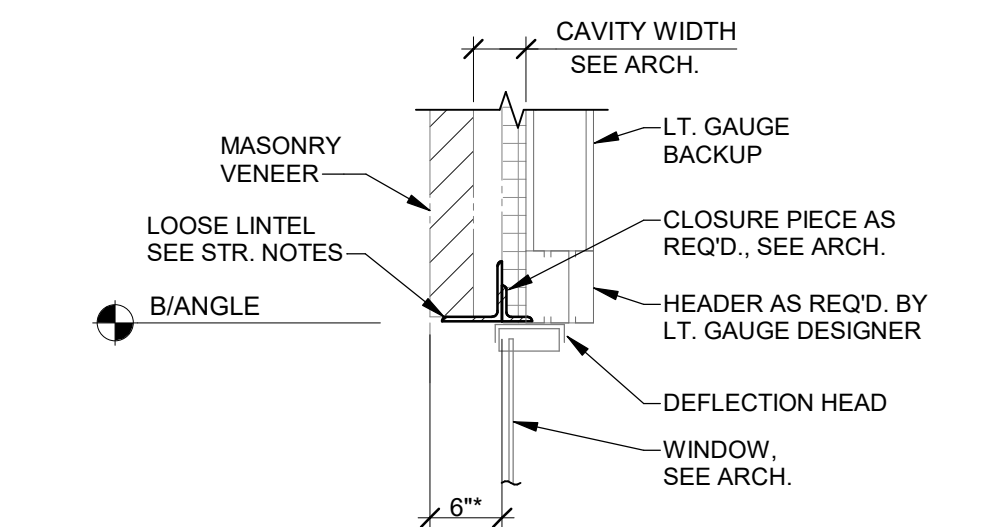


DETAIL A

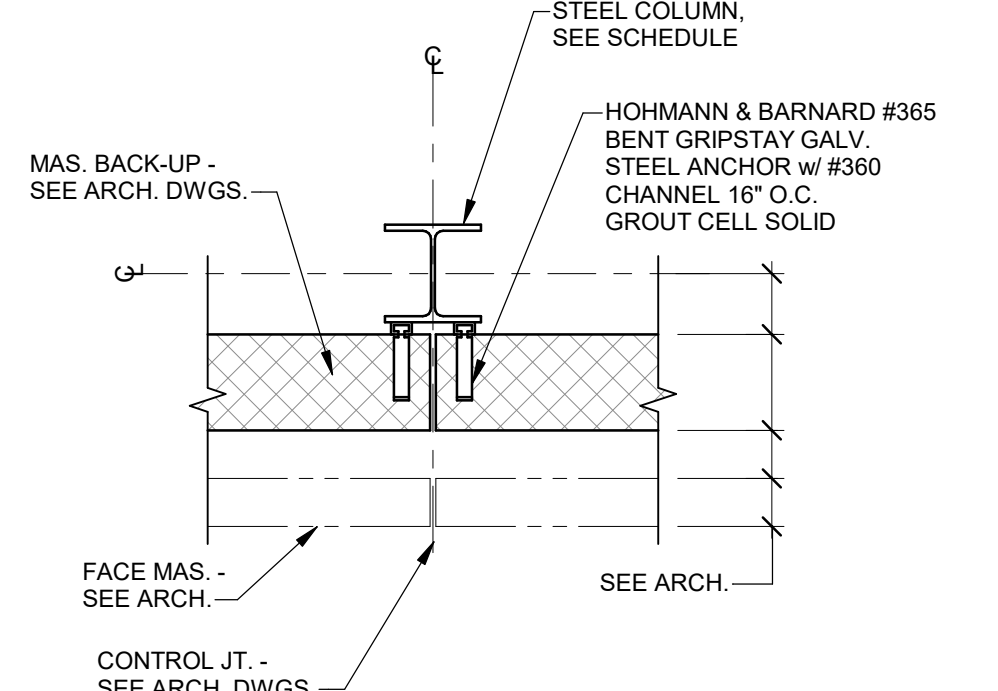


INTERIOR CMU PARTITION WALL REINFORCING SCHEDULE		
NOM. WALL THICKNESS	MAX WALL HEIGHT	VERT. REINF.
8"	10'-0"	UNREINFORCED*
10"	25'-0"	#4@48"
12"	30'-0"	#4@48"

- NOTES:
- INTERIOR CMU PARTITION REINFORCING IS REQUIRED AT WALLS NOT EXPOSED TO WIND LOADS, UNO.
 - WHERE INTERIOR WALLS ARE ALSO VEHICLE BARRIERS WALLS, THE WORST CASE REINFORCING SHALL CONTROL.
 - PROVIDE TRIM REINFORCING AT WALL ENDS, INTERSECTIONS AND EACH SIDE OF CONTROL JOINTS. TRIM REINFORCING SIZE SHALL MATCH TYPICAL WALL REINFORCING SIZE. MINIMUM #16 WHERE NONE IS INDICATED IN SCHEDULE.
 - PROVIDE ONE-HALF OF THE INTERRUPTED REINFORCING EACH SIDE OF OPENINGS W/ BARS @ 8" O.C. (MINIMUM #14), UNLESS NOTED OTHERWISE IN PLAN OR DETAILS.
 - ALL CELLS WITH REINFORCING TO BE GROUTED. GROUT CMU SOLID BELOW GRADE. REFER TO ARCHITECTURE FOR ADDITIONAL GROUT REQUIRED FOR DURABILITY AND LIFE SAFETY REQUIREMENTS.
 - WHERE STRAIGHT DEVELOPMENT OF REINFORCING CANNOT BE ACCOMMODATED, HOOK REINFORCING IN BEAM OR SLAB. EMBED HOOK 6" MIN INTO CONCRETE.
 - FOR PARTITIONS SUPPORTED ON SLAB-ON-GRADE PROVIDE THICKENED SLAB. SEE TYPICAL DETAIL.
 - UNREINFORCED WALLS ARE NOT PERMITTED AT ELEVATOR SHAFT WALLS. USE #4@8" O.C. MIN.
 - HORIZONTAL JOINT REINFORCING NOT SHOWN FOR CLARITY. PROVIDE IN ACCORDANCE WITH GENERAL NOTES AND SPECIFICATIONS.



- NOTES:
- SEE STRUCTURAL NOTES FOR LOOSE LINEL ANGLE BASED ON OPENING SIZE AND CAVITY DEPTH.
 - SUPPORTED HEIGHT OF MASONRY SHALL NOT EXCEED 4'-0". U.N.O.
 - WHERE CAVITY WIDTH EXCEEDS 2" MOVE ANGLE OUTBOARD OF INSULATION LAYER. PROVIDE LOOSE LINEL SIZED FOR A 1" CAVITY, AND SET (1) DIMENSION TO 5" MAX.



- NOTES:
- FACING TIES AND MASONRY BACK-UP HORIZONTAL REINF. NOT SHOWN.
 - REINFORCE MASONRY BACK-UP PER NOTES AND DETAILS.

TYPICAL INTERIOR CMU WALL BRACING

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TYPICAL INTERIOR CMU PARTITION WALL REINFORCING (NON LOAD-BEARING)

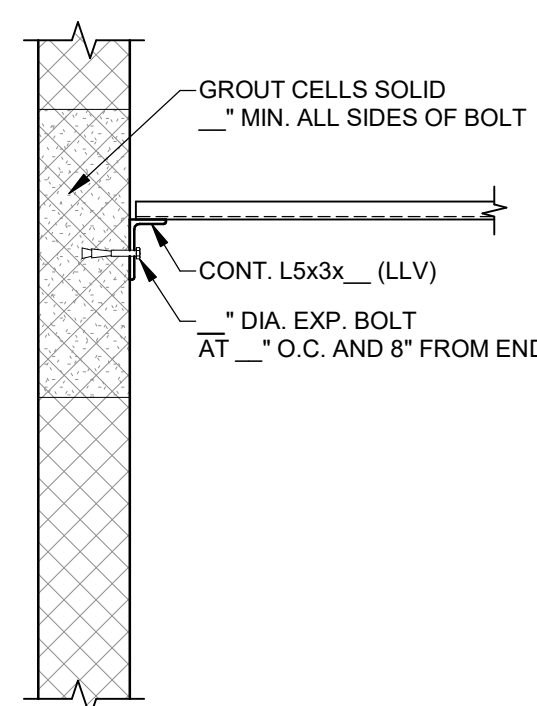
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S0005

TYPICAL CAVITY WALL LOOSE LINEL

C
S0005

TYPICAL MASONRY CONTROL JOINT AT COLUMN

D
S0005



TYPICAL DECK SUPPORT ANGLE

E
S0005

TYPICAL MASONRY WALL BRACING PARALLEL TO BEAM

F
S0005

TYPICAL BOND BEAM DETAILS

G
S0005

TYPICAL STEEL BEAM FRAMING INTO BOND BEAM

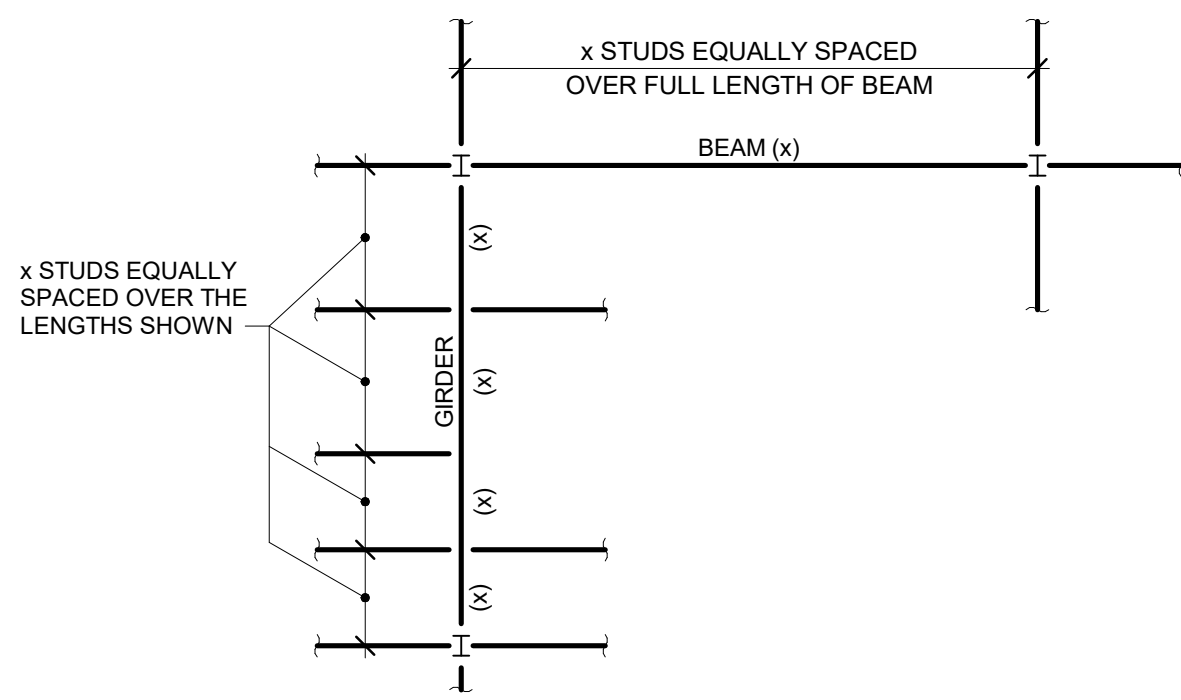
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MASONRY LAP SPICE LENGTHS

J
S0005

BAR SIZE	MASONRY LAP SPICE LENGTHS (Fm = 2,000 PSI)				
	REINFORCING CENTERED IN CMU CELL	6" CMU	8" CMU	10" CMU	12" CMU
#4	18	13	12	12	22
#5	28	20	16	13	35
#6	53	38	29	24	54
#7	NP	52	40	33	63
#8	NP	72	61	50	72
#9	NP	NP	79	64	82

- NOTES:
- PRIOR TO GROUTING, ALL MORTAR FINS TO BE REMOVED FROM 6" CMU CELLS WITH REINFORCING.
 - WHEN EPOXY COATED REINFORCING IS USED, INCREASE LAP SPICE LENGTH BY 50%.
 - MECHANICAL SPICES MAY BE USED IN LIEU OF LAP SPICES. MECHANICAL SPICES SHALL DEVELOP 125% OF F_y.
 - BARS SPICED BY NON-CONTACT LAP SPICES SHALL BE SPACED NO FARTHER APART THAN ONE-FIFTH THE REQUIRED LENGTH OF LAP AND NO MORE THAN 8".
 - "NP" INDICATES BAR SIZE THAT IS NOT PERMITTED FOR THE GIVEN CMU THICKNESS.



NOTES:
GENERAL:
PLACE FIRST STUD 12" FROM CENTERLINE OF SUPPORTS, EACH END.
PLACE STUDS AT SIDE OF DECK RIB CLOSEST TO SUPPORT (STRONG POSITION).
MAXIMUM STUD SPACING = 24"
MINIMUM LONGITUDINAL STUD SPACING = 4-1/2"
MINIMUM TRANSVERSE SPACING = 3"

DECK PERPENDICULAR TO SUPPORTS:
FEWER STUDS THAN VALLEYS: START AT EACH END OF BEAM (COLUMN OR WALL) AND PLACE A STUD IN ALTERNATING FLUTES FOR LENGTH OF BEAM (EVERY FOURTH FLUTE IF RIBS ARE 6" O.C.), WORKING TOWARD CENTER OF BEAM SPAN. IF STUDS REMAIN, PLACE STUD IN EMPTY FLUTES STARTING AT BEAM ENDS AND WORKING TOWARD CENTER.

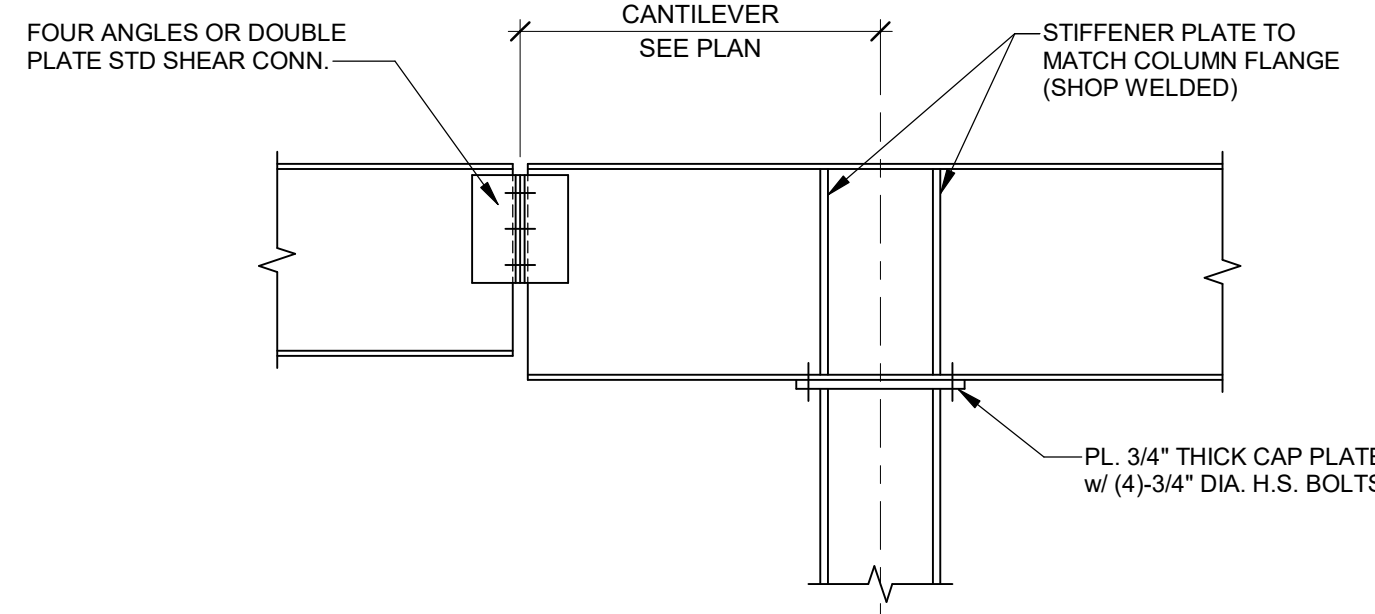
MORE STUDS THAN VALLEYS: PLACE A STUD IN EACH AVAILABLE FLUTE. IF STUDS REMAIN, STARTING AT BEAM ENDS, PLACE A SECOND STUD IN EACH FLUTE WORKING TOWARD CENTER OF SPAN. IF STUDS REMAIN, PLACE A THIRD STUD IN EACH FLUTE IN THE SAME SEQUENCE AS ABOVE.

DECK PARALLEL TO SUPPORTS:
SEGMENTAL LAYOUT: PLACE A SINGLE ROW OF STUDS IN THE END SEGMENT OF THE GIRDER. IF SPACING IS LESS THAN 4-1/2", USE DOUBLE ROW OR TRIPLE ROW AS REQUIRED. CONTINUE TO INTERIOR SEGMENTS AND REPEAT LAYOUT AS ABOVE.

UNIFORM LAYOUT: PLACE A SINGLE ROW OF STUDS ALONG FULL LENGTH OF GIRDER AT EQUAL SPACING. IF SPACING IS LESS THAN 4-1/2", USE DOUBLE ROW OR TRIPLE ROW AS REQUIRED.

TYPICAL SHEAR STUD LAYOUT

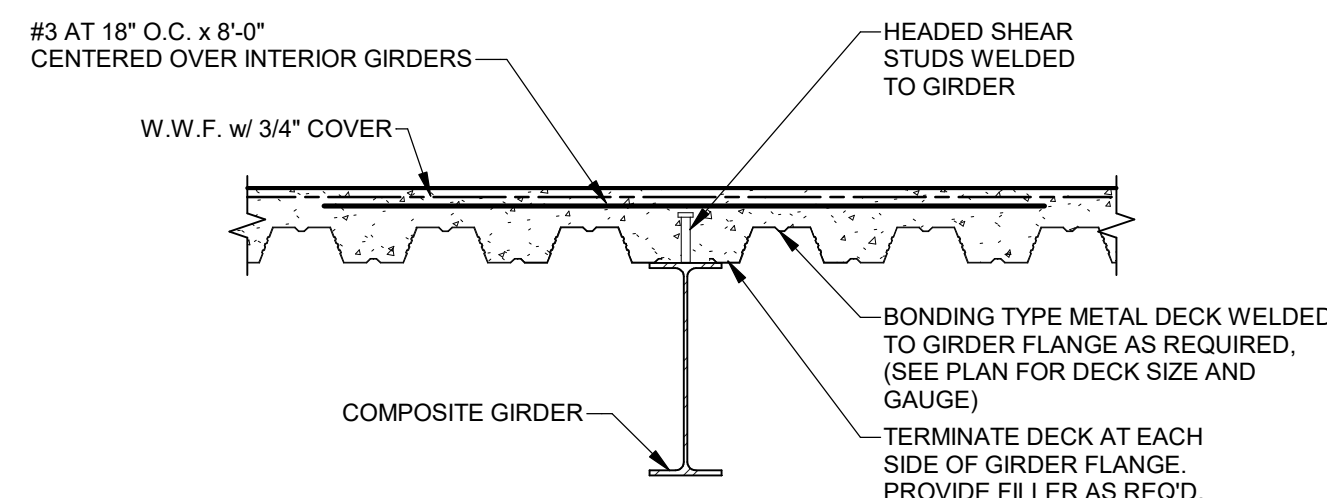
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NOTE:
1. MEMBERS SPANNING IN OPPOSITE DIRECTION NOT SHOWN FOR CLARITY.

TYPICAL CANTILEVERED BEAM

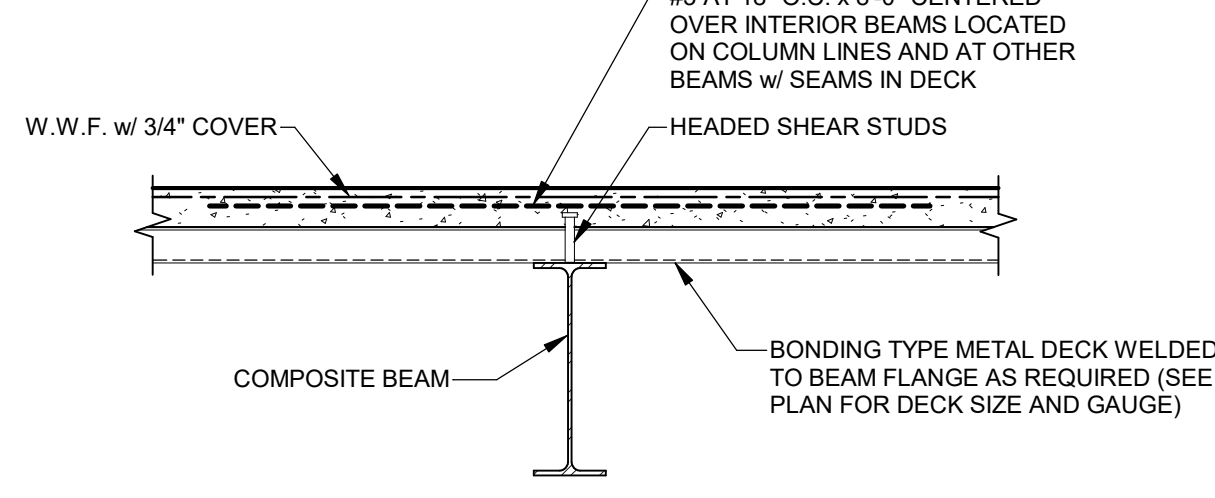
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NOTES:
1. NUMBER OF SHEAR STUDS SHALL BE AS SHOWN ON PLANS AND SPACED ACCORDING TO TYPICAL SHEAR STUD LAYOUT DETAIL.
2. FOR W12X16 AND SMALLER, WELD STUDS DIRECTLY OVER THE WEB TO AVOID DAMAGE TO THE FLANGE.
3. FOR W12X18 AND LARGER, STUDS MAY BE STAGGERED, WITH THE CENTERLINES OF THE STUDS A MINIMUM OF 1-3/8" FROM THE EDGE OF FLANGE.
4. LOCATE PARALLEL CONSTRUCTION JOINTS, AT MID-SPAN BETWEEN FILLER BEAMS, AS REQUIRED. W.W.F. LAP+WIRE SPACING PLUS 2"

TYPICAL COMPOSITE GIRDER

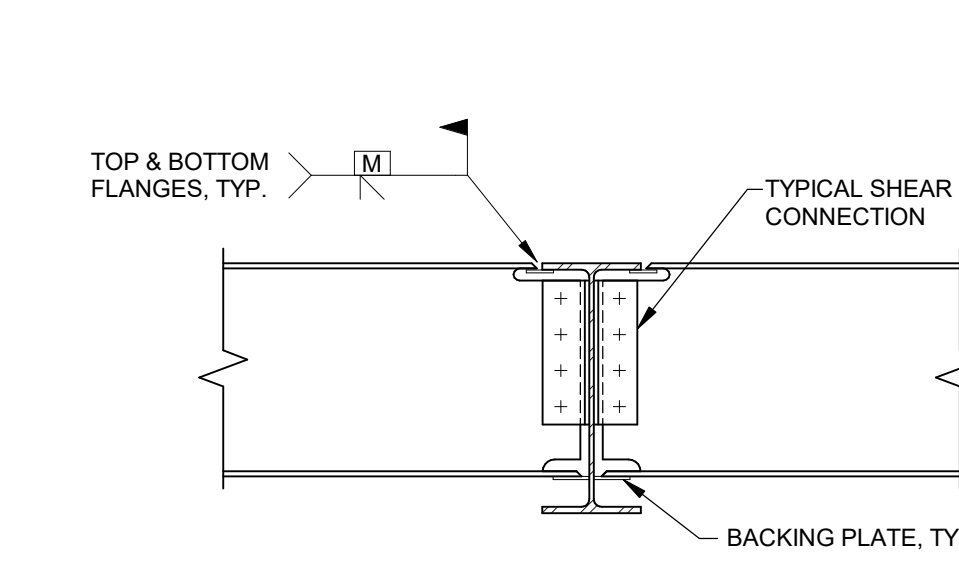
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NOTES:
1. NUMBER OF SHEAR STUDS SHALL BE AS SHOWN ON PLANS AND SPACED ACCORDING TO TYPICAL SHEAR STUD LAYOUT DETAIL.
2. FOR W12X16 AND SMALLER, WELD STUDS DIRECTLY OVER THE WEB TO AVOID DAMAGE TO THE FLANGE.
3. FOR W12X18 AND LARGER, STUDS MAY BE STAGGERED, WITH THE CENTERLINES OF THE STUDS A MINIMUM OF 1-3/8" FROM THE EDGE OF FLANGE.
4. LOCATE PARALLEL CONSTRUCTION JOINTS, AT MID-SPAN BETWEEN FILLER BEAMS, AS REQUIRED. W.W.F. LAP+WIRE SPACING PLUS 2"

TYPICAL COMPOSITE BEAM

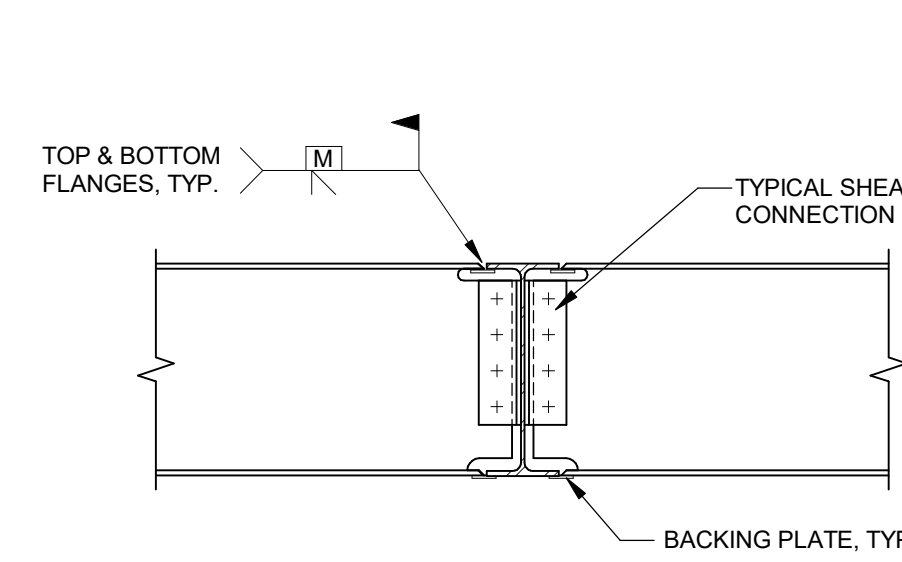
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NOTE:
1. WELDS SHALL BE PRE-QUALIFIED FULL PENETRATION.

TYPICAL BEAM TO BEAM MOMENT CONNECTION

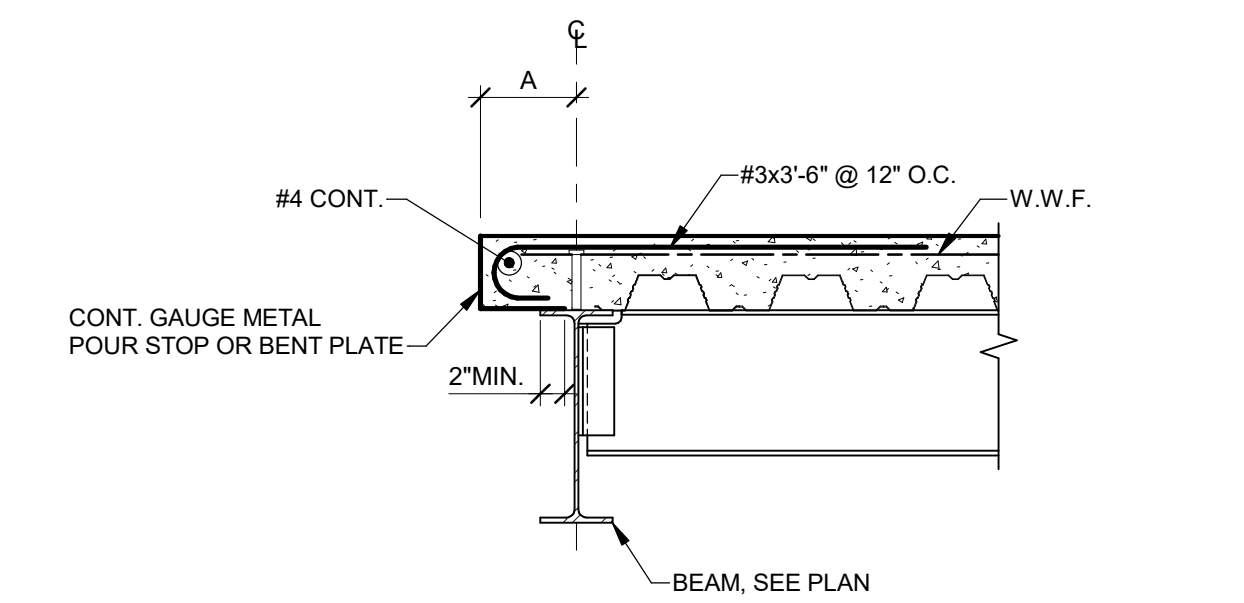
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NOTE:
1. WELDS SHALL BE PRE-QUALIFIED FULL PENETRATION.

TYPICAL BEAM TO BEAM MOMENT CONNECTION

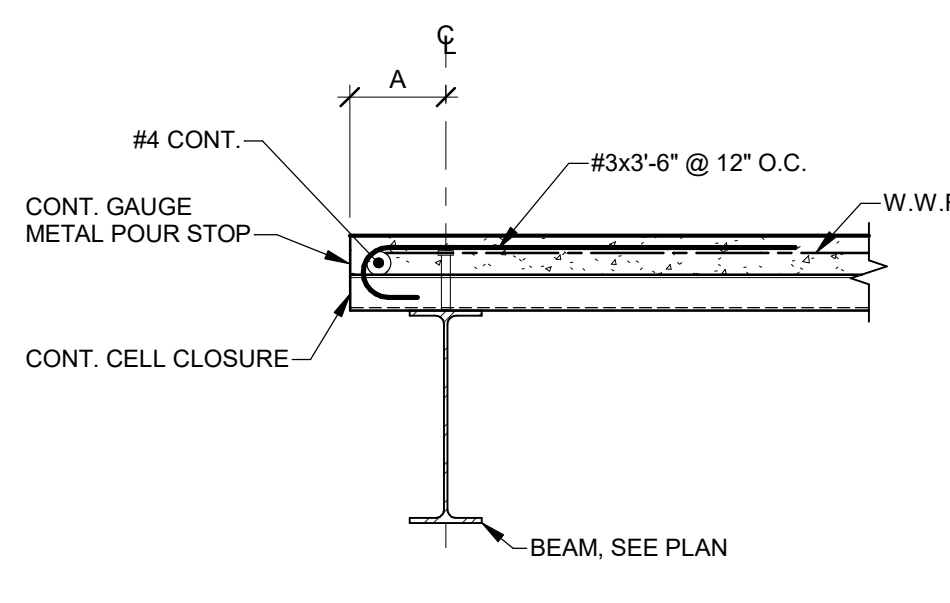
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S0006



NOTES:
1. SLAB REINFORCING MAY BE OMITTED WHERE "A" IS LESS THAN SLAB THICKNESS.
2. GAUGE METAL POUR STOP THICKNESS AND OVERHANG PER SDI POUR STOP SELECTION TABLE.
3. PROVIDE BENT PL. 1/4" FOR OVERHANGS GREATER THAN SDI TABLE, UP TO 10".
4. PROVIDE HORIZONTAL ADJUSTMENT IN BENT PLATE, FIELD WELD TOE WITH 3/16"x1" FILLET AT 12" O.C.
5. ENDS OF STAIR STRINGERS SHALL BE CONFIGURED TO CONNECT DIRECTLY TO WEBS OF EDGE BEAMS. CONNECTIONS AT EDGE OF SLAB ARE NOT PERMITTED.

TYPICAL DETAIL AT SHAFTS AND STAIRS (DECK PARALLEL TO BEAM)

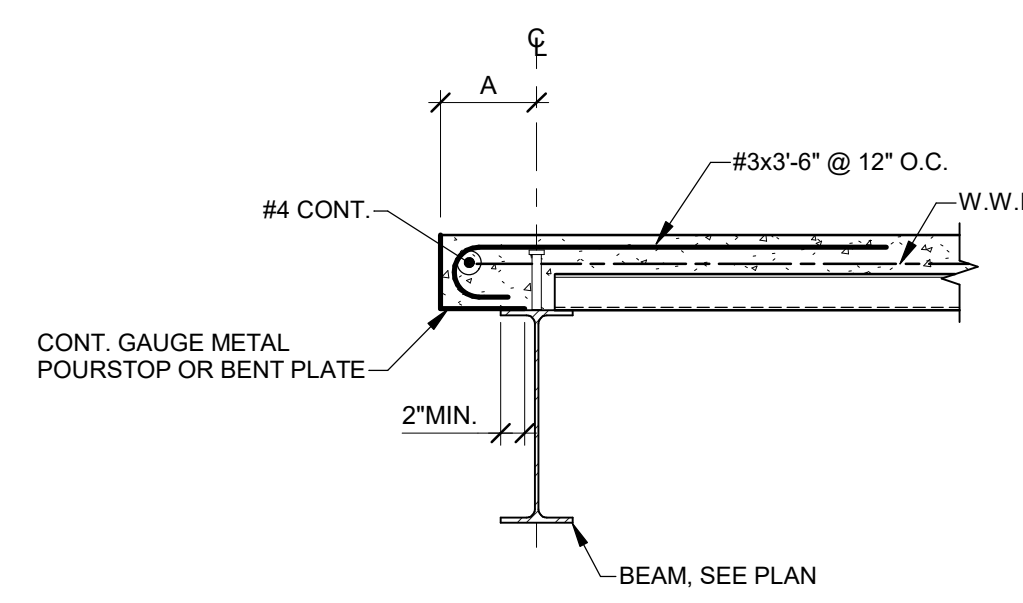
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NOTES:
1. SLAB REINFORCING MAY BE OMITTED WHERE "A" IS LESS THAN 10".
2. GAUGE METAL POUR STOP THICKNESS AND OVERHANG PER SDI POUR STOP SELECTION TABLE.
3. ENDS OF STAIR STRINGERS SHALL BE CONFIGURED TO CONNECT DIRECTLY TO WEBS OF EDGE BEAMS. CONNECTIONS AT EDGE OF SLAB ARE NOT PERMITTED.

TYPICAL DETAIL AT SHAFTS AND STAIRS (DECK PERPENDICULAR TO BEAM)

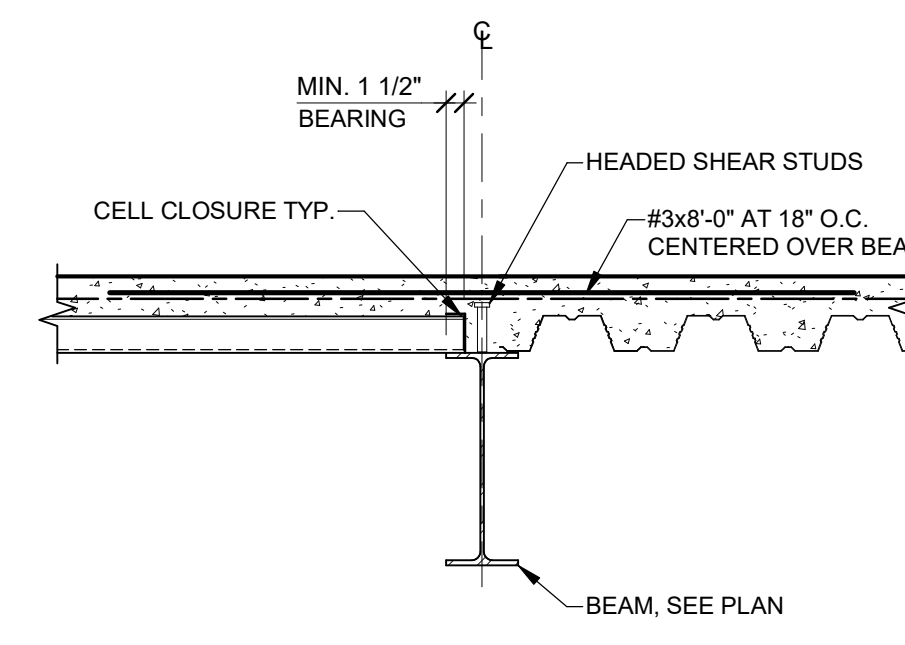
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NOTES:
1. SLAB REINFORCING MAY BE OMITTED WHERE "A" IS LESS THAN SLAB THICKNESS.
2. GAUGE METAL POUR STOP THICKNESS AND OVERHANG PER SDI POUR STOP SELECTION TABLE.
3. PROVIDE BENT PL. 1/4" FOR OVERHANGS GREATER THAN SDI TABLE, UP TO 10".
4. PROVIDE HORIZONTAL ADJUSTMENT IN BENT PLATE, FIELD WELD TOE WITH 3/16" x 1" FILLET AT 12" O.C.
5. ENDS OF STAIR STRINGERS SHALL BE CONFIGURED TO CONNECT DIRECTLY TO WEBS OF EDGE BEAMS. CONNECTIONS AT EDGE OF SLAB ARE NOT PERMITTED.

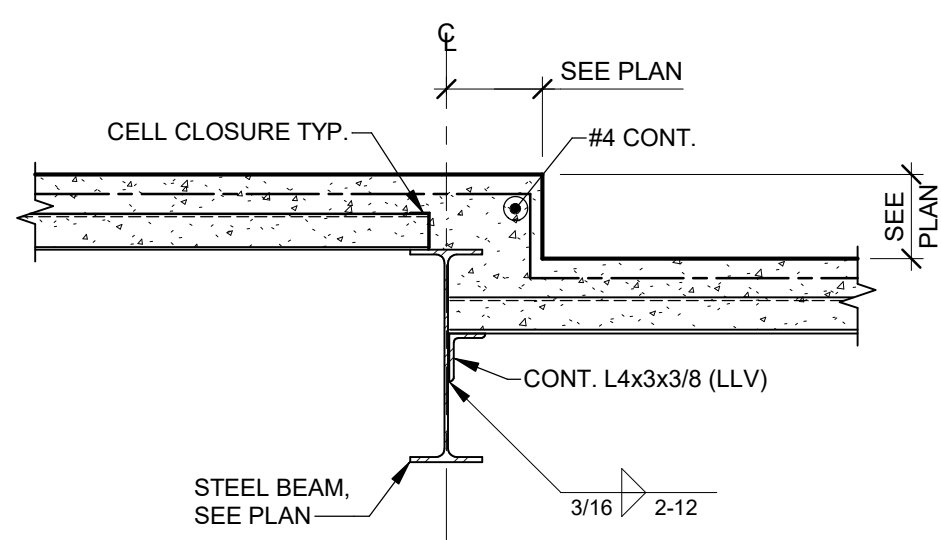
TYPICAL DETAIL AT SHAFTS AND STAIRS (DECK PERPENDICULAR TO BEAM)

J
S0006



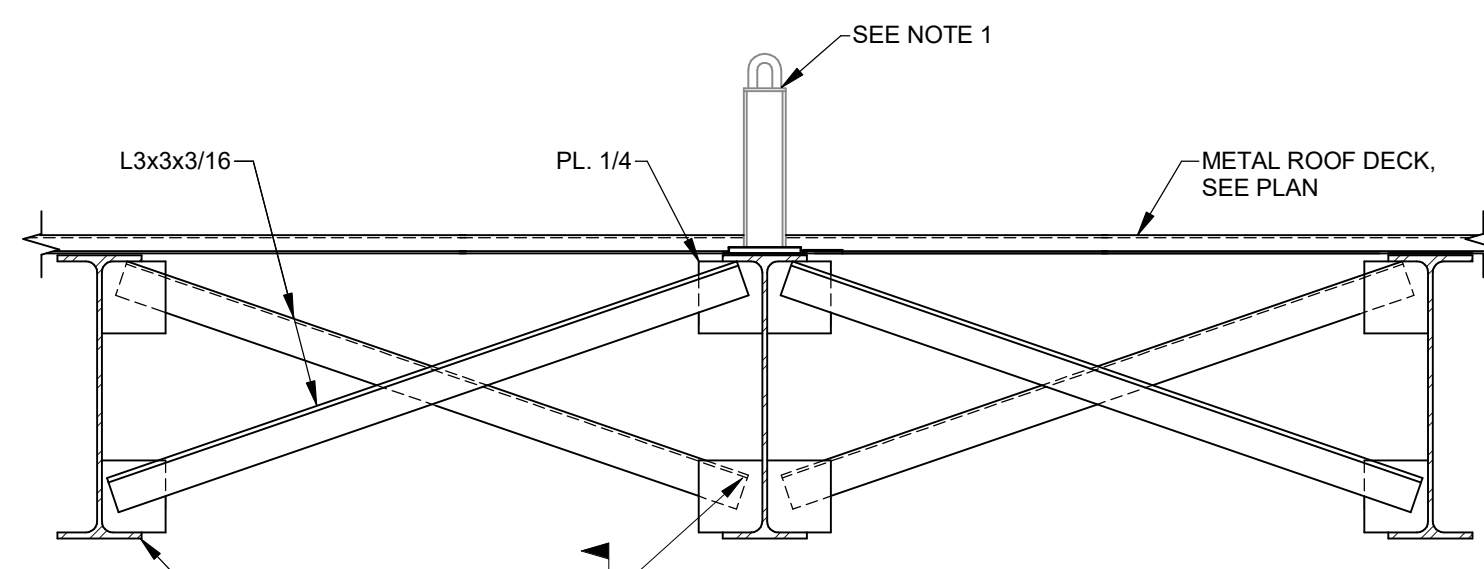
TYPICAL SLAB ON DECK CHANGE IN DIRECTION

K
S0006



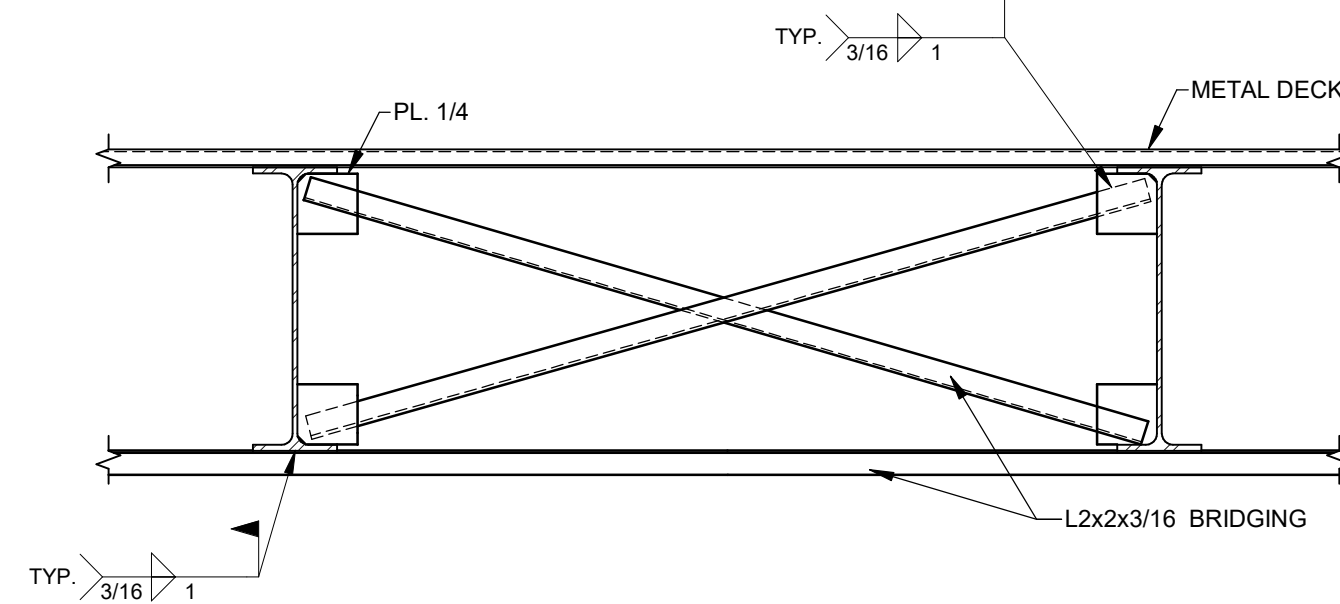
TYPICAL SLAB-ON-DECK STEP

L
S0006



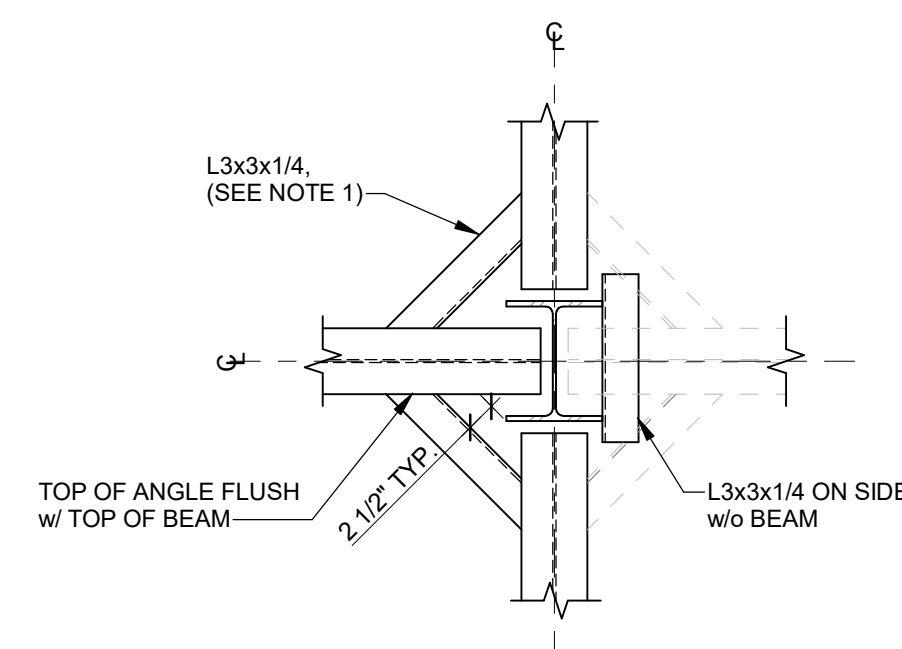
TYPICAL FACADE ACCESS EQUIPMENT ANCHOR

M
S0006



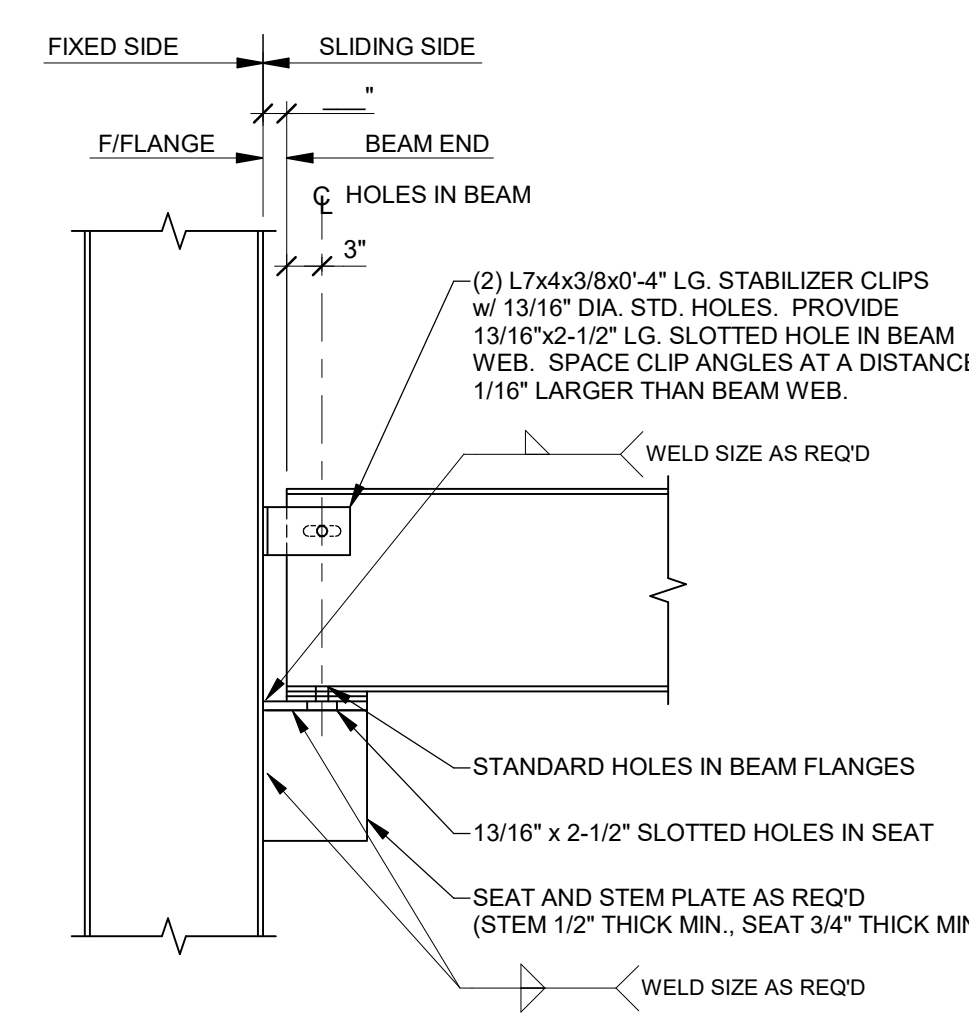
TYPICAL BRIDGING AT WF ROOF FRAMING

N
S0006



TYPICAL FLOOR DECK SUPPORT AT COLUMN

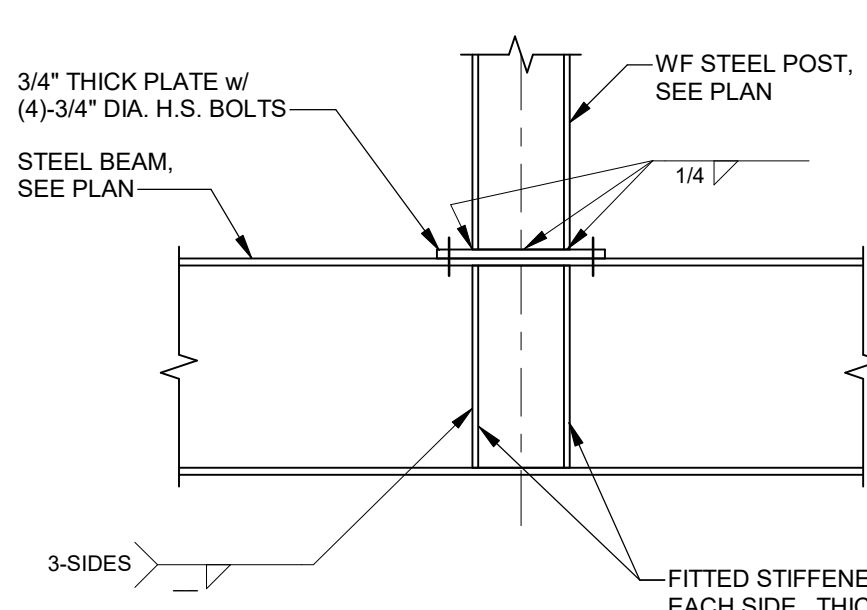
P
S0006



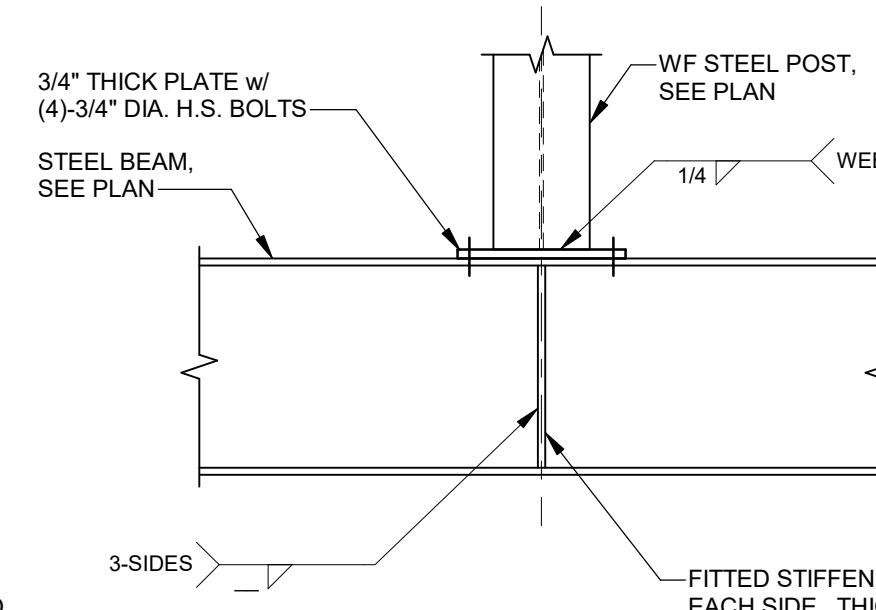
NOTES:
1. PROVIDE TEFLOON PLATES BONDED TO BEAM AND SEAT. TYPICAL.
2. ALL BOLTS SHALL BE 3/4" DIA. H.S. w/ NYLON WASHERS, HAND TIGHTENED.

TYPICAL BEAM TO COLUMN FLANGE EXPANSION JOINT CONNECTION

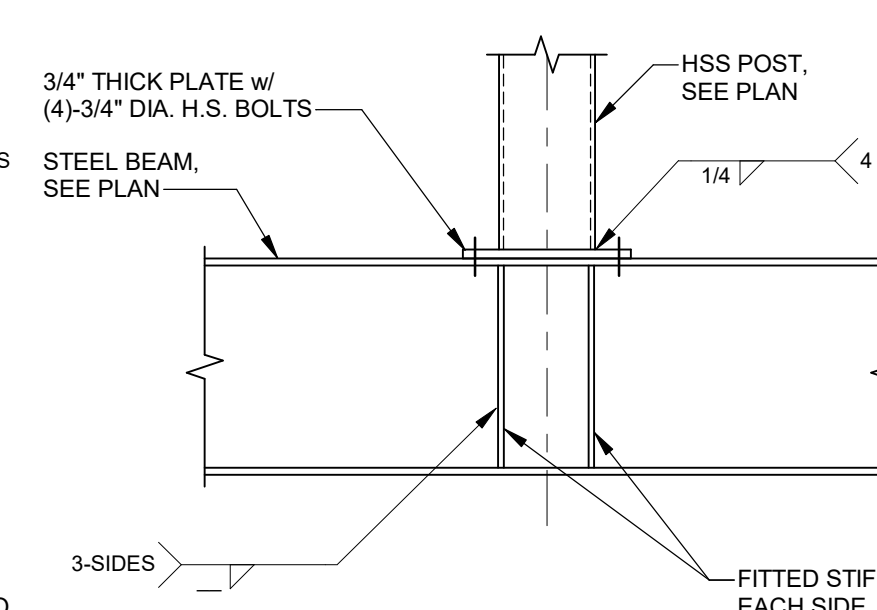
Q
S0006



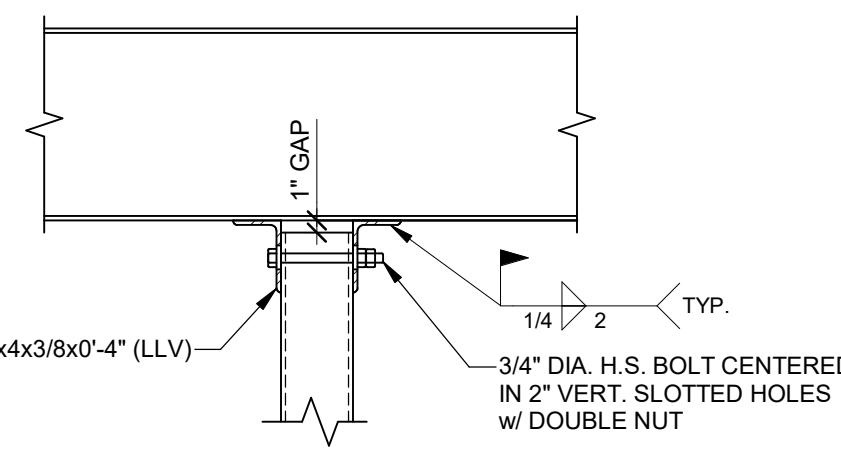
WEBS PARALLEL



WEBS PERPENDICULAR



HSS POST



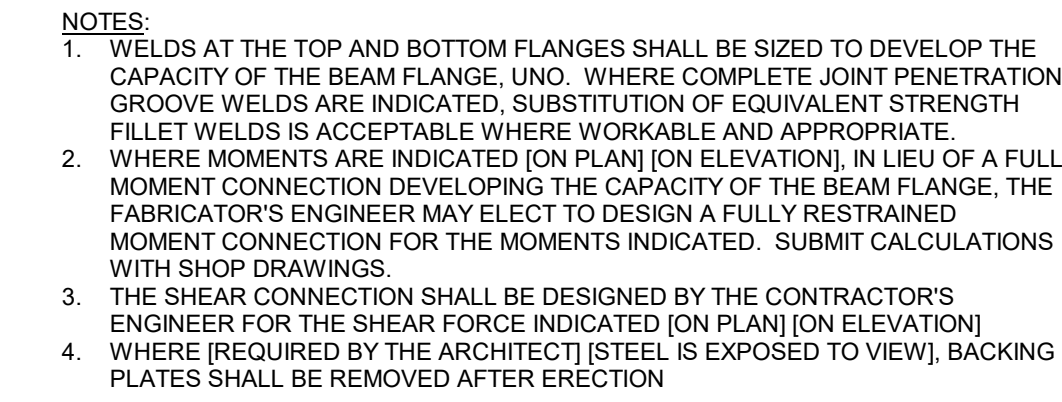
TYPICAL VERTICAL SLIP CONNECTION

S
S0006

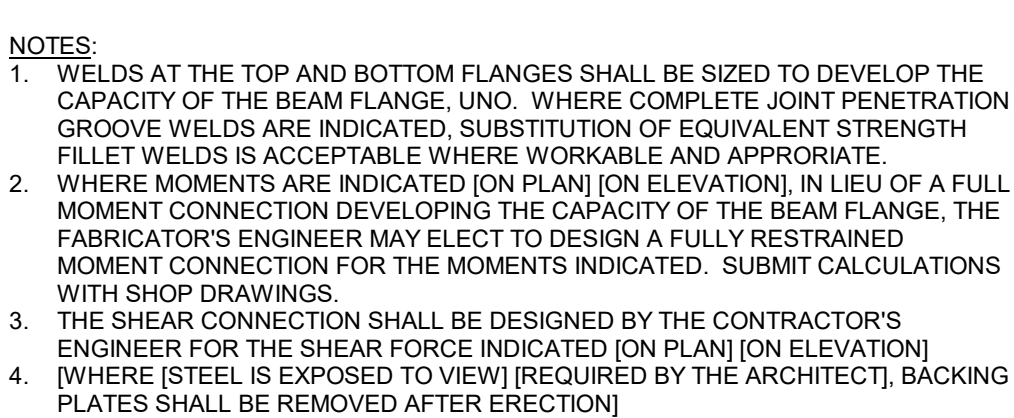
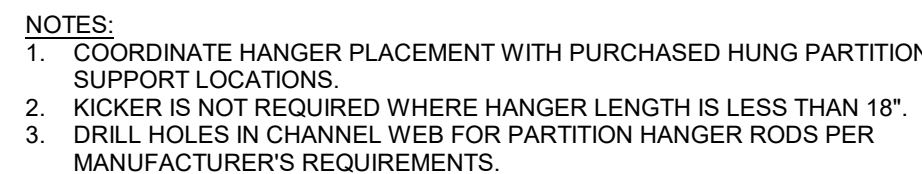
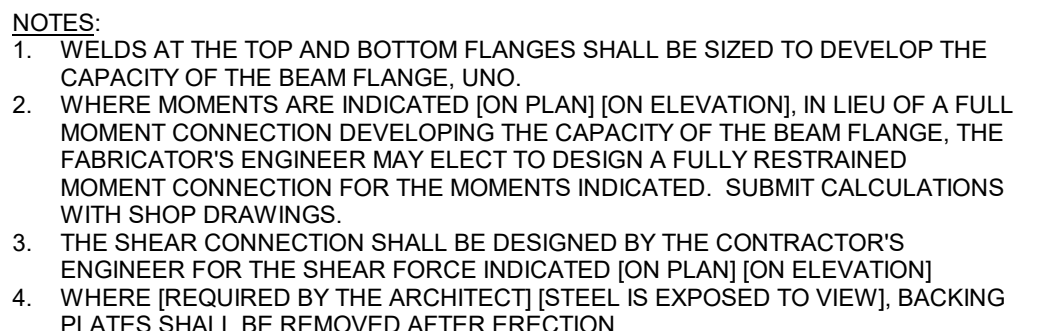
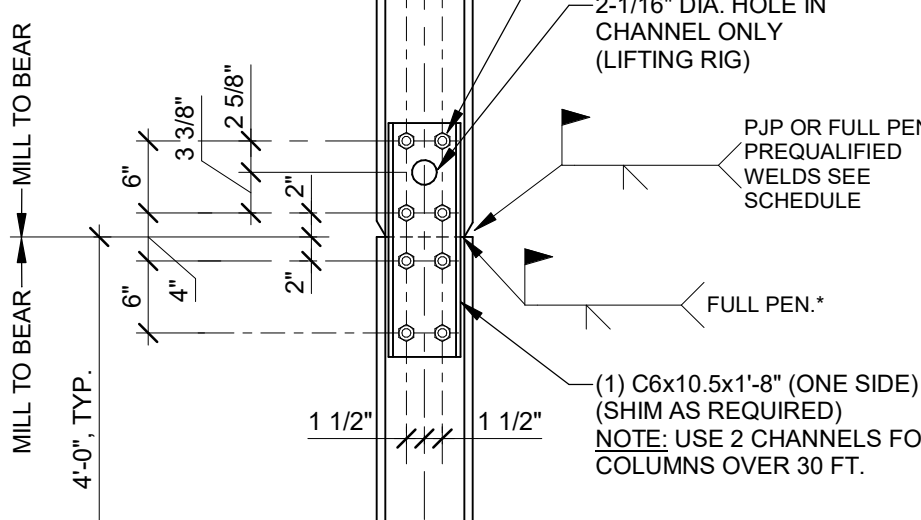
TYPICAL POST ON STEEL BEAM

R
S0006

NOTE:
1. CENTERLINE OF POST TO ALIGN WITH CENTERLINE OF BEAM WEB.
2. MEMBERS SPANNING IN OPPOSITE DIRECTION NOT SHOWN FOR CLARITY.

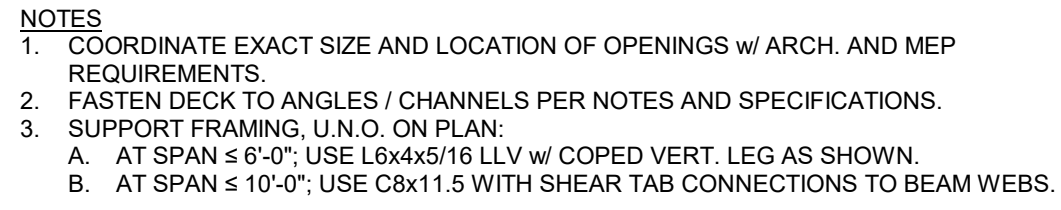




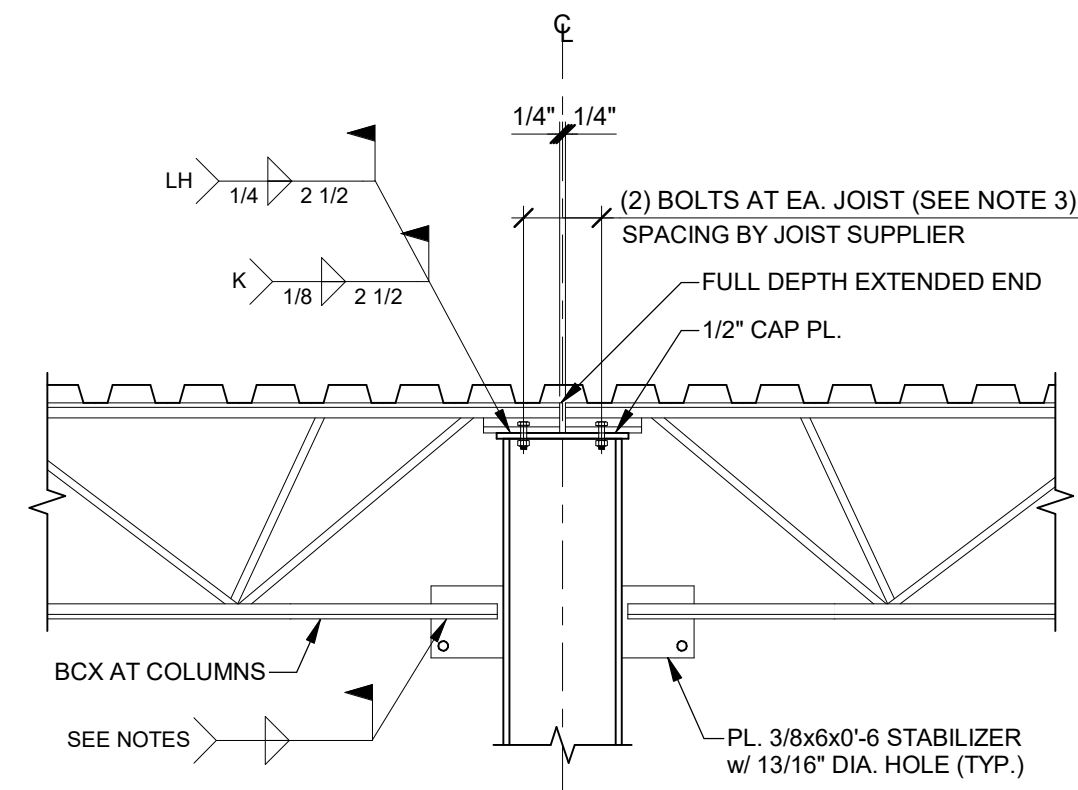
A circular logo with a thick black border. Inside the circle, the letter 'H' is centered at the top, and the alphanumeric string 'S0007' is centered below it.

NOTES:

1. USE FULL PENETRATION WELDS AT BRACED OR MOMENT FRAMES.
2. BTC-P4 & TC-P8 ARE PREQUALIFIED WELDS PER AISC STEEL CONSTRUCTION MANUAL.
3. FOR BRACED FRAMES AND MOMENT FRAMES: WELD WEB IN ADDITION TO FLANGES. WELD COLUMN FLANGES FIRST, REMOVE CHANNELS. THEN WELD COLUMN WEB.

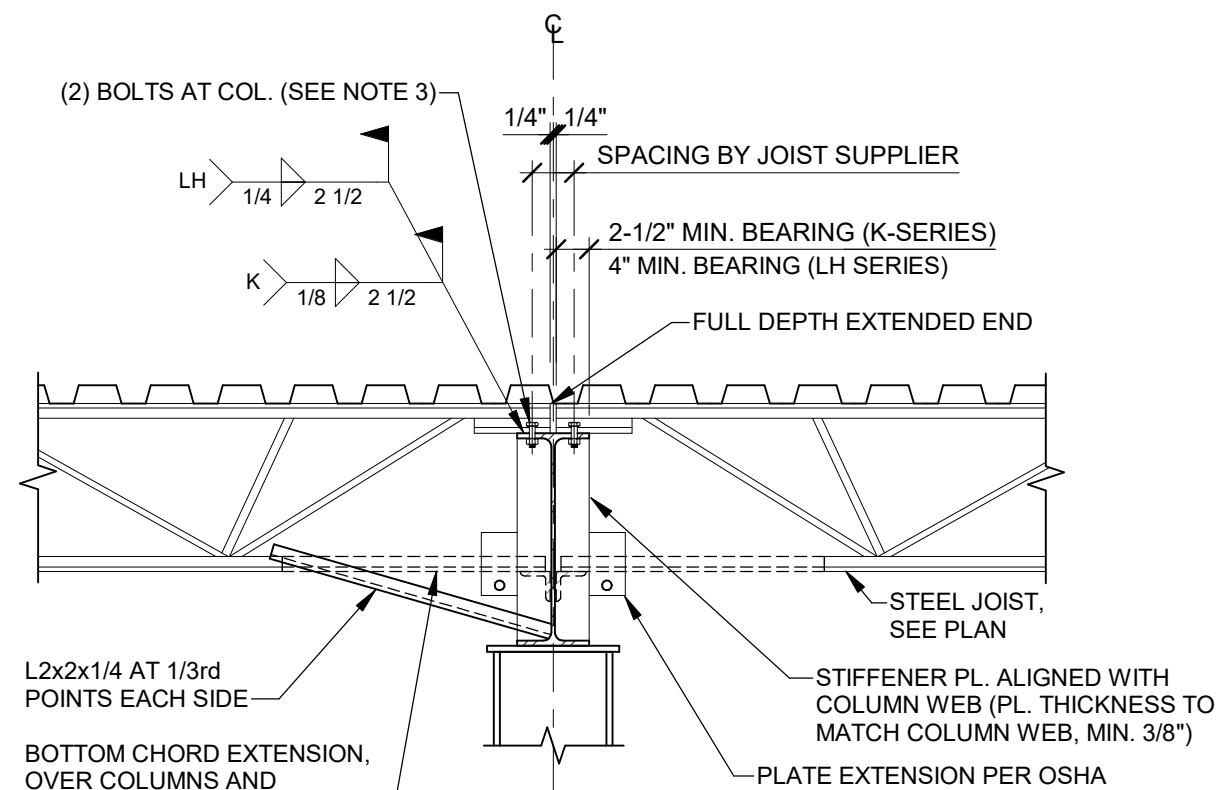


K
S0007



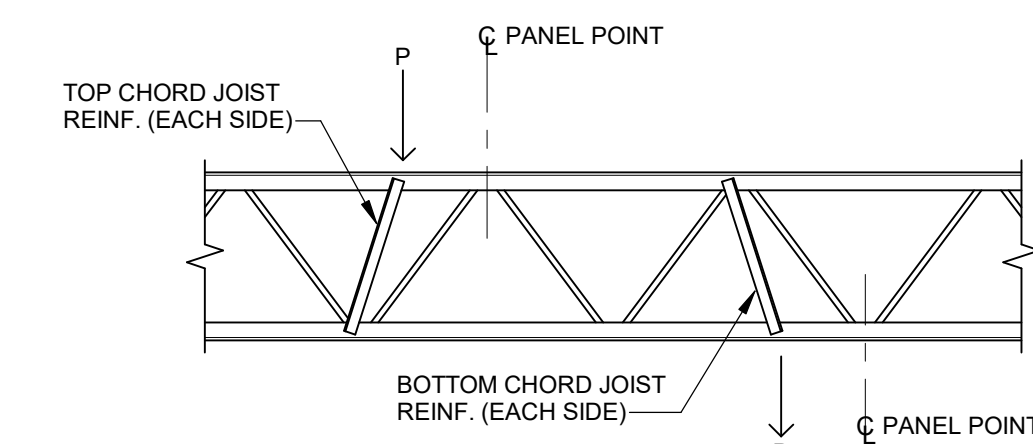
TYPICAL JOIST BEARING
AT INTERIOR COLUMN

A
S0008



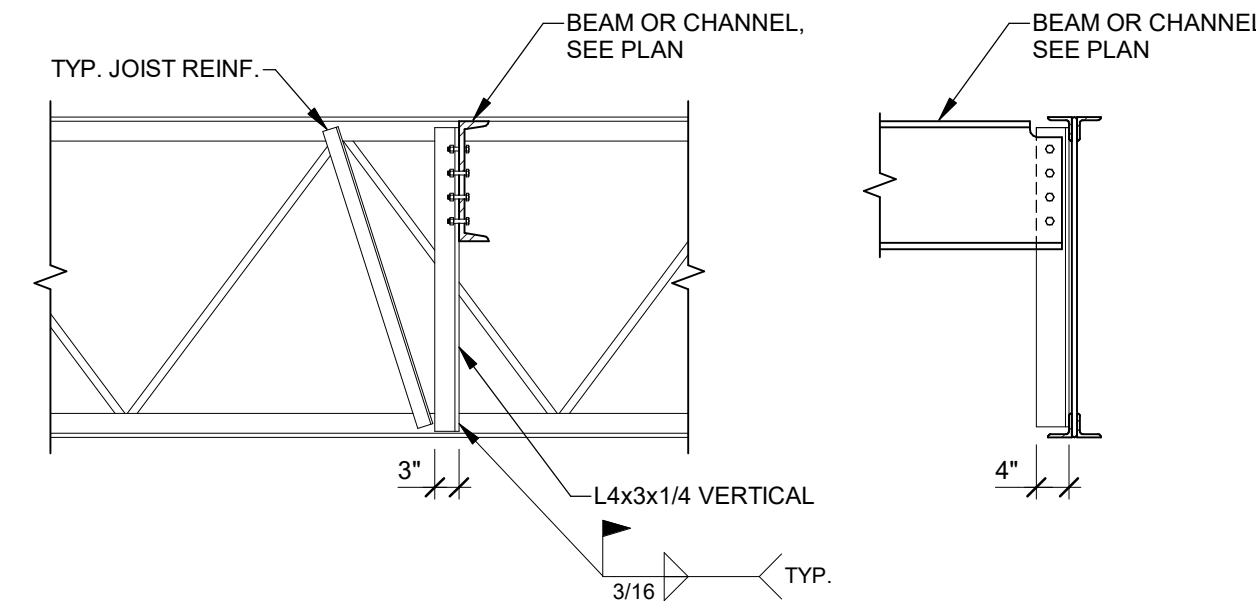
TYPICAL JOIST BEARING
AT BEAM ON COLUMN

B
S0008



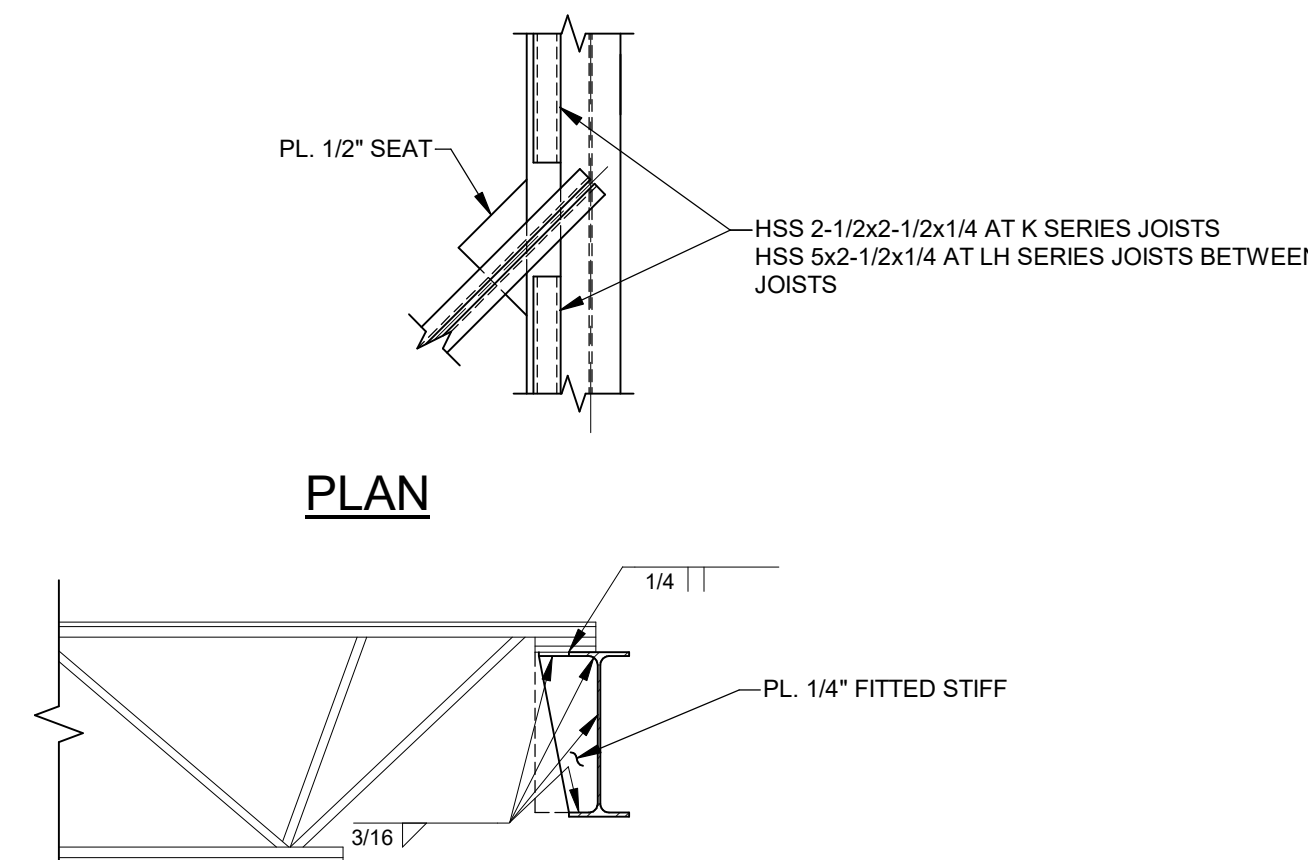
TYPICAL JOIST
REINFORCEMENT

C
S0008



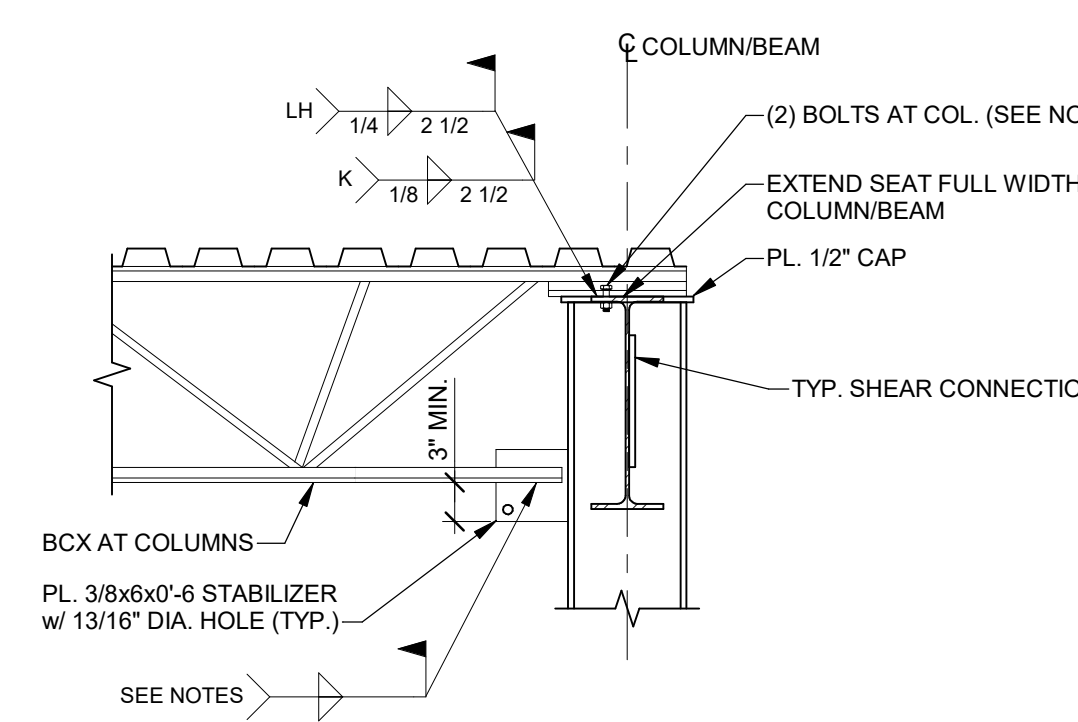
TYPICAL JOIST WITH HEADER

D
S0008



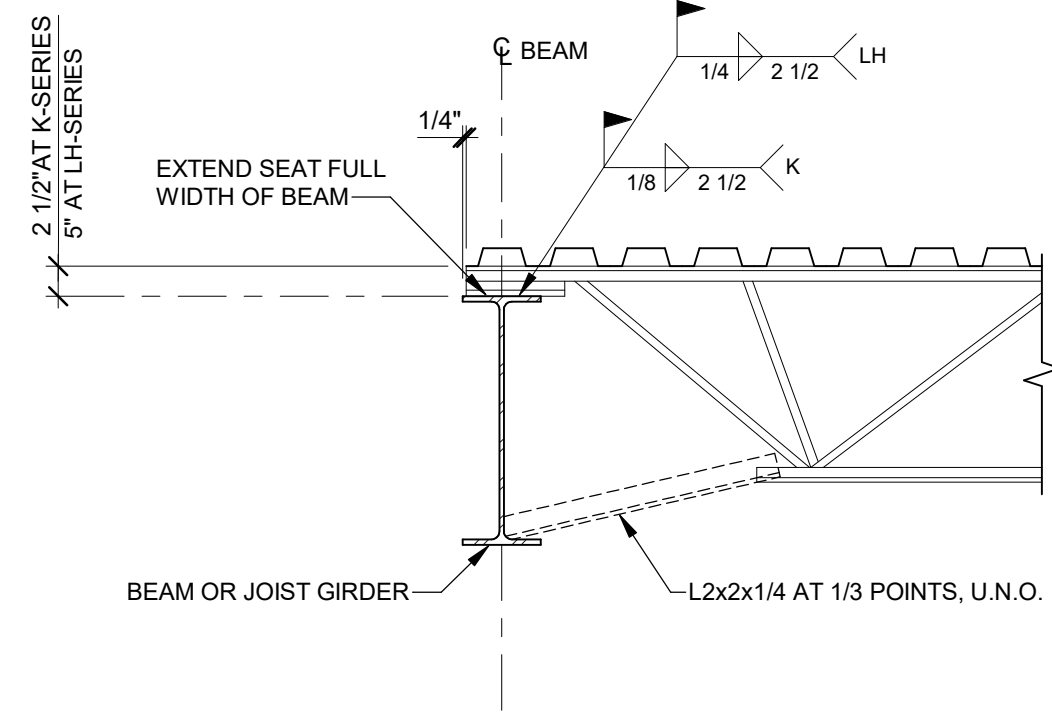
TYPICAL SKEWED JOIST
BEARING

E
S0008



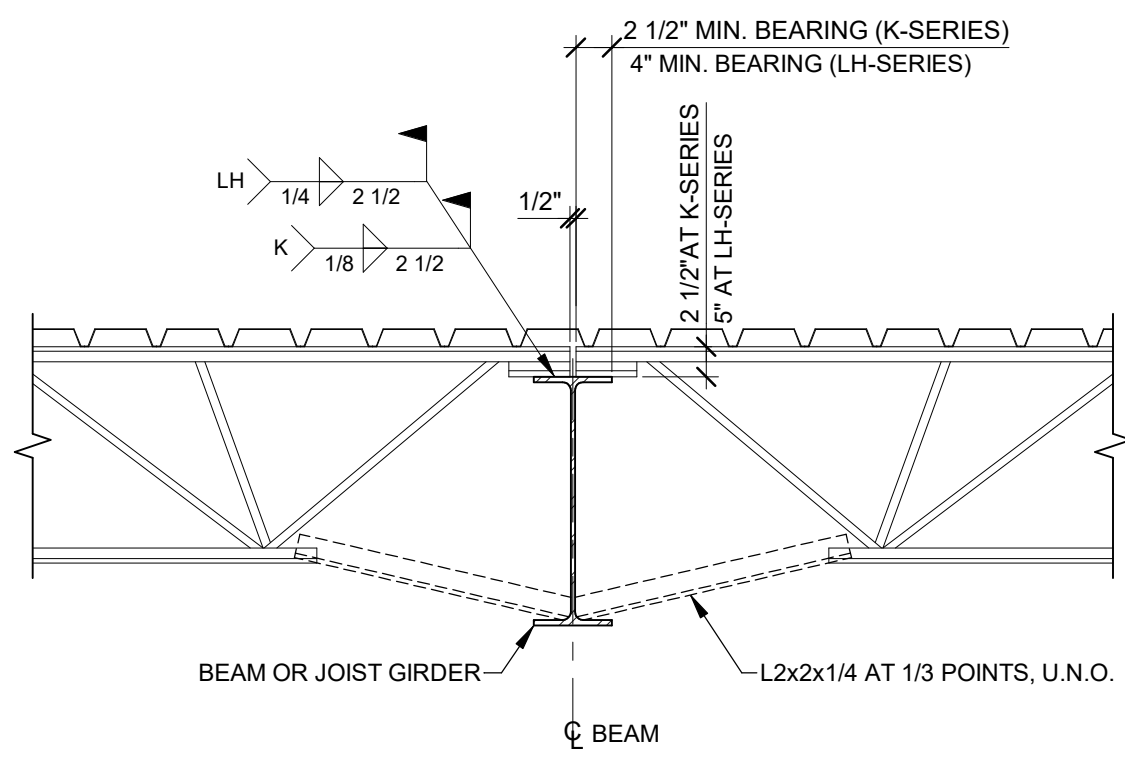
TYPICAL JOIST BEARING
AT EDGE COLUMN

F
S0008



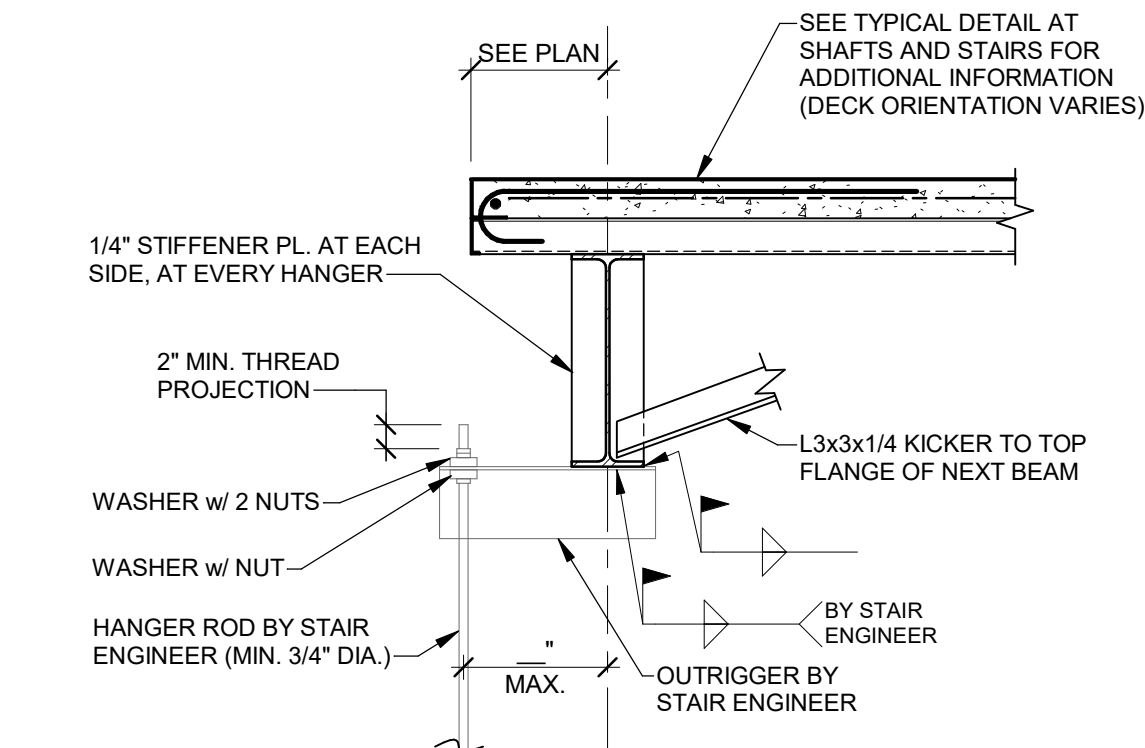
TYPICAL JOIST BEARING
ONE SIDE OF GIRDER

G
S0008



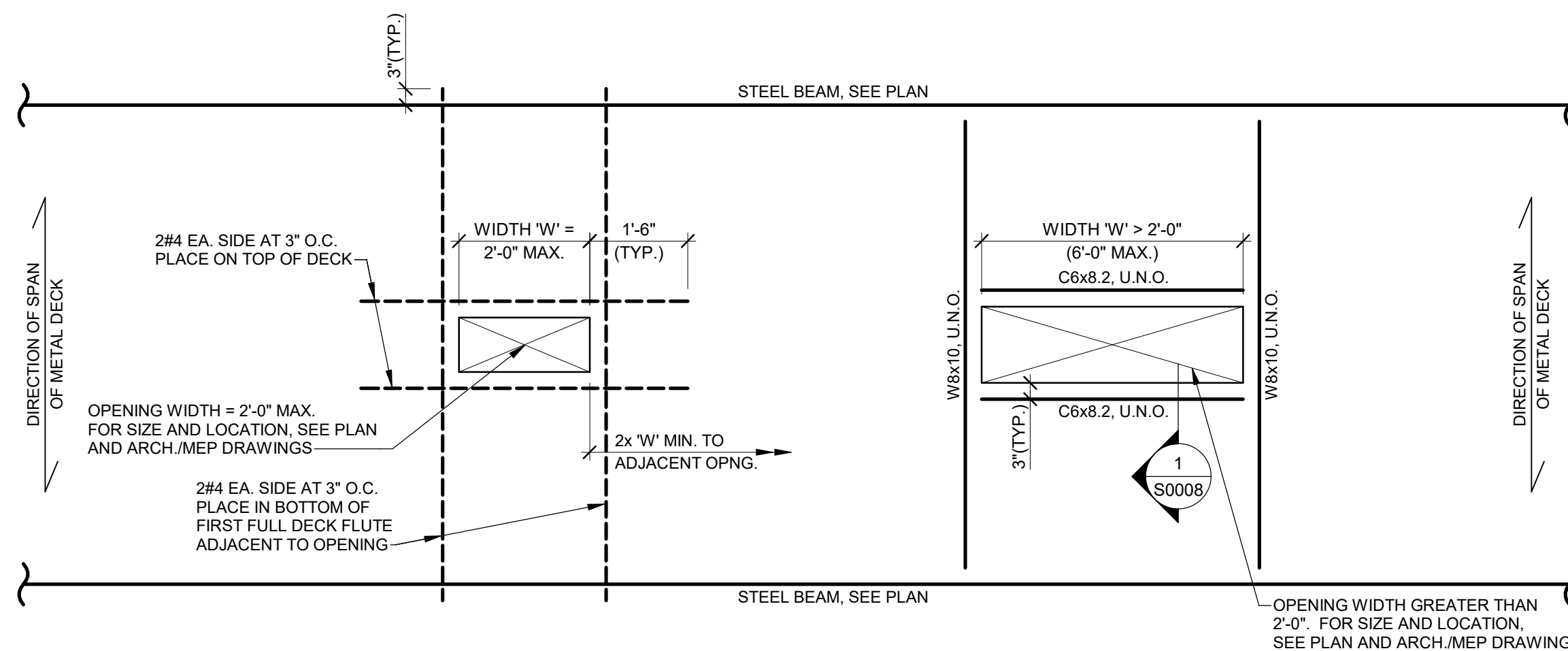
TYPICAL JOIST BEARING
ON GIRDER

H
S0008



TYPICAL STAIR HANGER

J
S0008



SLAB REINFORCED CONDITION
(2'-0" MAX. OPENING WIDTH)

STEEL FRAMED CONDITION
(OPENING WIDTH > 2'-0")

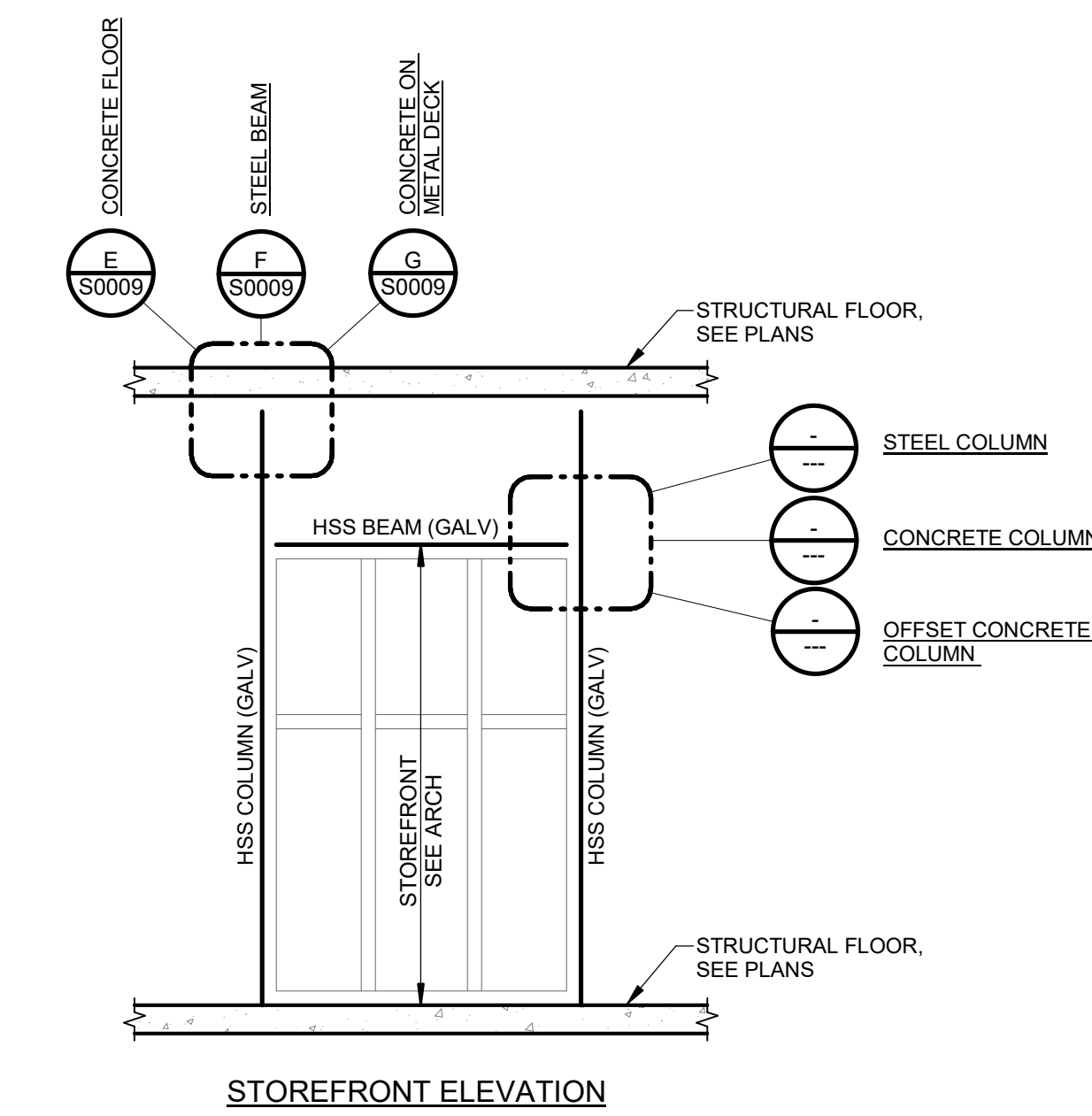
TYPICAL DETAIL AT OPENING
THROUGH SLAB OVER COMPOSITE
METAL DECK

K
S0008

SECTION

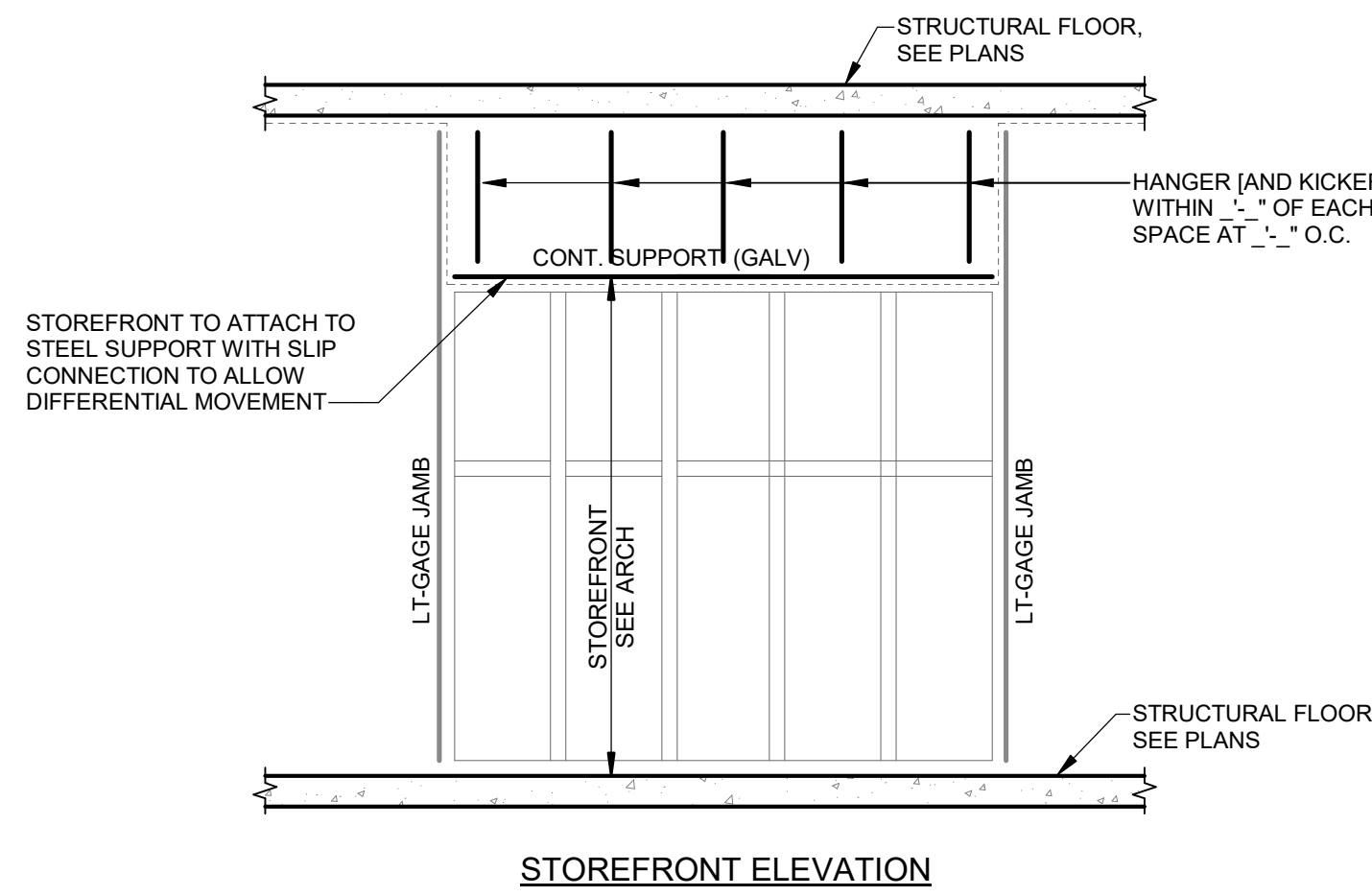
SCALE: 3/4" = 1'-0"

1
S0008



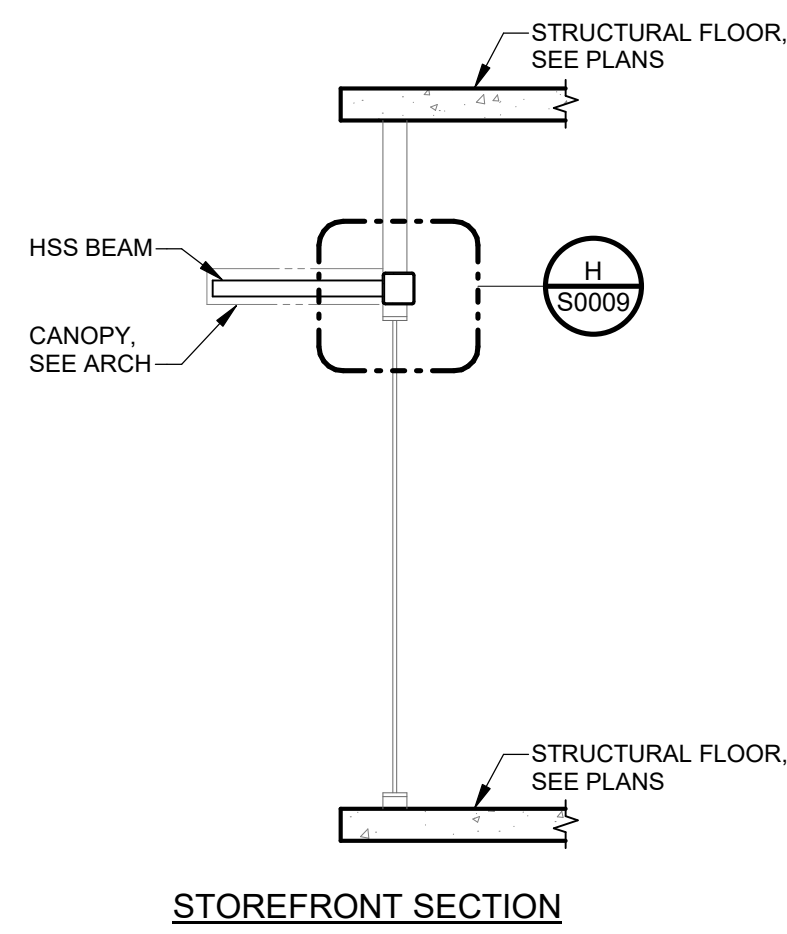
ELEVATION: BEAM+POST SUPPORT

A
S0009



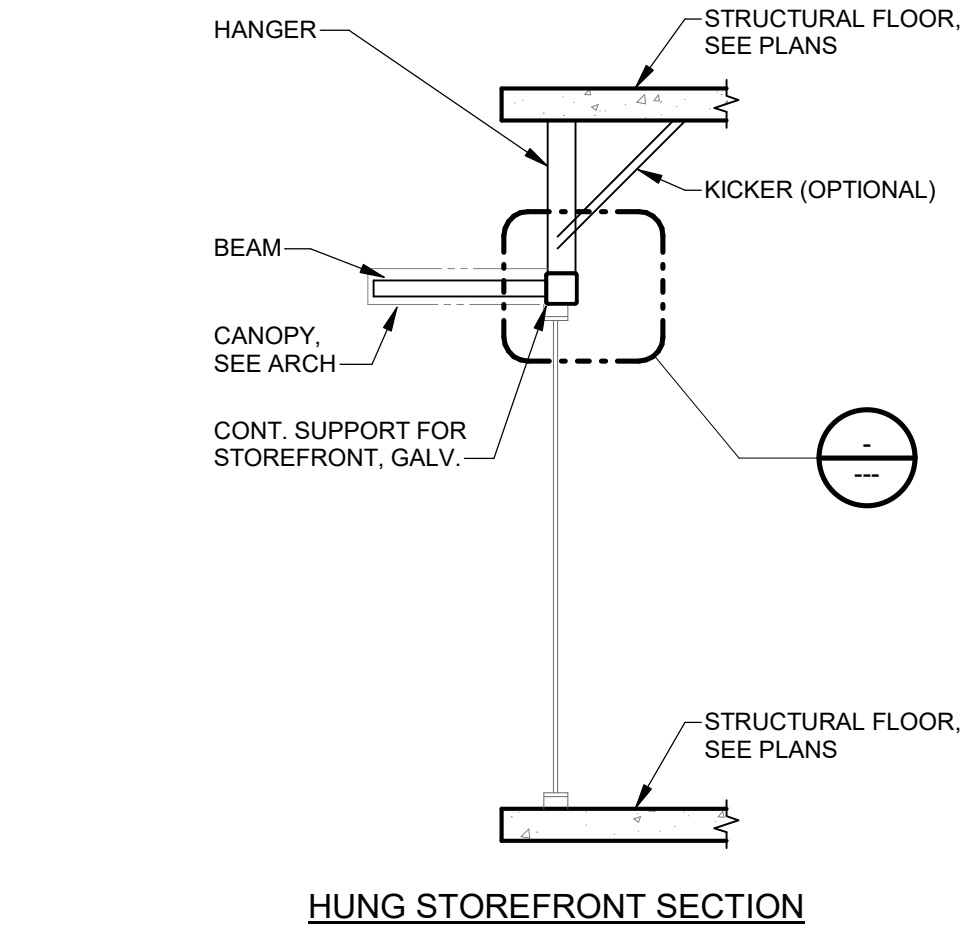
ELEVATION: HUNG SUPPORT

B
S0009



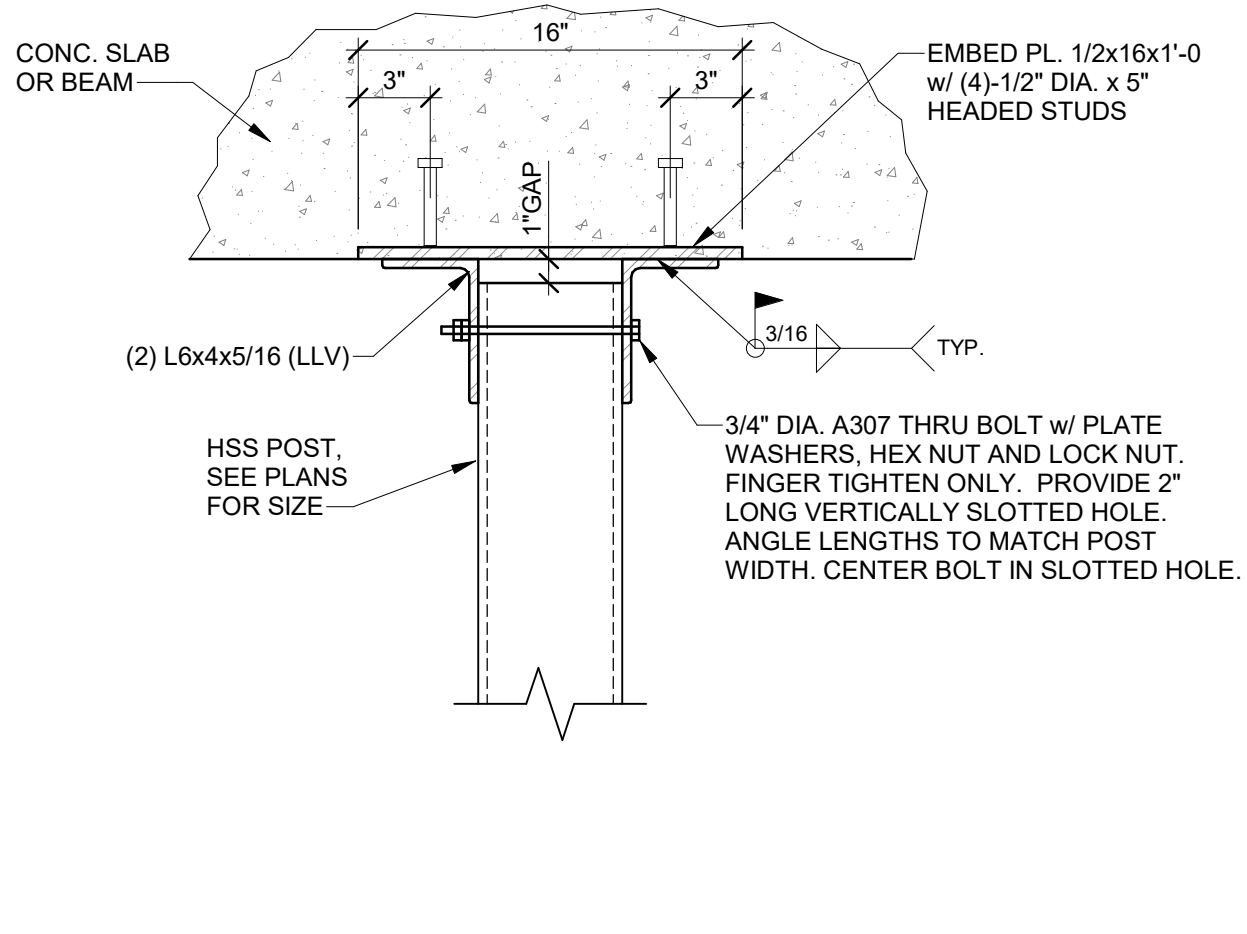
STOREFRONT SECTION WITH CANOPY

C
S0009



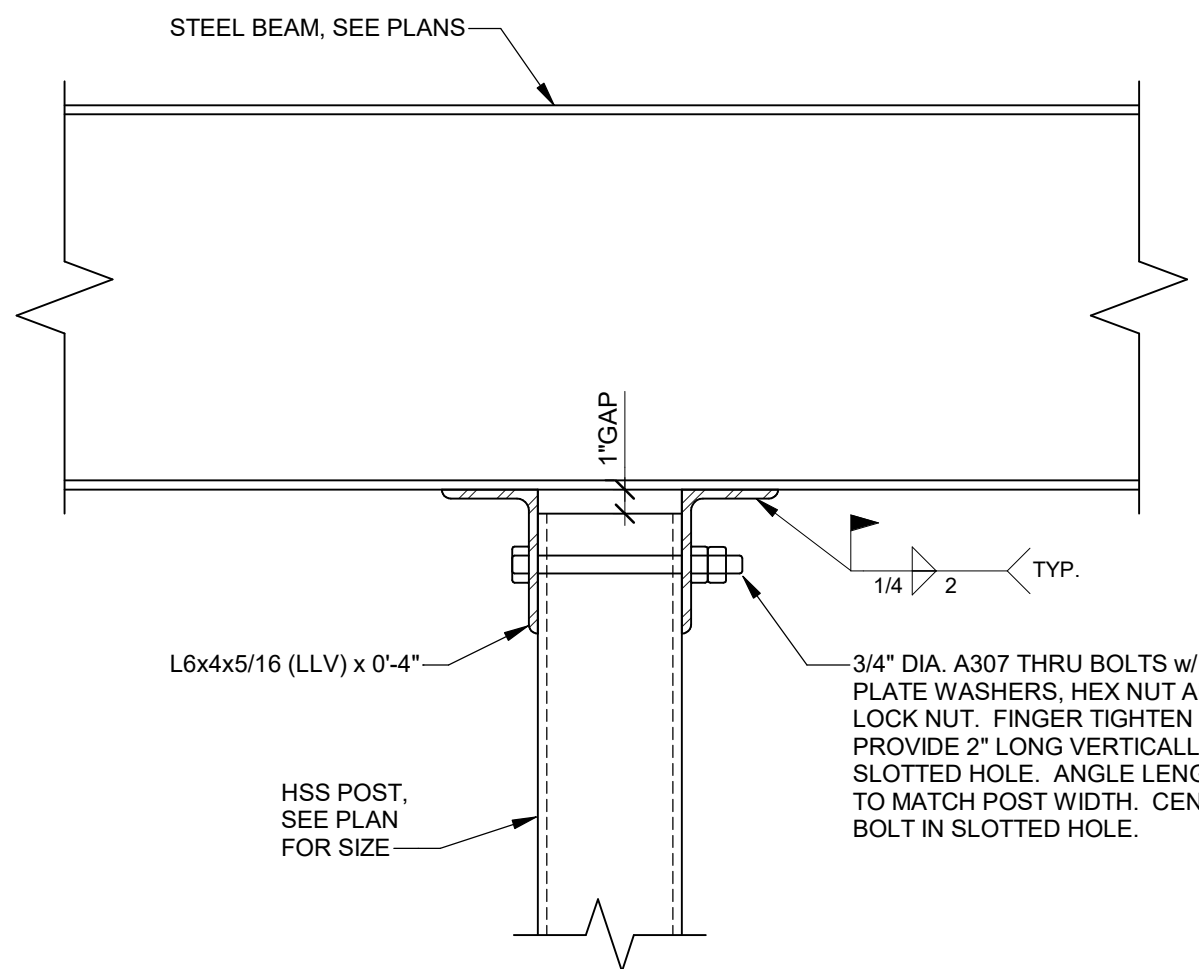
STOREFRONT SECTION WITH CANOPY

D
S0009



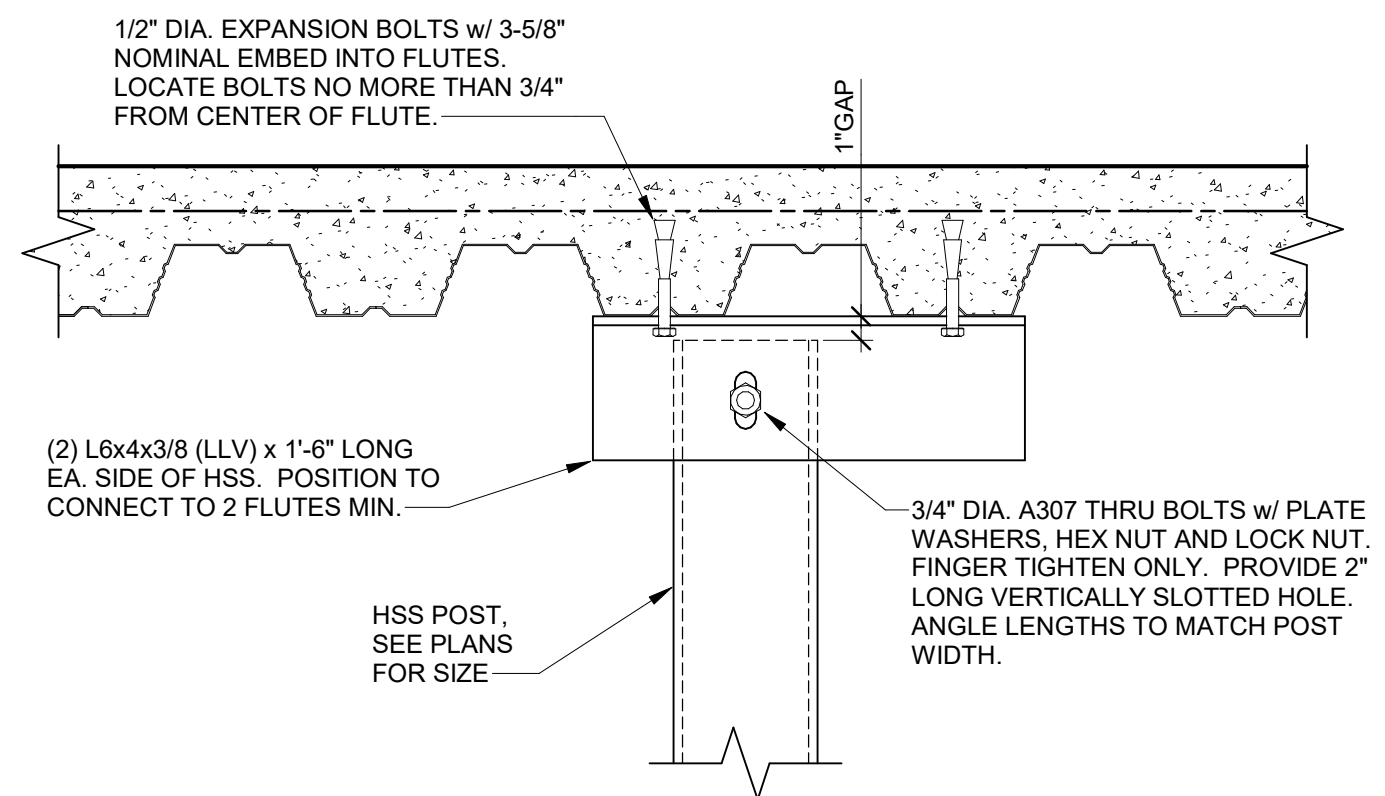
TYPICAL HSS TO CONCRETE SOFFIT

E
S0009



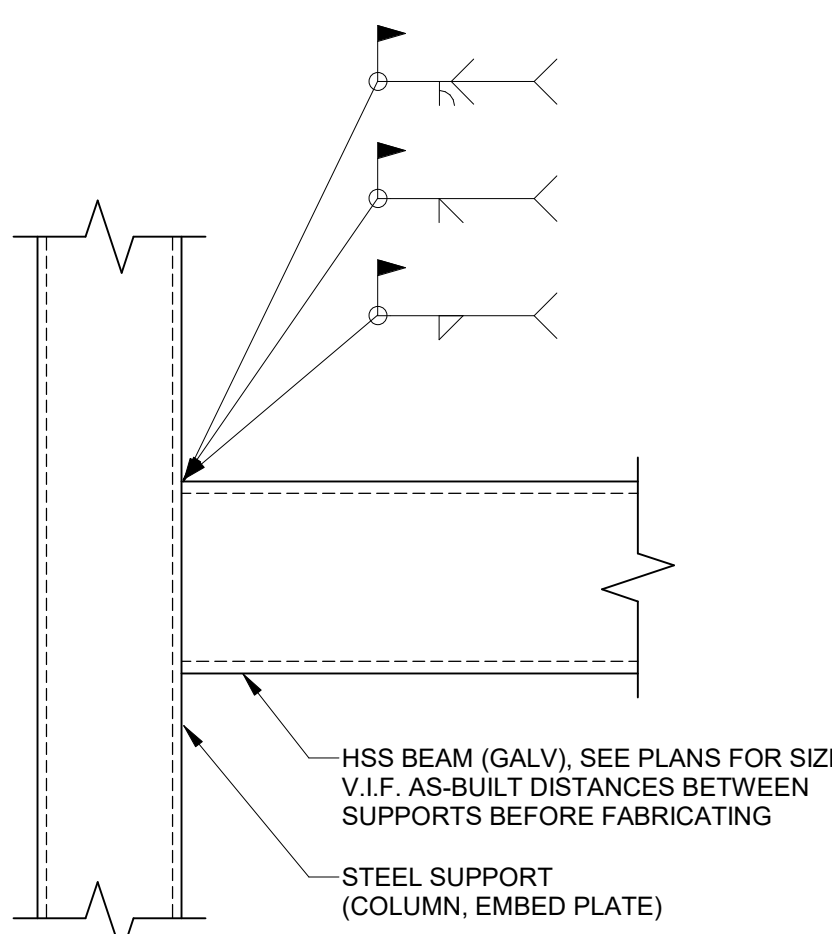
TYPICAL VERTICAL SLIP CONNECTION

F
S0009



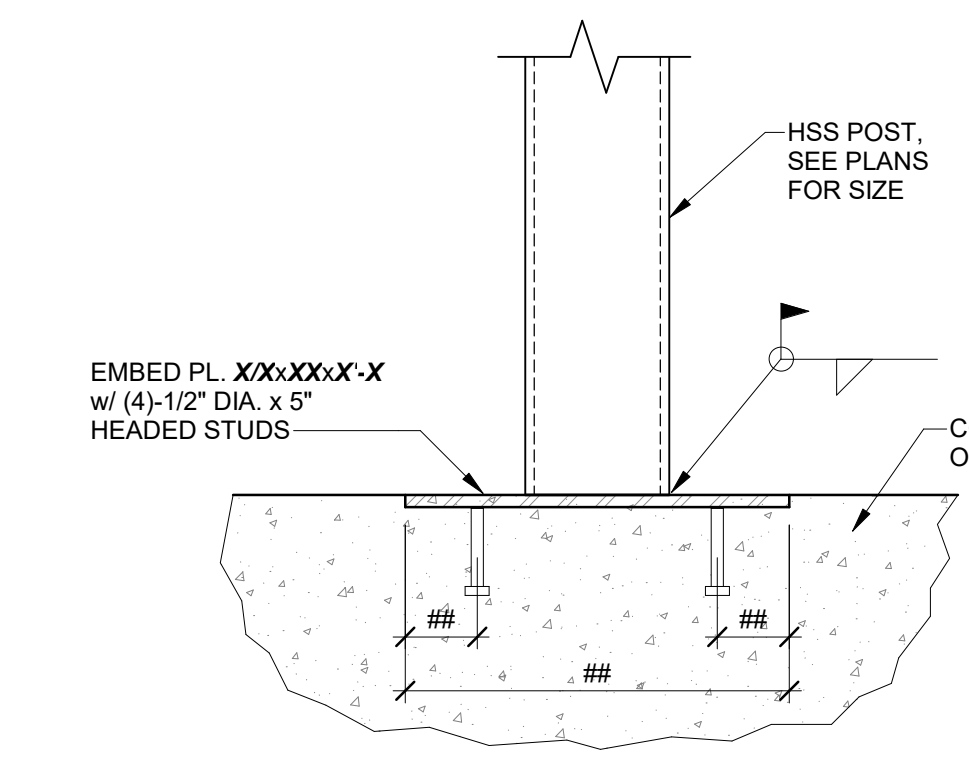
TYPICAL HSS TO CONCRETE ON METAL DECK

G
S0009



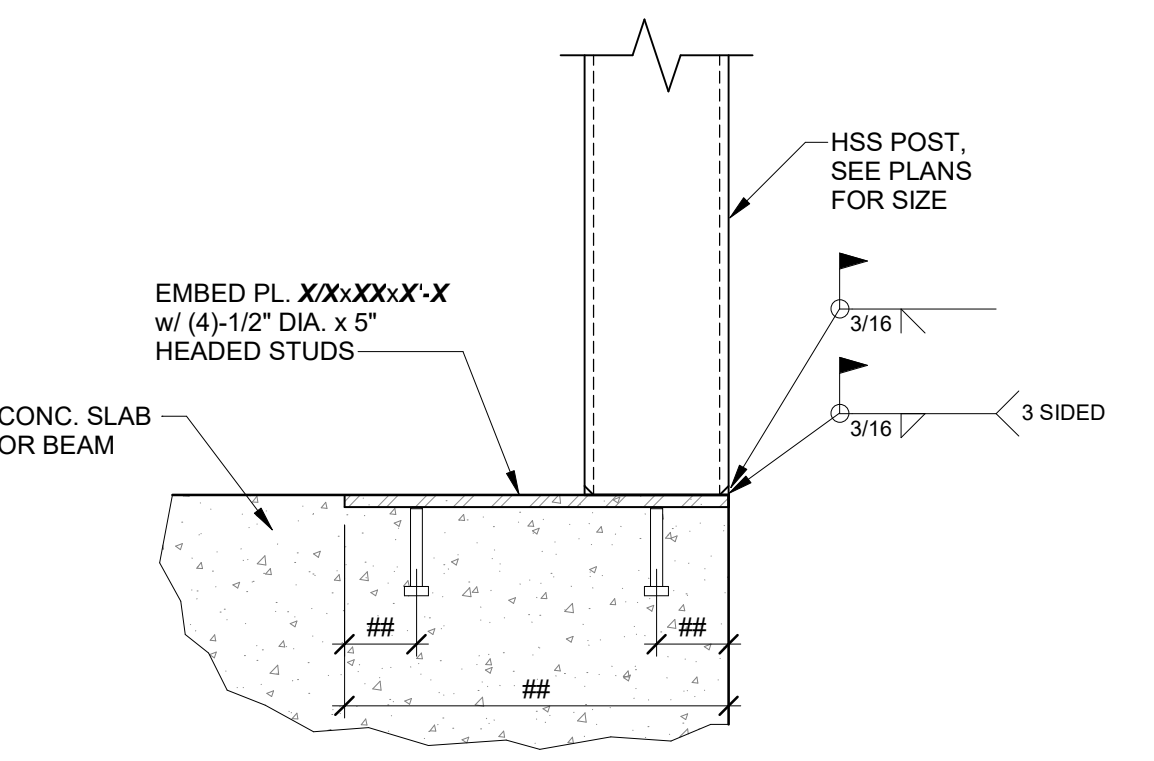
TYPICAL HSS CANTILEVER TO HSS POST

H
S0009



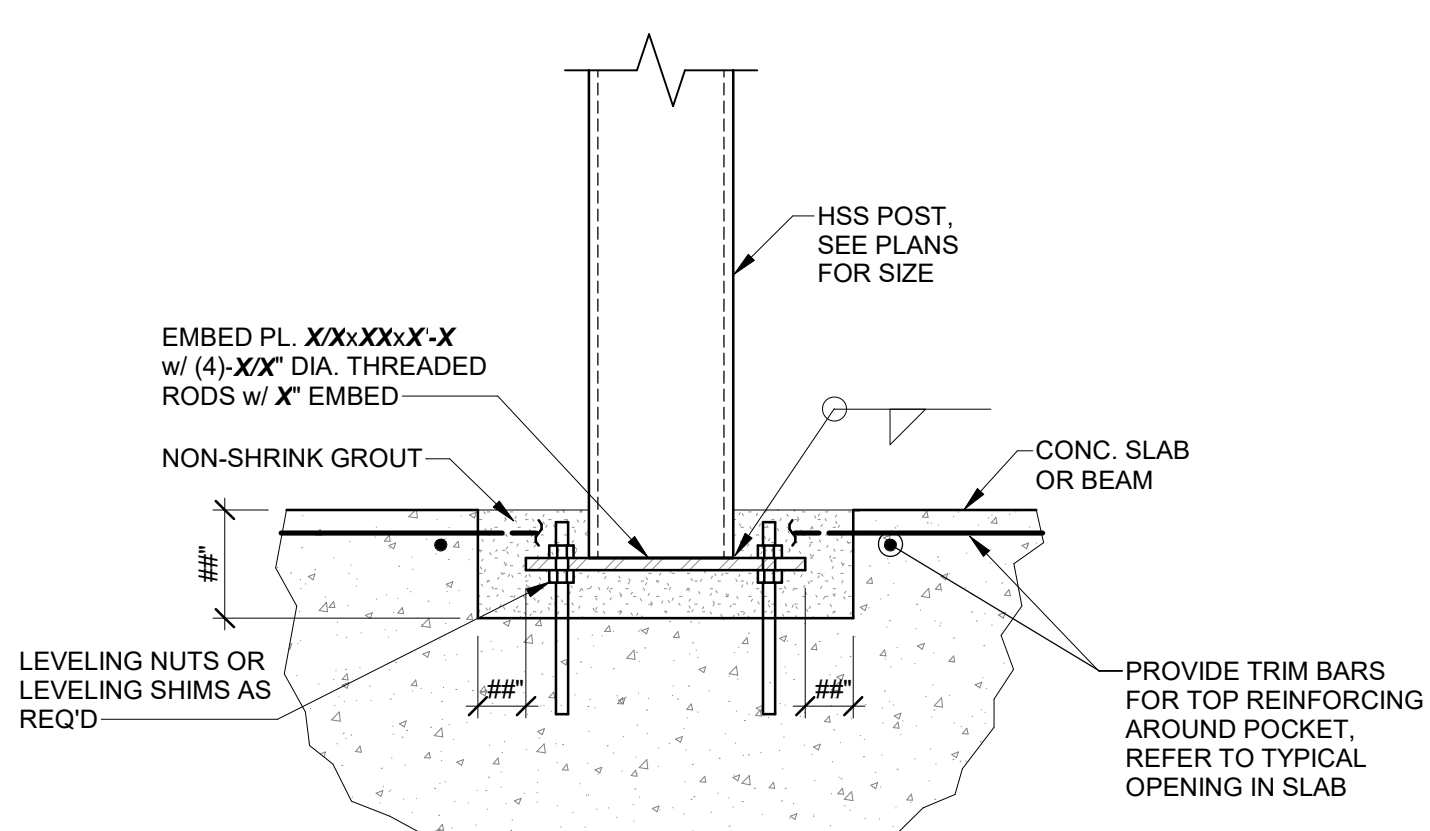
TYPICAL HSS BASE PLATE

J
S0009



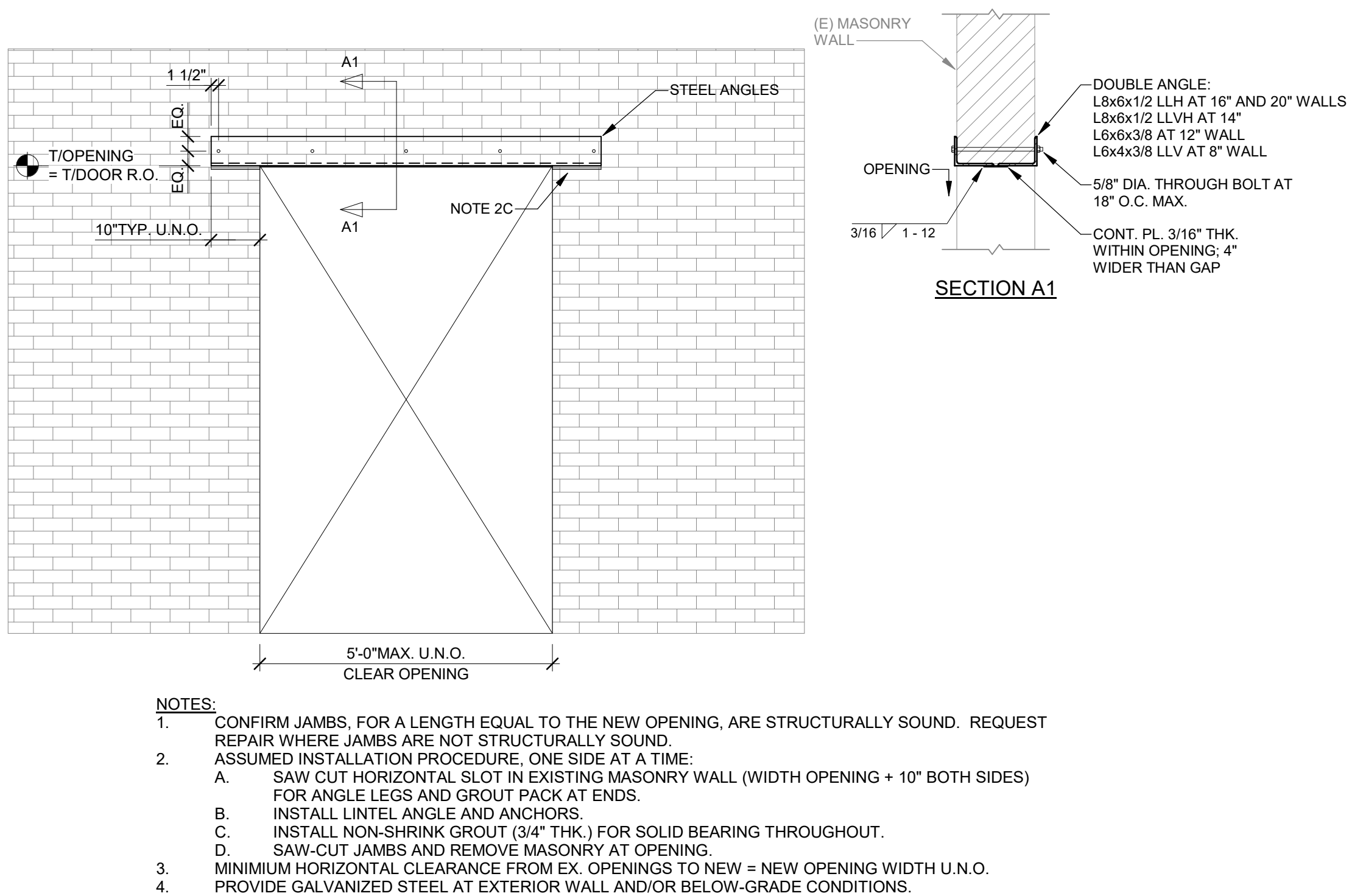
TYPICAL HSS ECCENTRIC BASE PLATE

K
S0009



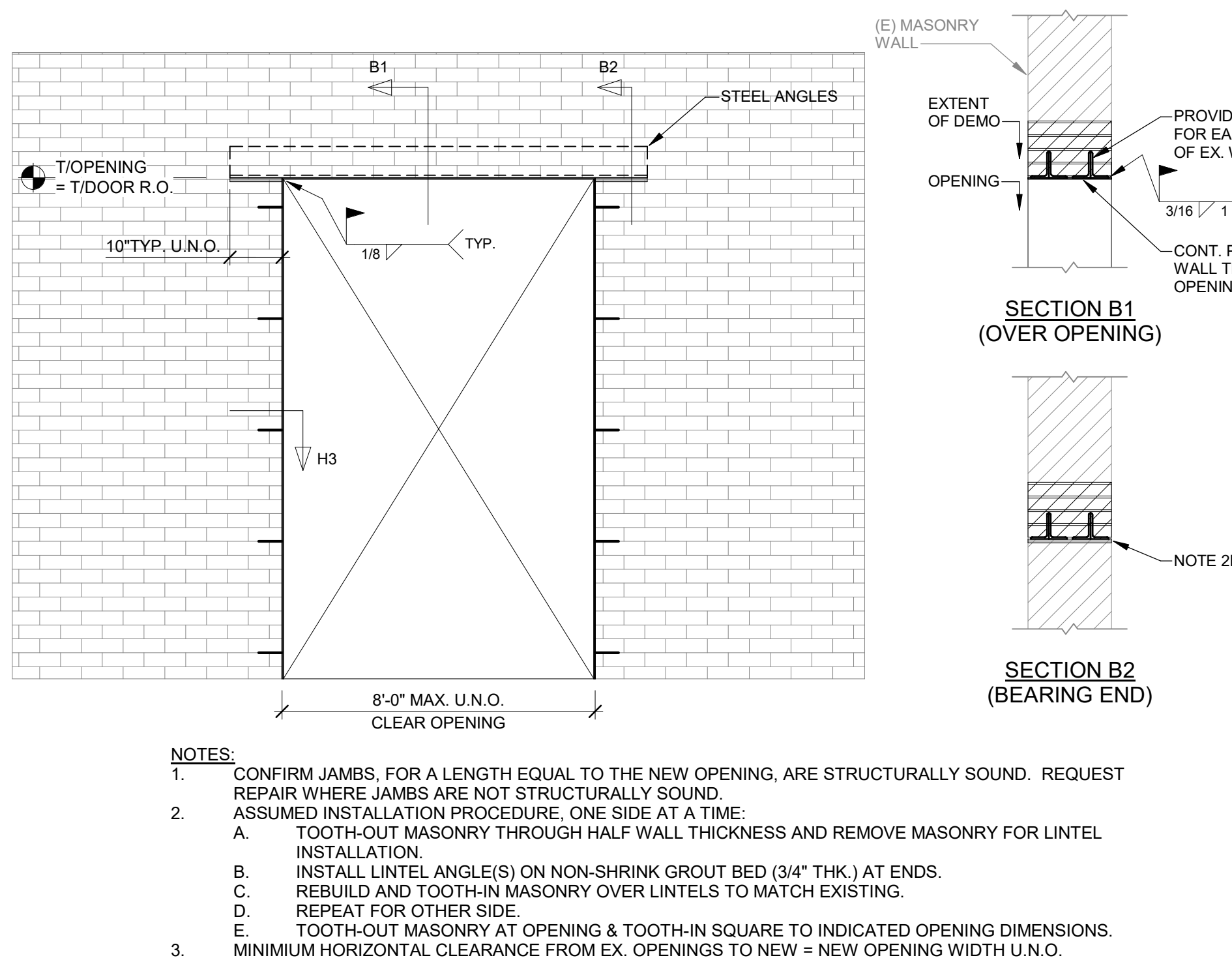
TYPICAL HSS RECESSED BASE PLATE

L
S0009



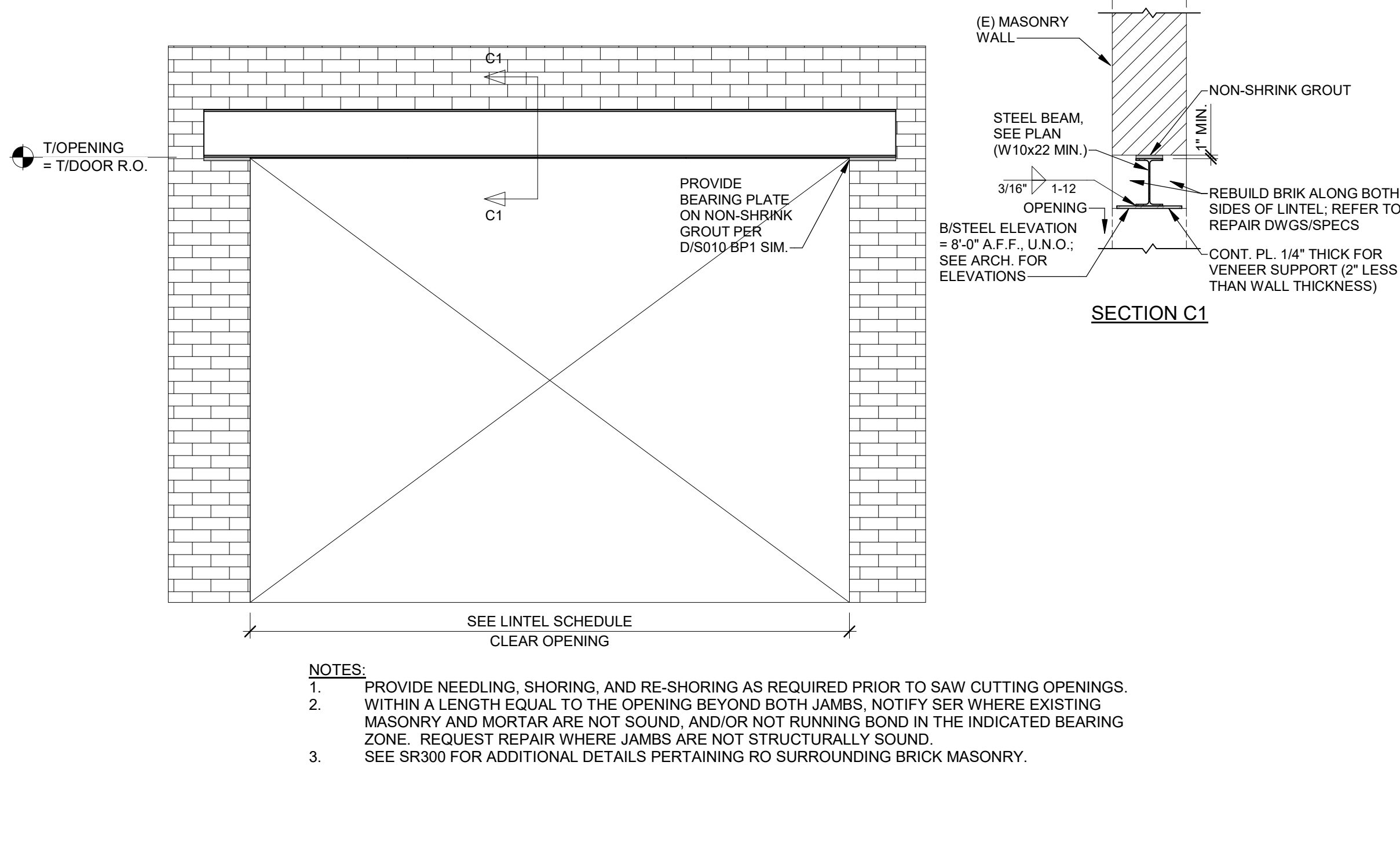
NEW DOOR/WINDOW OPENING IN EXISTING BRICK WALL - DOUBLE ANGLE LINTELS

A
S0010



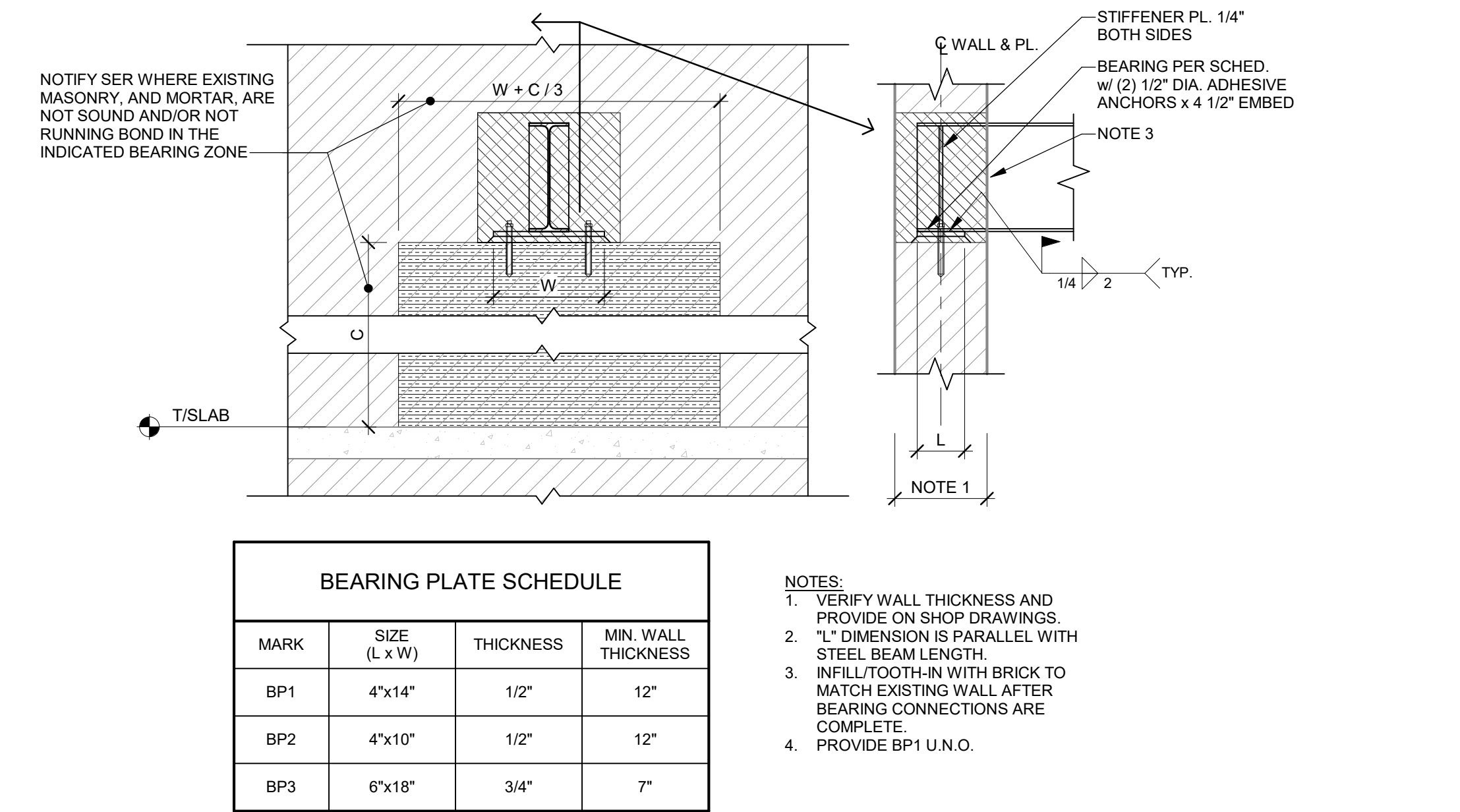
NEW DOOR/WINDOW OPENING IN EXISTING BRICK WALL - CONCEALED LINTEL

B
S0010



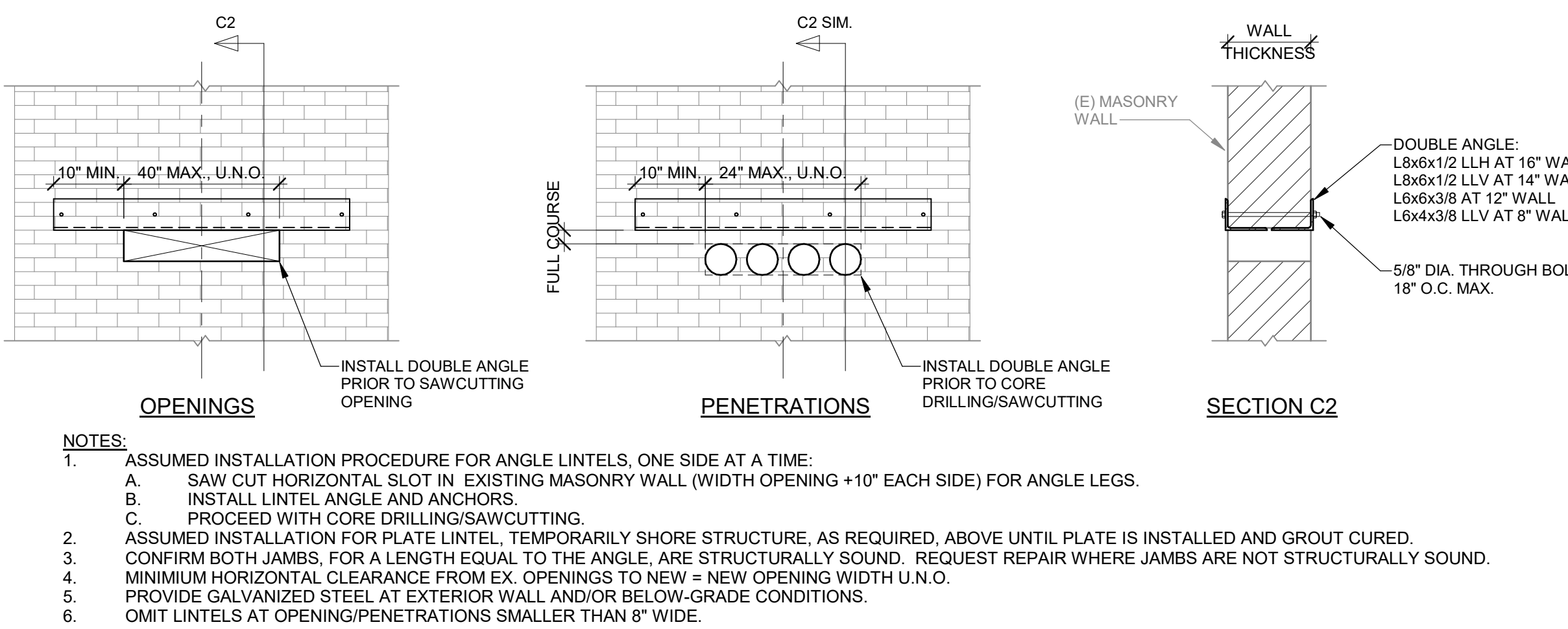
NEW DOOR/WINDOW OPENING IN EXISTING BRICK WALL - WIDE FLANGE LINTEL

C
S0010



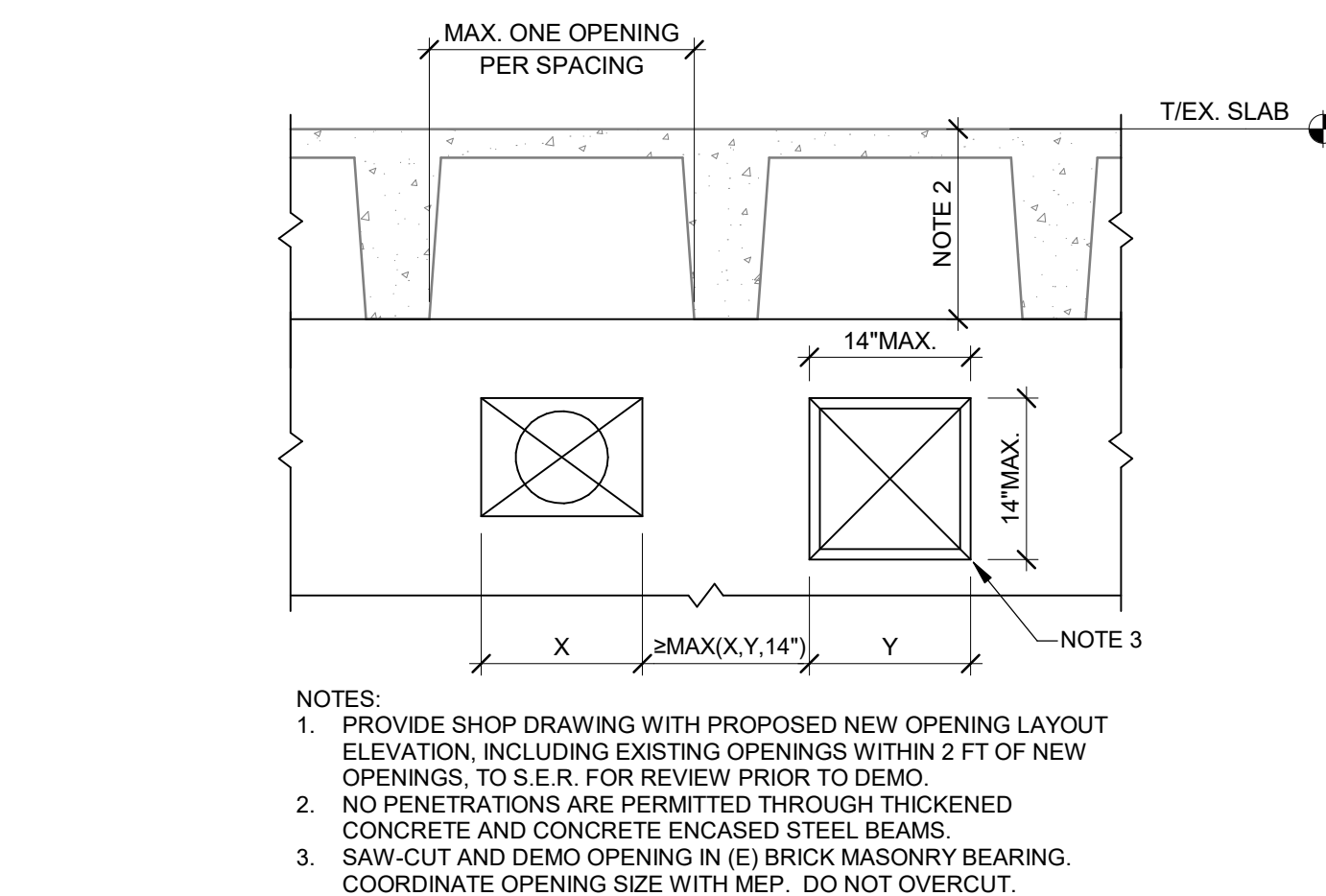
TYPICAL BEAM BEARING ON EXISTING MASONRY WALL

D
S0010



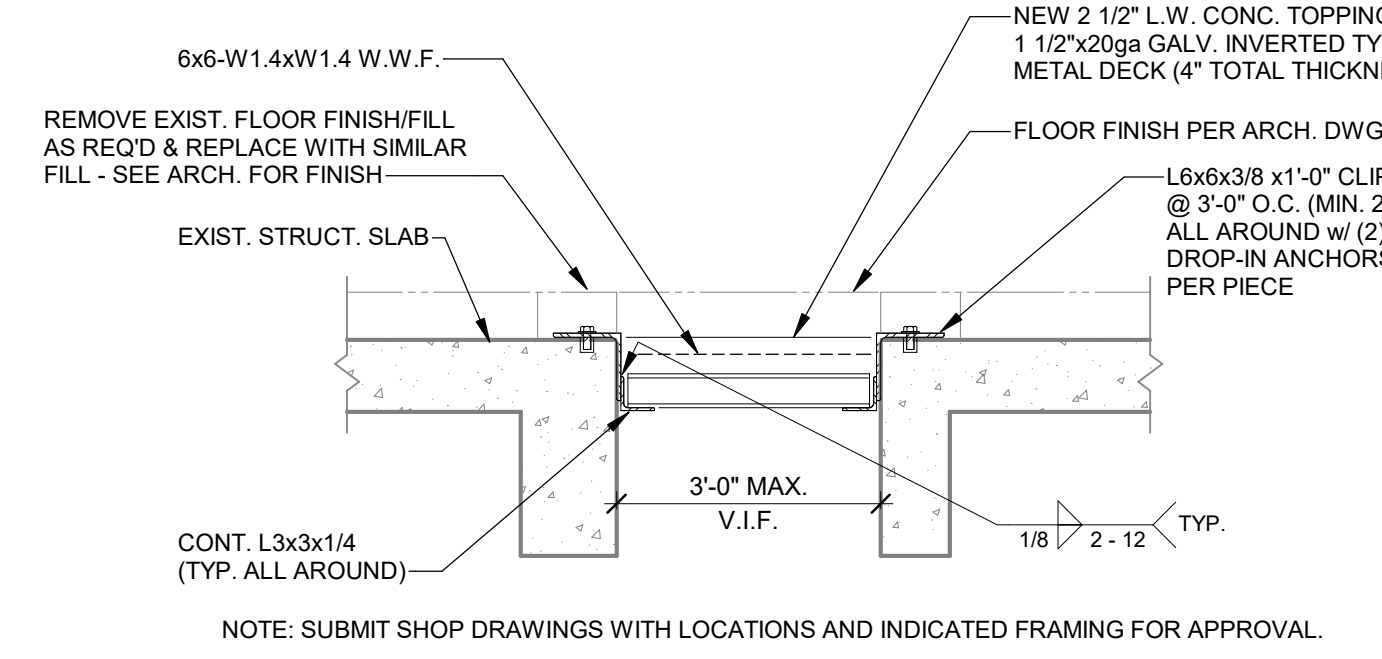
NEW MECHANICAL OPENINGS AND PENETRATIONS IN EX. BRICK WALL

E
S0010



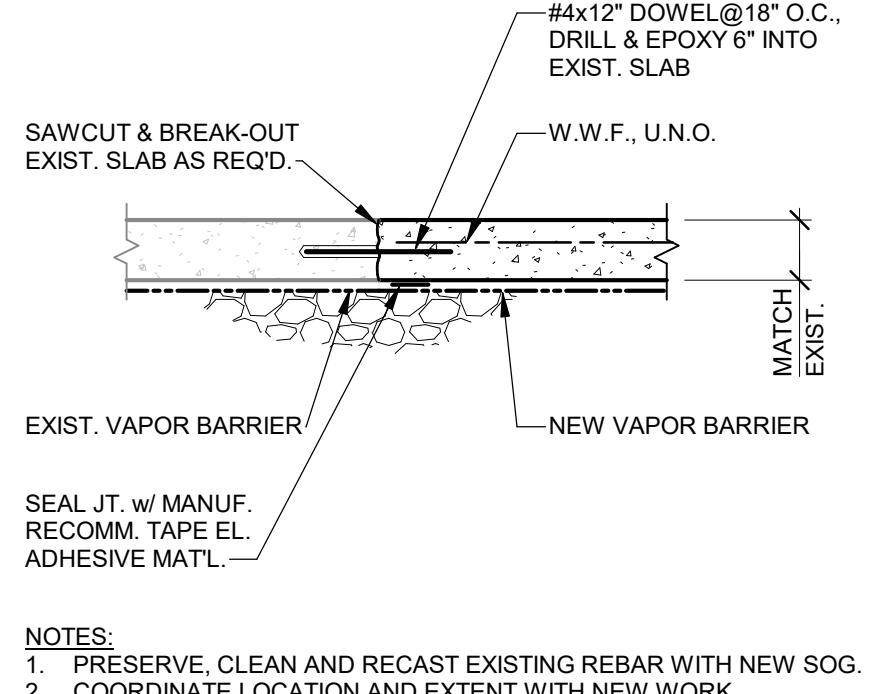
NEW MEP OPENING IN EXISTING BRICK WALL

F
S0010



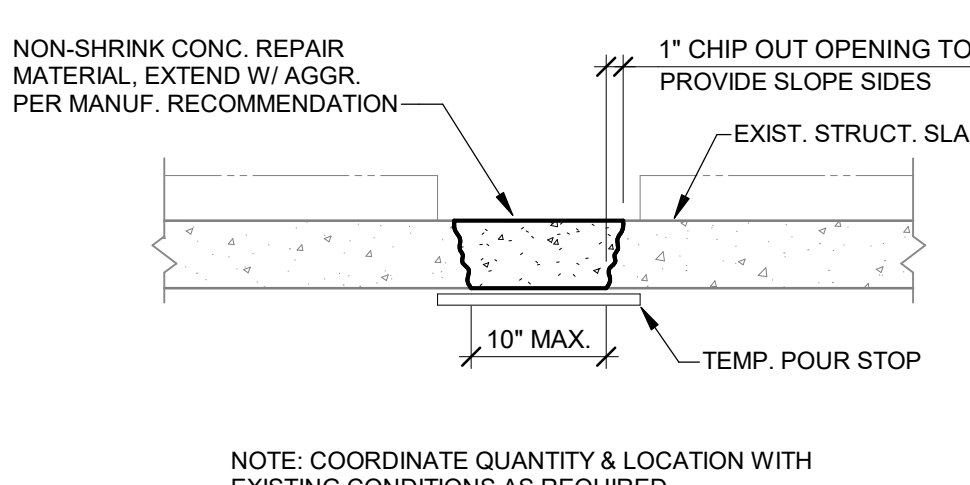
TYPICAL EXISTNG SLAB INFILL DETAIL

G
S0010



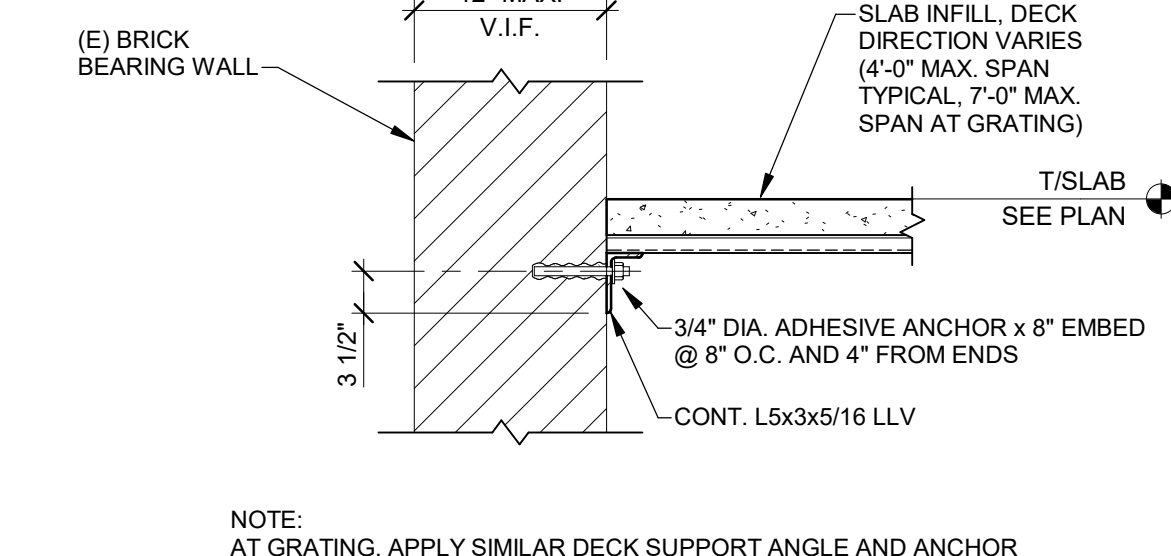
TYPICAL SLAB-ON-GRADE REPAIR

H
S0010



TYPICAL PIPE PENETRATION INFILL

J
S0010



TYPICAL FLOOR SUPPORT AT EXISTING BRICK WALL

K
S0010

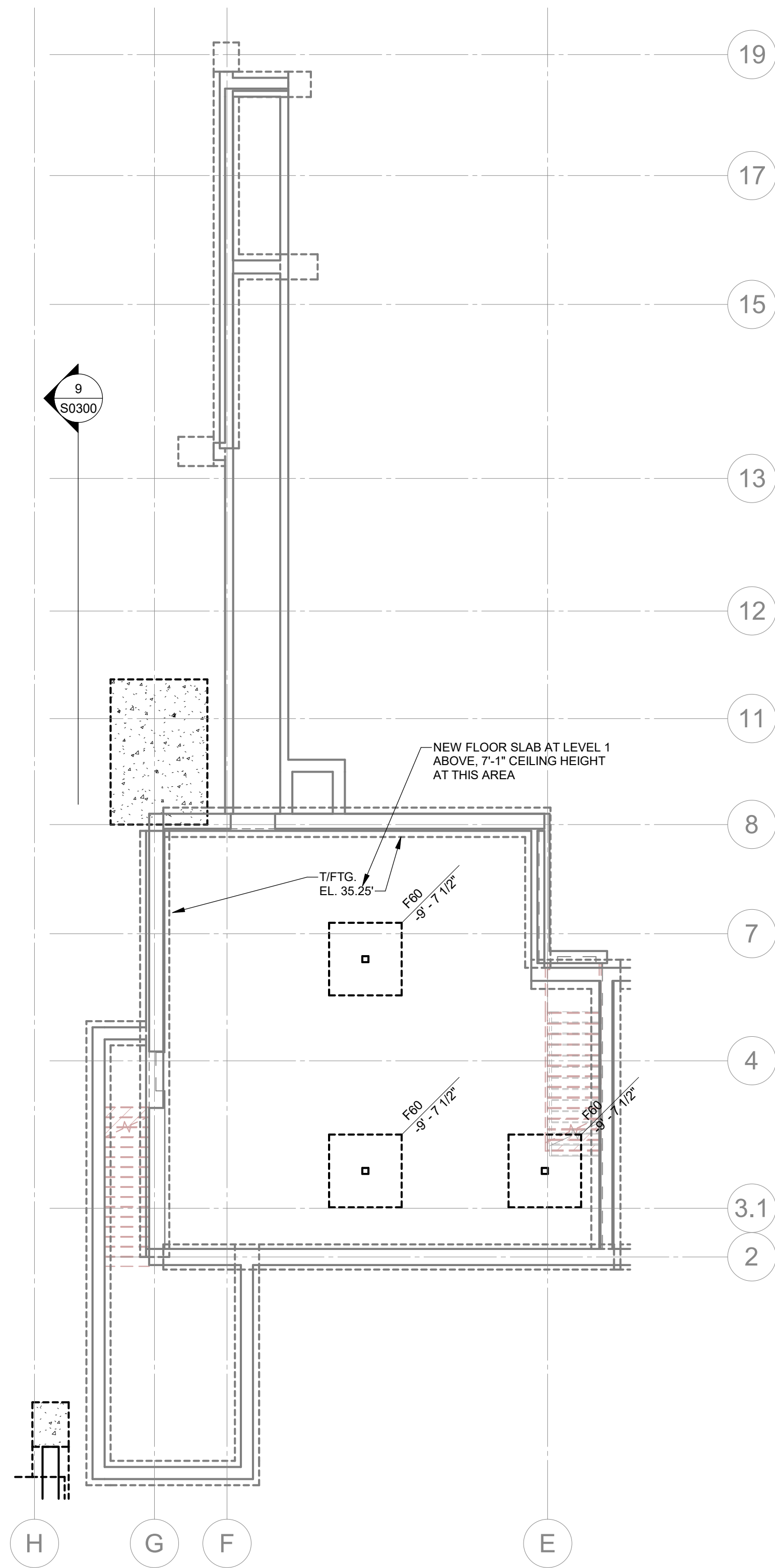
Autodesk Docu/55-25106-00 DC DGS - Ketcham ES Design-Build/Ketcham E.S. STRUCT R24.rvt
5/8/2020 10:39:35 AM

LEVEL 00 FOUNDATION PLAN

1/8" = 1'-0"

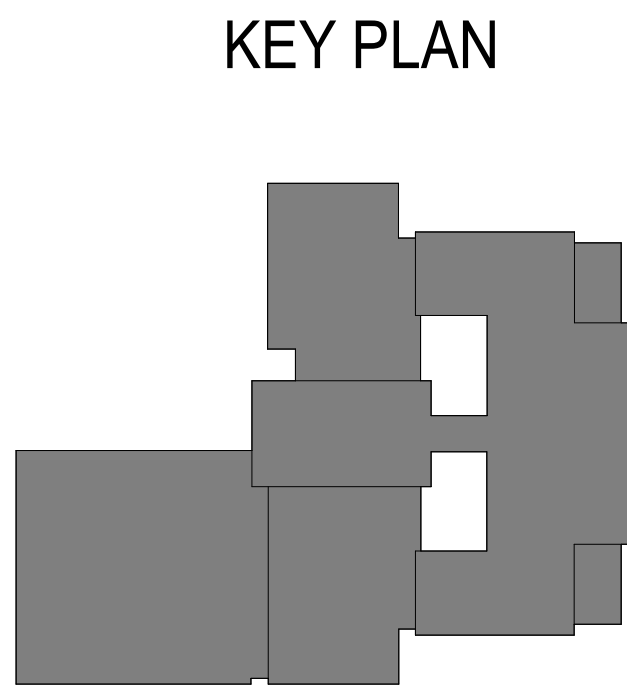
FOUNDATION PLAN NOTES:

1. SLAB ELEVATION = SEE PLAN UNLESS NOTED OTHERWISE. COORDINATE ELEVATIONS WITH ARCHITECTURAL DRAWINGS.
2. SLAB-ON-GRADE SHALL BE 5" 5000 PSI NORMAL WEIGHT CONCRETE PLACED OVER VAPOR BARRIER AND 10" MIN. GRAVEL FILL. REINFORCE SLAB WITH (8X6-W2-DW2-D) WELDED WIRE MESH PLACED 2" BELOW TOP OF SLAB. MAINTAIN UNIFORM SLAB THICKNESS AT DRAINS, VALLEYS, RIDGES, AND SLOPES.
3. SHALLOW FOUNDATIONS ARE DESIGNED FOR A NET ALLOWABLE BEARING PRESSURE OF 4000 PSF ON NATURAL SOILS OR SOIL IMPROVED WITH RAMMED AGGREGATE PIER (GEOPIER). GROUND IMPROVEMENT. AGGREGATE PIERS SHALL BE PROVIDED DIRECTLY BENEATH ALL COLUMN AND WALL FOOTINGS. SUBGRADE SHALL BE OBSERVED AND APPROVED BY THE GEOTECHNICAL ENGINEER ON RECORD PRIOR TO CONCRETE PLACEMENT. FOOTING ELEVATIONS SHALL BE LOWERED AS DIRECTED BY THE GEOTECHNICAL ENGINEER OF RECORD TO BEAR ON INDICATED SUBGRADE AND ACHIEVE THE NOTED BEARING CAPACITY.
4. TOP OF FOOTINGS TO BE 1'-0" BELOW SLAB-ON-GRADE UNLESS NOTED OTHERWISE.
5. SEE GEOTECHNICAL REPORT AND PLUMBING DRAWINGS FOR UNDERSLAB DRAINAGE REQUIREMENTS.
6. FOOTINGS SHALL BE CENTERED UNDER COLUMNS U.N.O. ON PLAN.
7. SEE STRUCTURAL NOTES AND TYPICAL DETAILS FOR ADDITIONAL INFORMATION.
8. SEE ARCHITECTURAL AND/OR MEP DRAWINGS FOR:
 - A. COLUMN AND WALL LOCATIONS
 - B. LOCATIONS OF RIDGES, VALLEYS, FLOOR DRAINS, AND SLOPES TO DRAIN
 - C. LOCATION AND EXTENT OF SLAB DEPRESSIONS AND RAMPS
 - D. LOCATION AND EXTENT OF EQUIPMENT PADS AND SUPPORTS, CURBS, ETC.
 - E. SIZE AND LOCATION OF NON-LOADBEARING MASONRY PARTITIONS. SEE TYPICAL DETAILS FOR REINFORCING REQUIREMENTS.



LEVEL 00/FOUNDATION PLAN

1/8" = 1'-0"



Ketcham Elementary School

1919 15TH ST SE WASHINGTON, DC 20020

1919 15th St SE
Washington, DC 20020

100% Design
Development
03/31/20
REVISIONS

55-25106-00

LEVEL
00/FOUNDATION
PLAN

S0100

NOT FOR
CONSTRUCTION

SK&A
Structural Engineers, LLC
1000 15th St NW, Suite 1000
Washington, DC 20004
P 202.638.2357 F 202.638.2358 W www.skand.com

Project
Logo

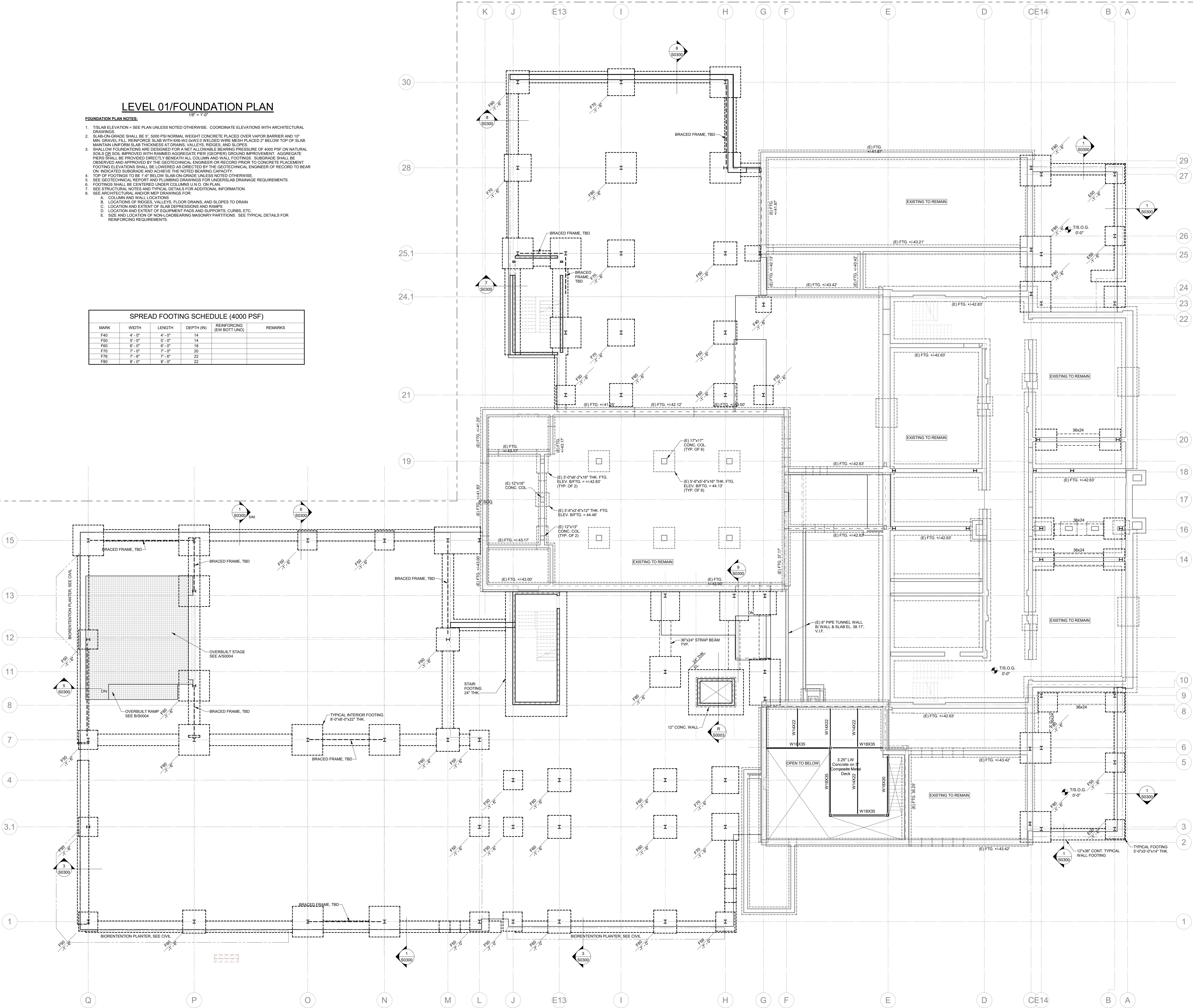
DLR GROUP
© DLR Group

LEVEL 01/FOUNDATION PLAN

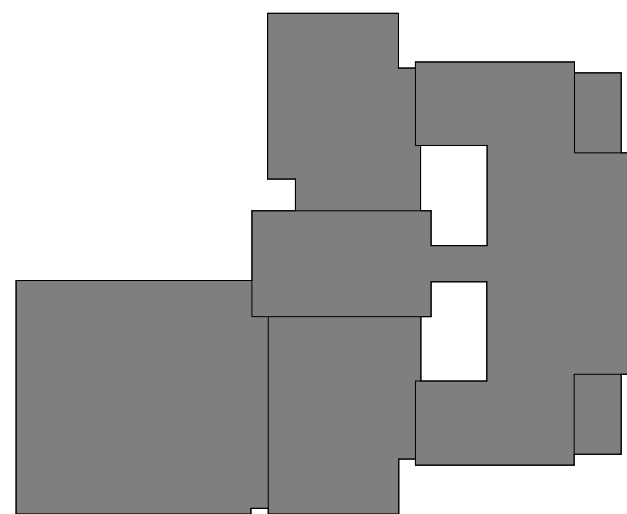
- FOUNDATION PLAN NOTES:
1. TISLAB ELEVATION + SEE PLAN UNLESS NOTED OTHERWISE. COORDINATE ELEVATIONS WITH ARCHITECTURAL DRAWINGS.
 2. SLAB-ON-GRADE SHALL BE 6" 5000 PSI NORMAL WEIGHT CONCRETE PLACED OVER VAPOR BARRIER AND 10" MIN. GRAVEL FILL. REINFORCE SLAB WITH 6X6-W2.0W2.0 WELDED WIRE MESH PLACED 2" BELOW TOP OF SLAB. MAINTAIN UNIFORM SLAB THICKNESS AT DRAINS, VALLEYS, RIDGES, AND SLOPES.
 3. SHALLOW FOUNDATIONS ARE DESIGNED FOR A NET ALLOWABLE BEARING PRESSURE OF 4000 PSF ON NATURAL SOILS OR SOIL IMPROVED WITH FRAIRED AGGREGATE PIER (GEOPYR) GROUND IMPROVEMENT. AGGREGATE PIERS SHALL BE PROVIDED DIRECTLY BENEATH ALL COLUMN AND WALL FOOTINGS. SUBGRADE SHALL BE OBSERVED AND APPROVED BY THE GEOTECHNICAL ENGINEER OR RECORD PRIOR TO CONCRETE PLACEMENT. FOOTING ELEVATIONS SHALL BE LOWERED AS DIRECTED BY THE GEOTECHNICAL ENGINEER OF RECORD TO BEAR ON INDICATED SUBGRADE AND ACHIEVE THE NOTED BEARING CAPACITY.
 4. TOP OF FOOTINGS TO BE 1'-6" BELOW SLAB-ON-GRADE UNLESS NOTED OTHERWISE.
 5. SEE GEOTECHNICAL REPORT AND PILING DRAWINGS FOR UNDERSLAB DRAINAGE REQUIREMENTS.
 6. FOOTINGS SHALL BE CENTERED UNDER COLUMNS UNLESS NOTED OTHERWISE.
 7. SEE STRUCTURAL NOTES AND TYPICAL DETAILS FOR ADDITIONAL INFORMATION.
 8. SEE ARCHITECTURAL AND/OR MEP DRAWINGS FOR:
 - A. COLUMN AND WALL LOCATIONS
 - B. LOCATIONS OF RIDGES, VALLEYS, FLOOR DRAINS, AND SLOPES TO DRAIN
 - C. LOCATION AND EXTENT OF SLAB DEPRESSIONS AND RAMPS
 - D. LOCATION AND EXTENT OF EQUIPMENT PADS AND SUPPORTS, CURBS, ETC.
 - E. SIZE AND LOCATION OF NON-LOADBEARING MASONRY PARTITIONS. SEE TYPICAL DETAILS FOR REINFORCING REQUIREMENTS.

SPREAD FOOTING SCHEDULE (4000 PSF)

MARK	WIDTH	LENGTH	DEPTH (IN)	REINFORCING (EW BOIT UNO)	REMARKS
F40	4'-0"	4'-0"	14		
F50	5'-0"	5'-0"	14		
F60	6'-0"	6'-0"	18		
F70	7'-0"	7'-0"	20		
F76	7'-6"	7'-6"	22		
F80	8'-0"	8'-0"	22		



KEY PLAN

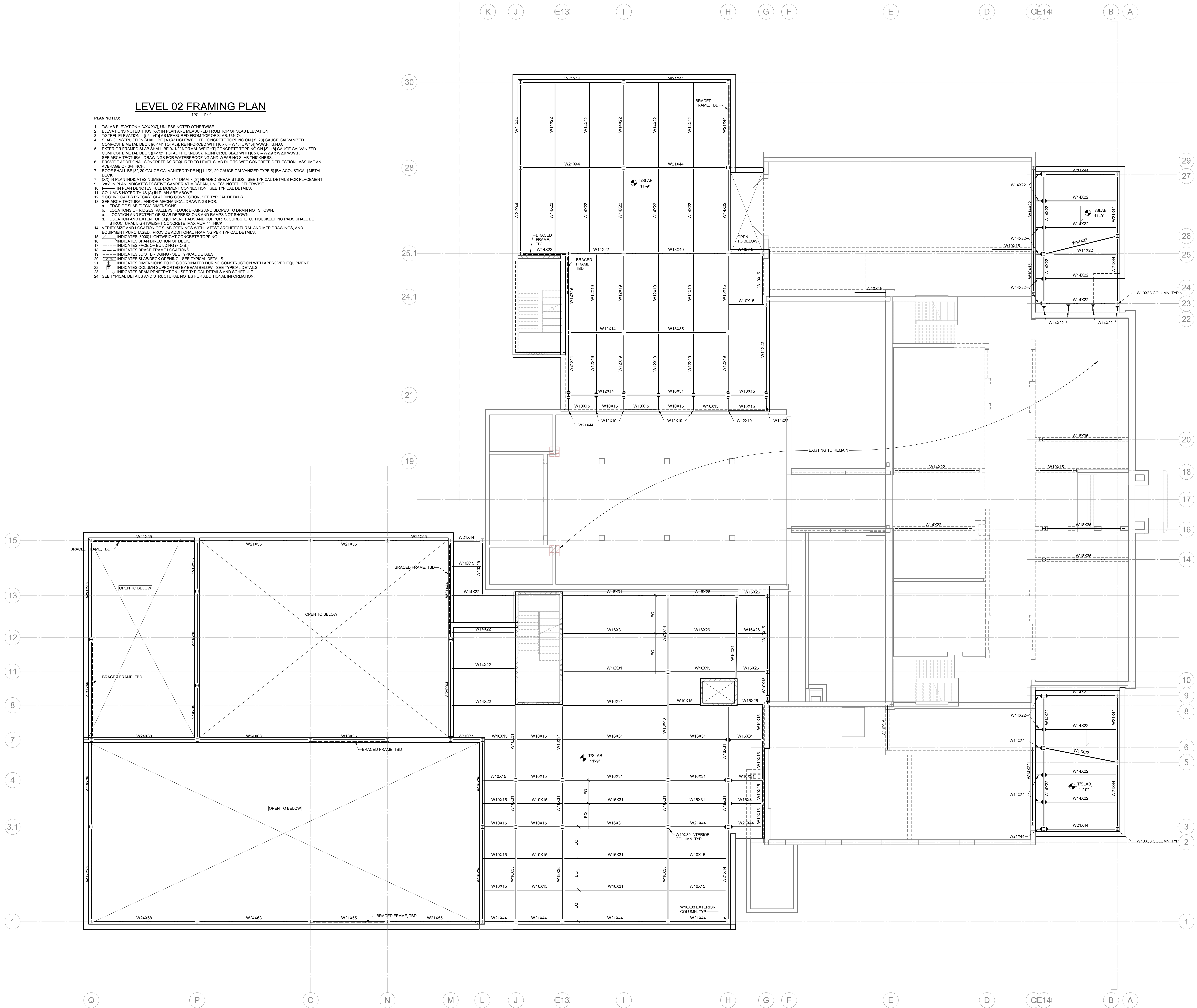


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LEVEL 02 FRAMING PLAN

PLAN NOTES:

1. T/S LAB ELEVATION = 1000.00, UNLESS NOTED OTHERWISE.
2. ELEVATIONS NOTED THUS (X) IN PLAN ARE MEASURED FROM TOP OF SLAB ELEVATION.
3. T/STEEL ELEVATION = 1000.00, UNLESS NOTED OTHERWISE.
4. SLAB CONSTRUCTION SHALL BE 1" (1" MINIMUM) CONCRETE TOPPING ON 3" (3" MINIMUM) GAUGE GALVANIZED COMPOSITE METAL DECK (8'-10" TOTAL), REINFORCED WITH 8'-10" W1.4 X W1.4 W.W.F., U.N.O.
5. EXTERIOR FRAMED SLAB SHALL BE 14" (14" MINIMUM) NORMAL WEIGHT CONCRETE TOPPING ON 3" (3" MINIMUM) GAUGE GALVANIZED COMPOSITE METAL DECK (7'-10" TOTAL), REINFORCED WITH 8'-10" W2.3 X W2.3 W.W.F.
6. SEE ARCHITECTURAL DRAWINGS FOR WATERPROOFING AND WEARING SLAB THICKNESS.
7. PROVIDE ADDITIONAL CONCRETE AS REQUIRED TO LEVEL SLAB DUE TO WET CONCRETE DEFLECTION. ASSUME AN AVERAGE OF 3/4" INCH.
8. ROOF SHALL BE 3" (3" MINIMUM) GAUGE GALVANIZED TYPE B (BA ACOUSTICAL) METAL DECK.
9. 000 IN PLAN INDICATES NUMBER OF 3/4" DIAM. X 8" (8" MINIMUM) HEADED SHEAR STUDS. SEE TYPICAL DETAILS FOR PLACEMENT.
10. 1/2" IN PLAN INDICATES POSITIVE CAMBER AT MIDSPAN, UNLESS NOTED OTHERWISE.
11. IN PLAN DEVOTES FULL MOMENT CONNECTION. SEE TYPICAL DETAILS.
12. COLUMNS NOTED THUS (A) IN PLAN ARE ABOVE.
13. SEE ARCHITECTURAL AND/OR MECHANICAL DRAWINGS FOR:
 - a. EDGE OF SLAB DECK DIMENSIONS.
 - b. LOCATIONS OF RIDGES, VALLEYS, FLOOR DRAINS AND SLOPES TO DRAIN NOT SHOWN.
 - c. LOCATION AND EXTENT OF SLAB DEPRESSIONS AND RAMPS NOT SHOWN.
 - d. LOCATION AND EXTENT OF EQUIPMENT PADS AND SUPPORTS, CURBS, ETC. HOUSEKEEPING PADS SHALL BE STRUCTURAL LIGHTWEIGHT CONCRETE, MAXIMUM 4" THICK.
14. VERIFY SIZE AND LOCATION OF SLAB OPENINGS WITH LATEST ARCHITECTURAL AND MEP DRAWINGS, AND EQUIPMENT PURCHASED. PROVIDE ADDITIONAL FRAMING PER TYPICAL DETAILS.
15. INDICATES (3000) LIGHTWEIGHT CONCRETE TOPPING.
16. INDICATES SPAN DIRECTION OF DECK.
17. INDICATES FACE OF BUILDING (F.O.B.).
18. INDICATES BRACE FRAME LOCATIONS.
19. INDICATES JOIST BRIDGING. SEE TYPICAL DETAILS.
20. INDICATES SLAB/DECK OPENING. SEE TYPICAL DETAILS.
21. INDICATES DIMENSIONS TO BE COORDINATED DURING CONSTRUCTION WITH APPROVED EQUIPMENT.
22. INDICATES COLUMN SUPPORTED BY BEAM BELOW. SEE TYPICAL DETAILS.
23. INDICATES BEAM PENETRATION. SEE TYPICAL DETAILS AND SCHEDULE.
24. SEE TYPICAL DETAILS AND STRUCTURAL NOTES FOR ADDITIONAL INFORMATION.



PLAN NOTES: 1/8" = 1'-0"

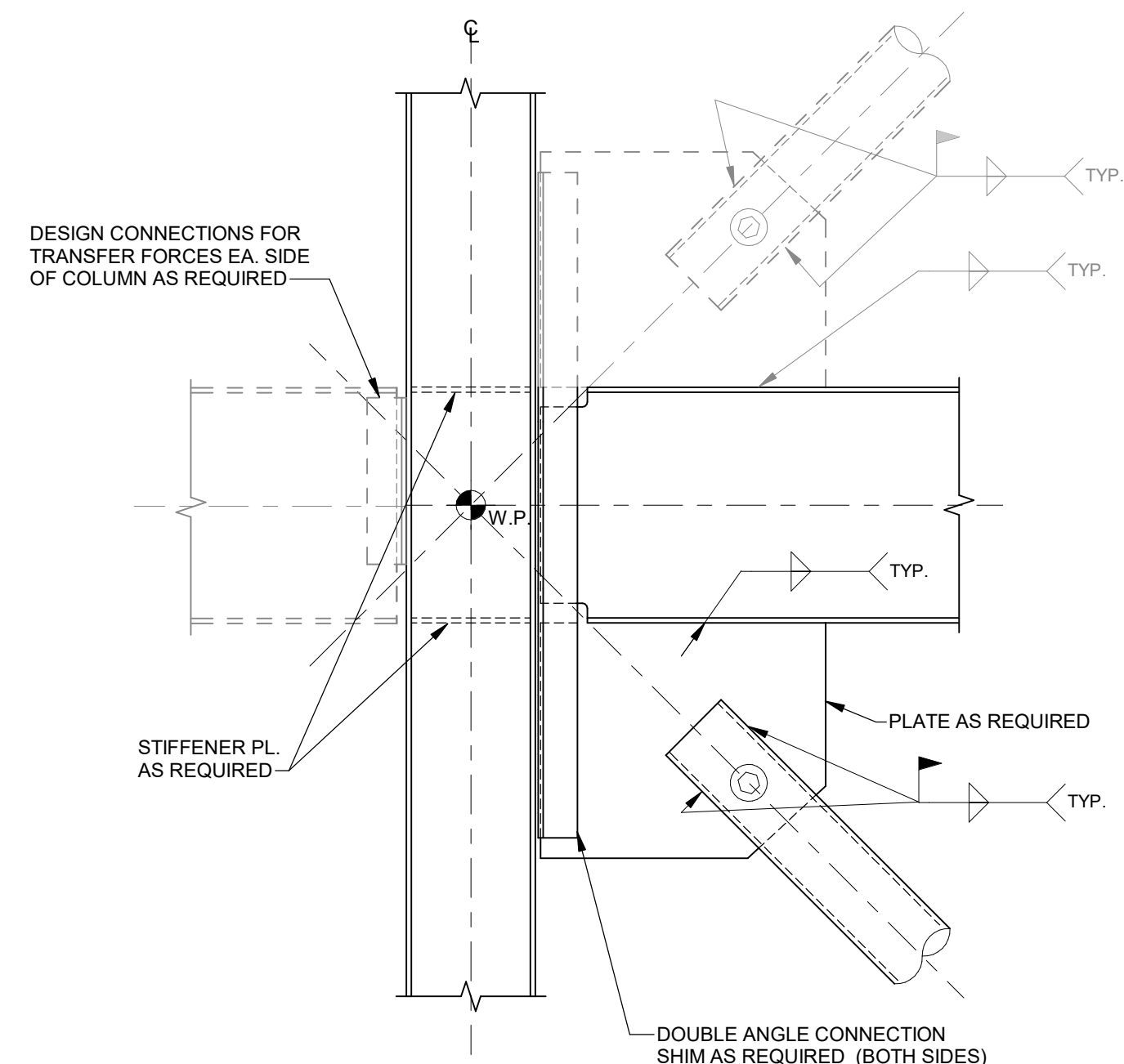
1. SEE S0102 FOR INFORMATION NOTE SHOWN HERE



Autodesk Docs: 75525106-00 DC DGS - Ketcham ES Design-Build\Ketcham ES_STRUCT_R24.rvt
5/8/2020 10:39:36 AM

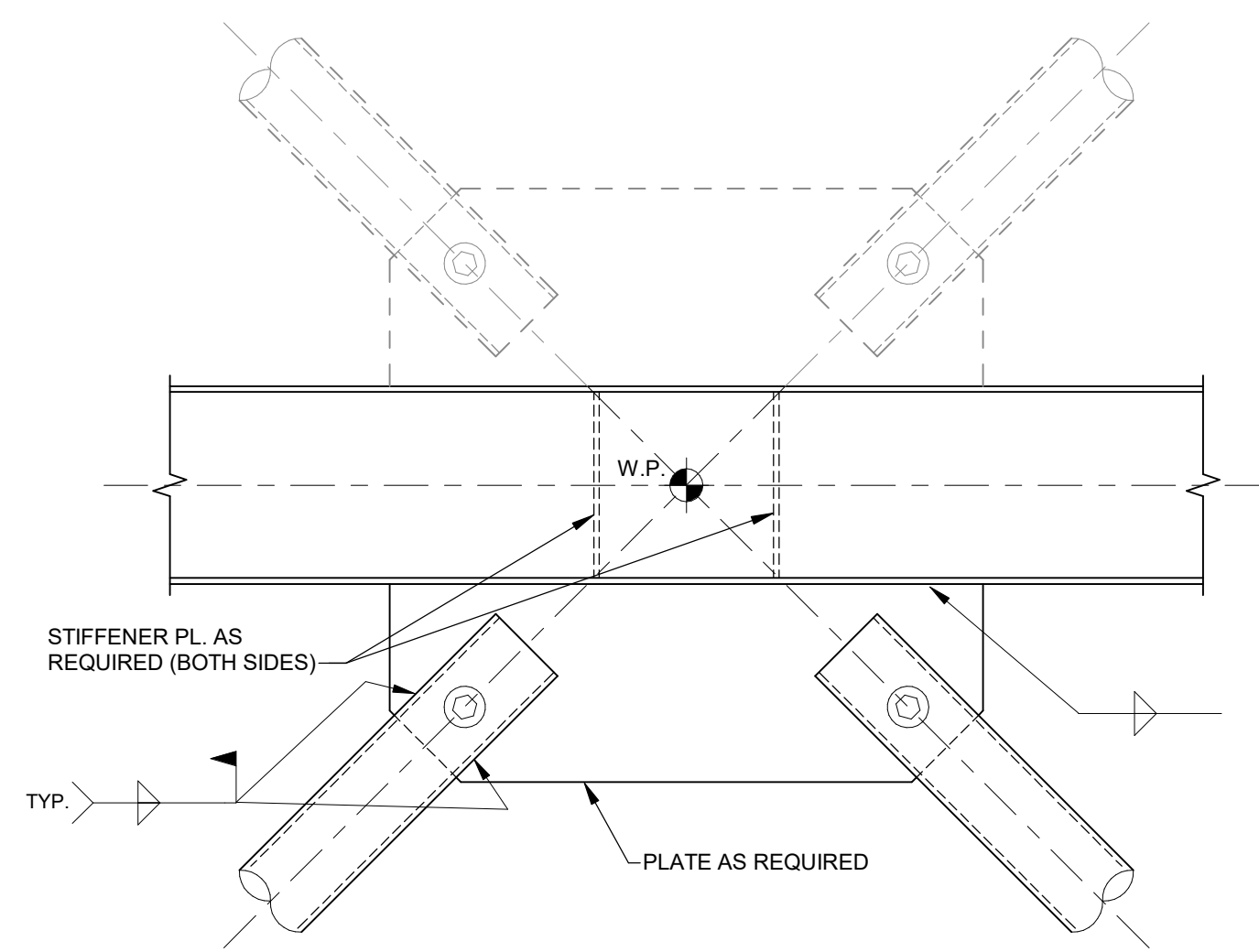


ATTIC FRAMING PLAN
1/8" = 1'-0"
1. SEE S0102 FOR INFORMATION NOTE SHOWN HERE.



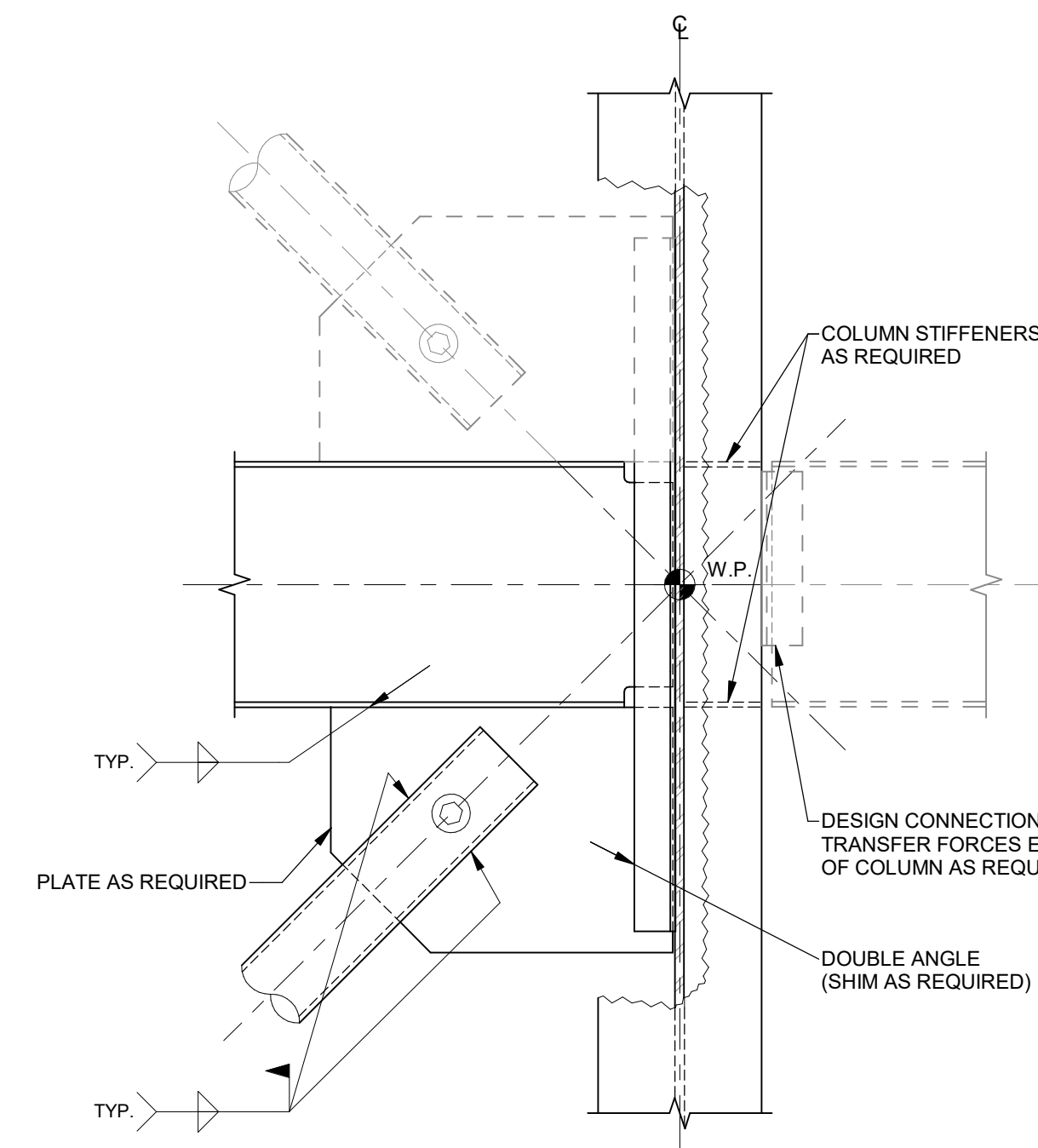
TYPICAL BRACED FRAME CONNECTION
(COLUMN FLANGE)

A
S0210



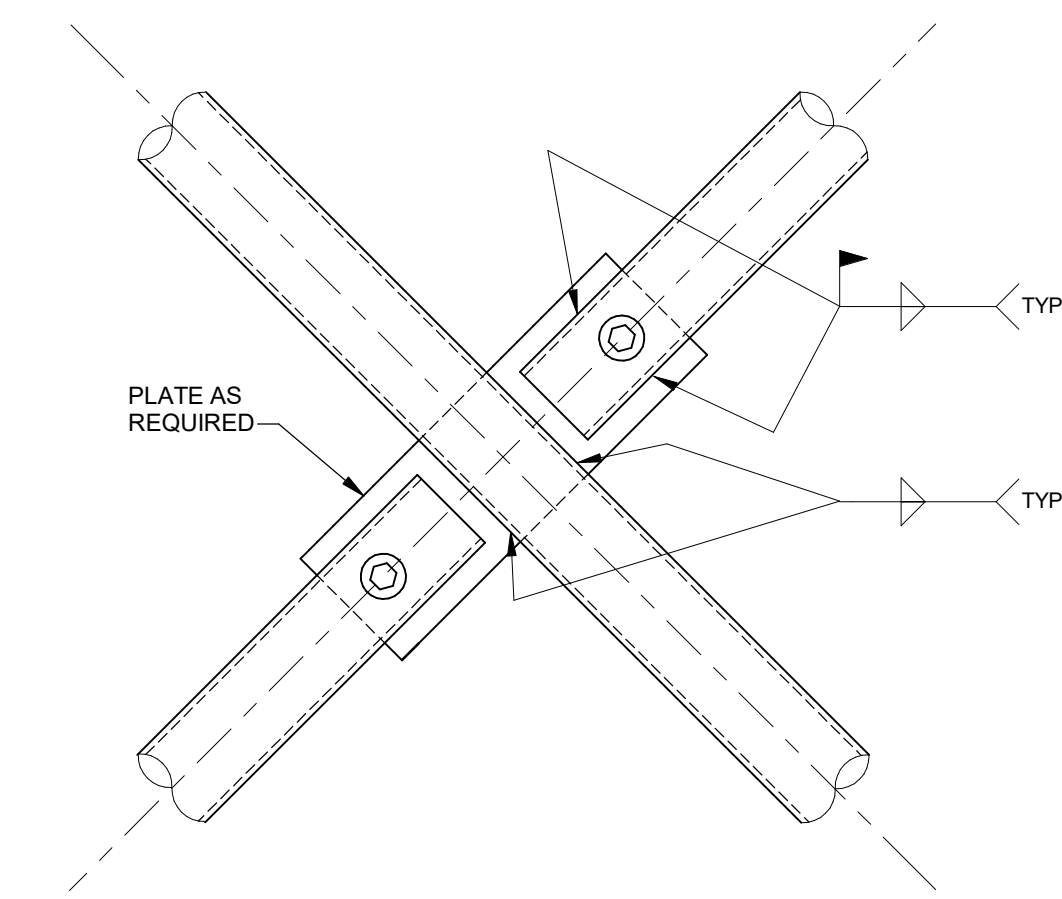
TYPICAL BRACED FRAME CONNECTION
(MID-BEAM)

B
S0210



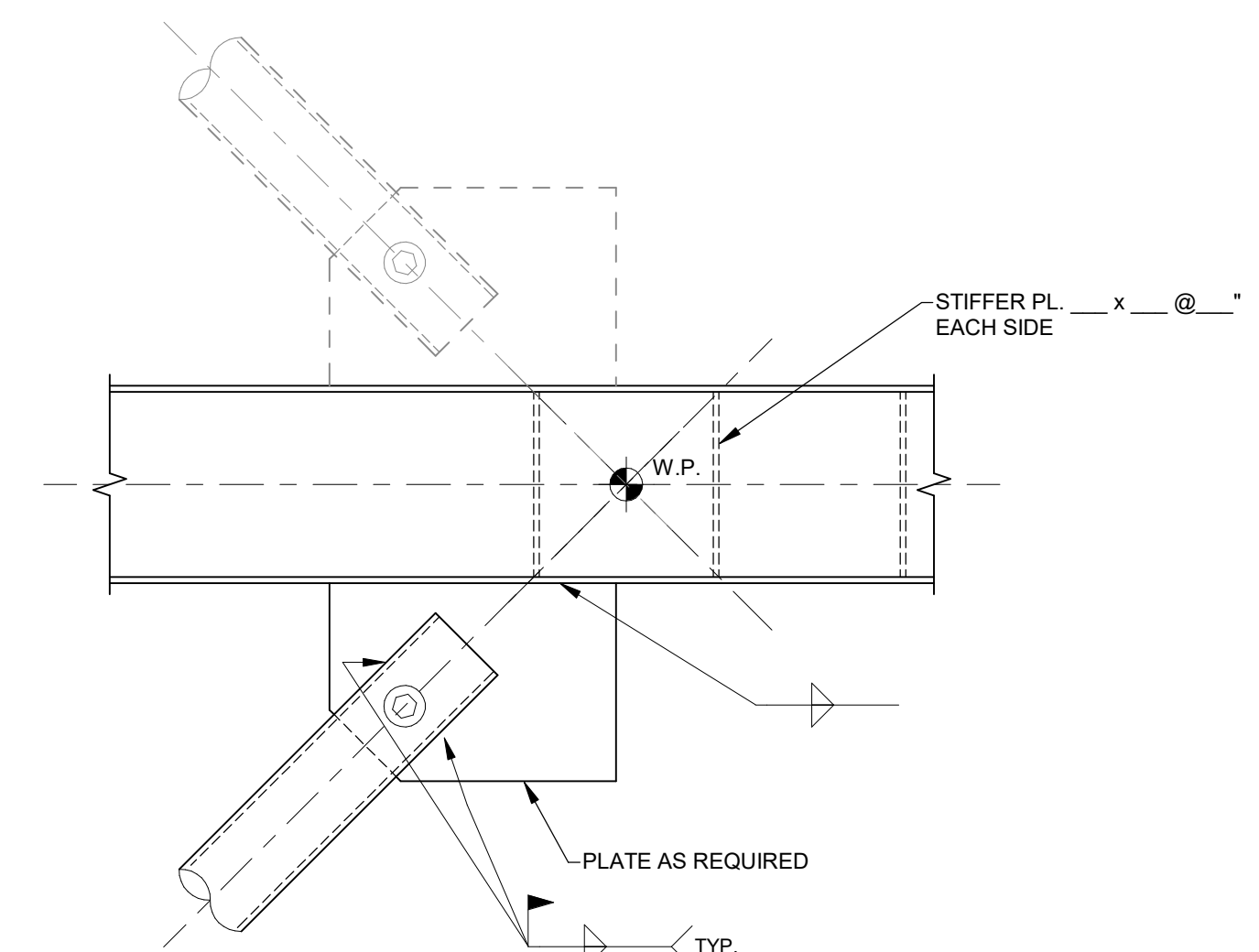
TYPICAL BRACED FRAME CONNECTION
(COLUMN WEB)

C
S0210



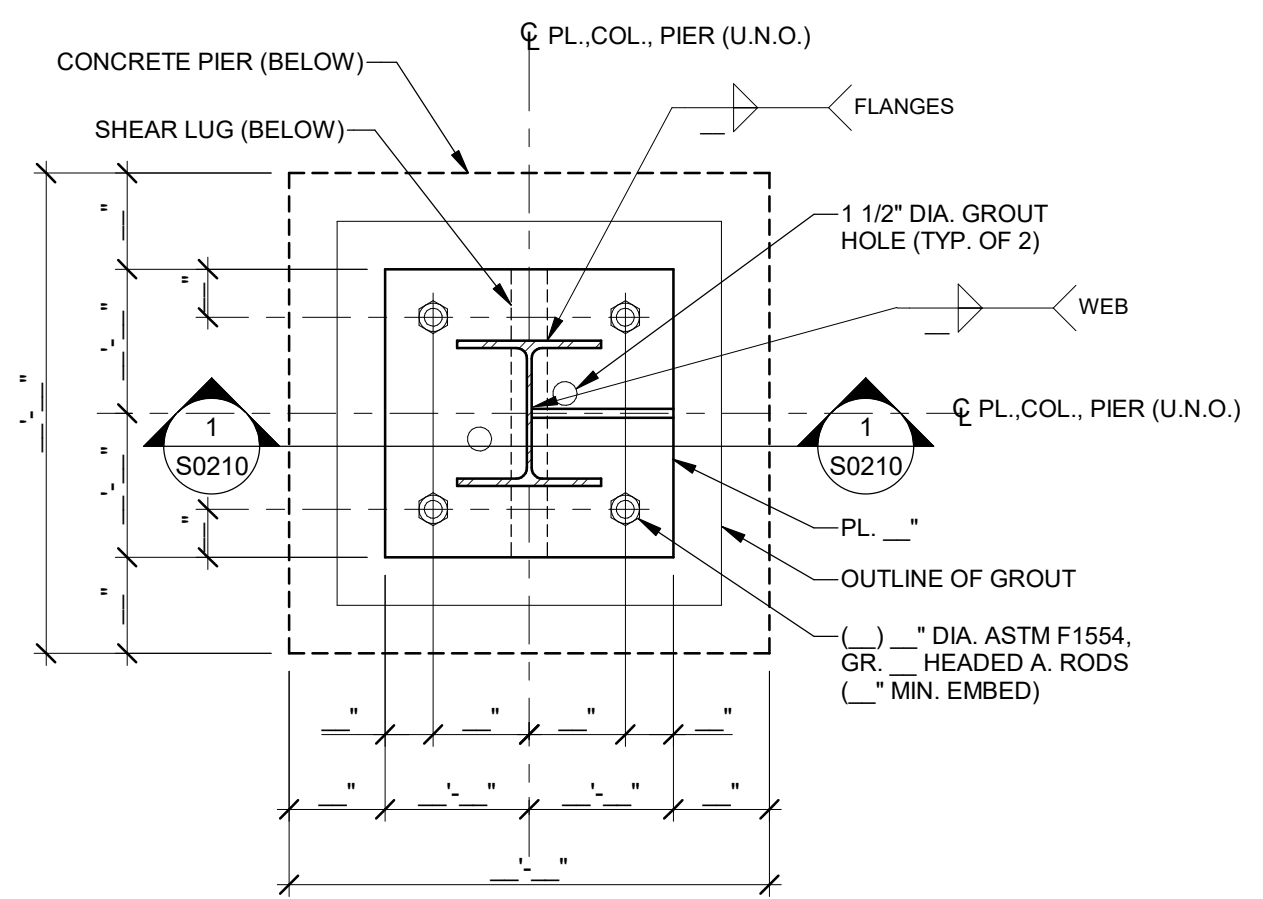
TYPICAL BRACE CROSSING

D
S0210



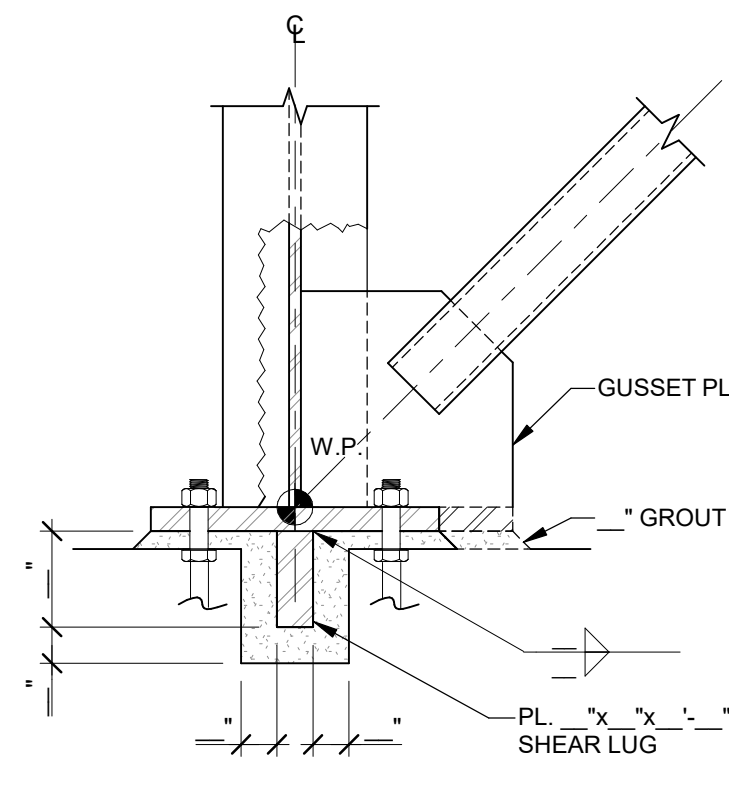
TYPICAL ECCENTRIC BRACED FRAME CONNECTION

E
S0210



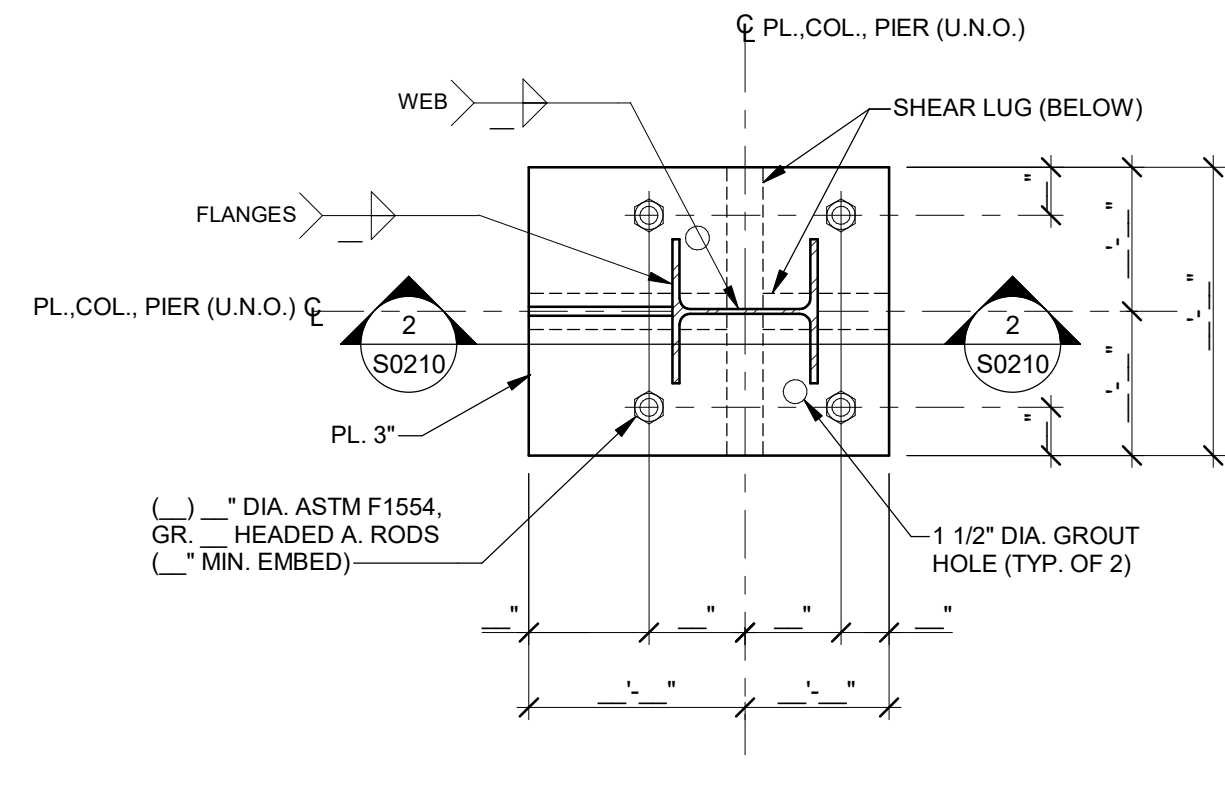
BASE PLATE DETAIL
(TYPE 1)

F
S0210



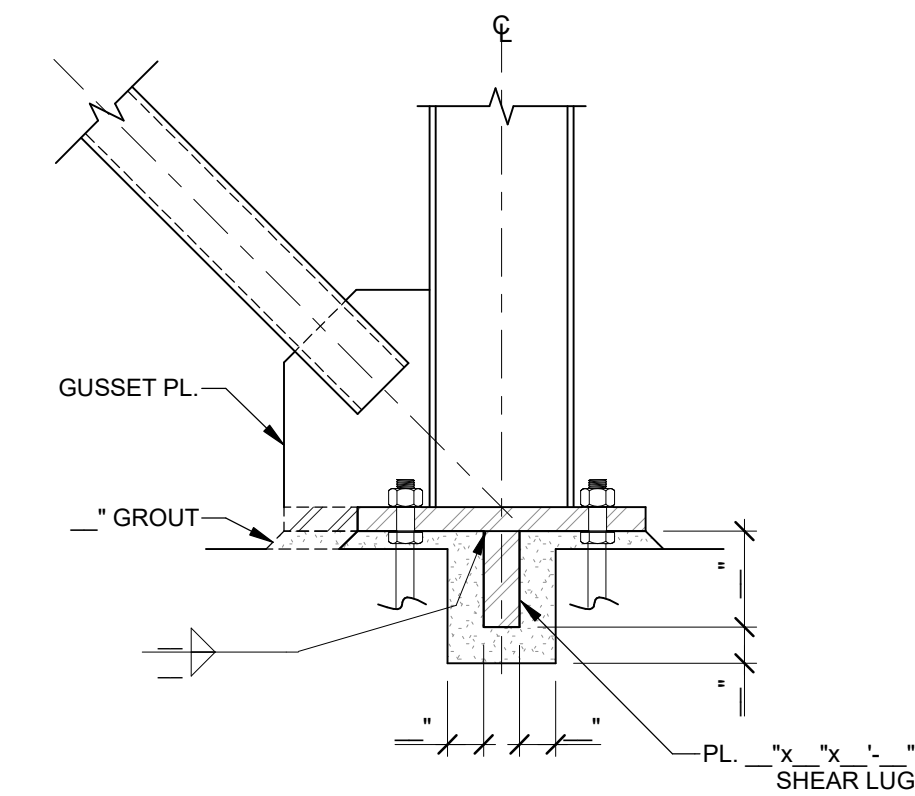
SECTION
SCALE: 3/4" = 1'-0"

1
S0210



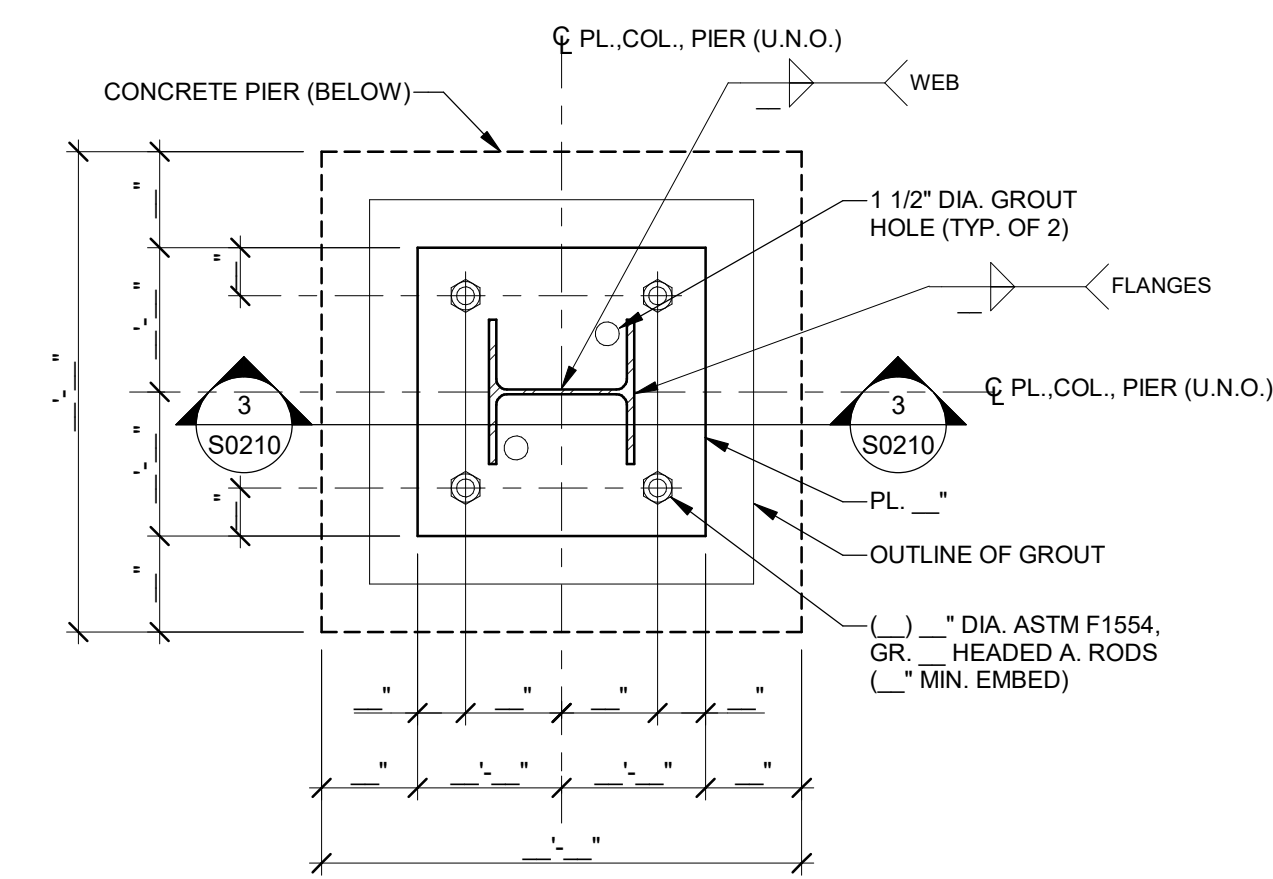
BASE PLATE DETAIL
(TYPE 2)

G
S0210



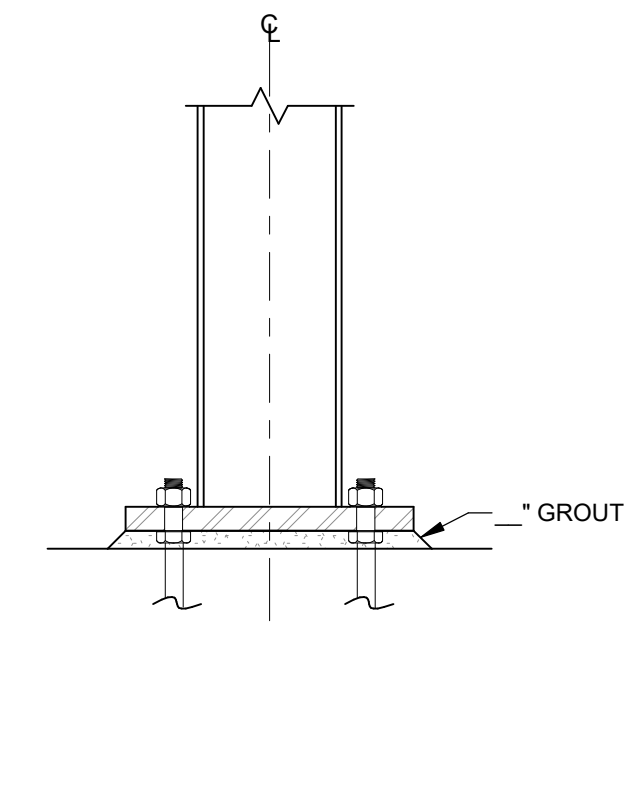
SECTION
SCALE: 3/4" = 1'-0"

2
S0210



BASE PLATE DETAIL
(TYPE 3)

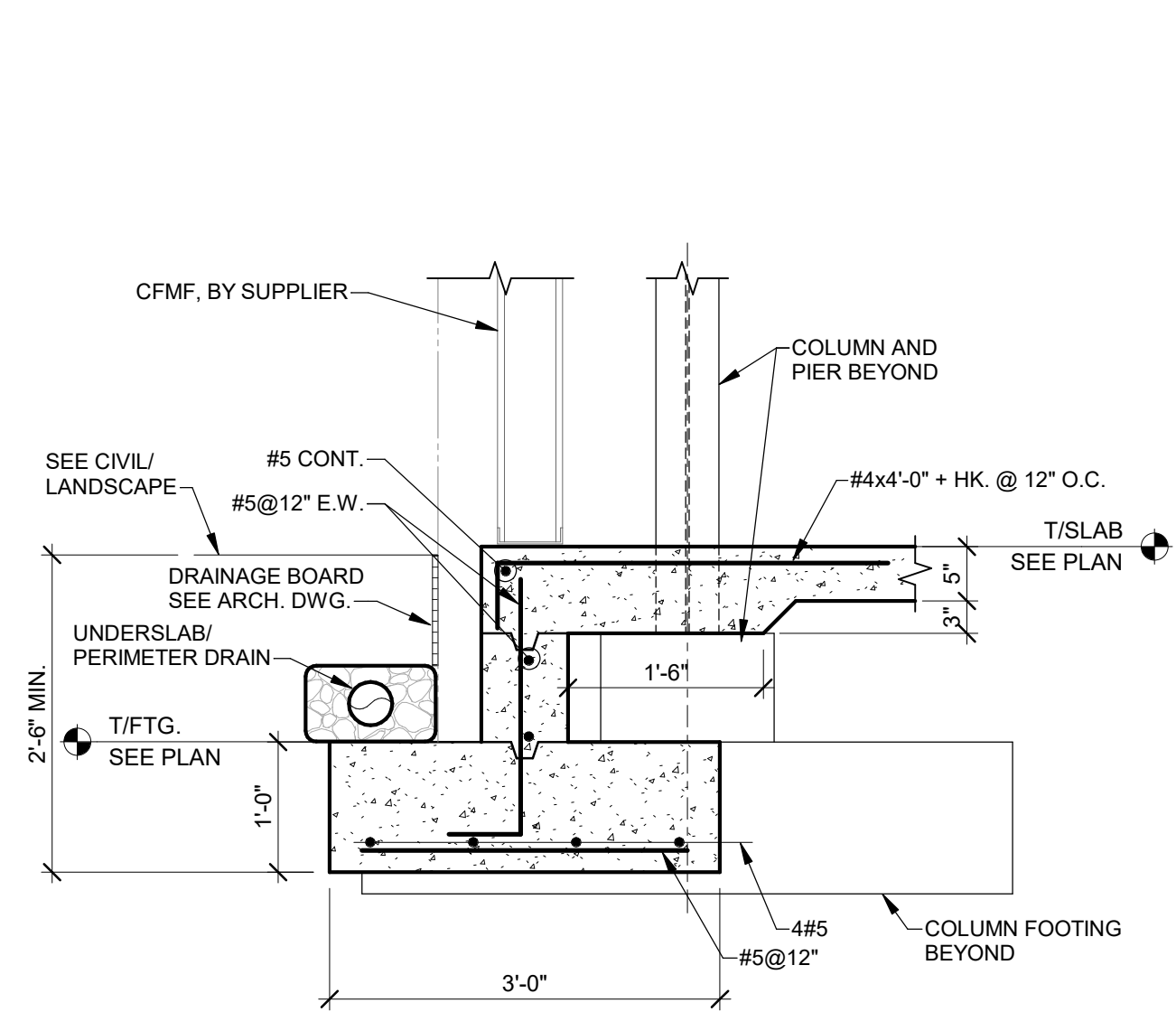
H
S0210



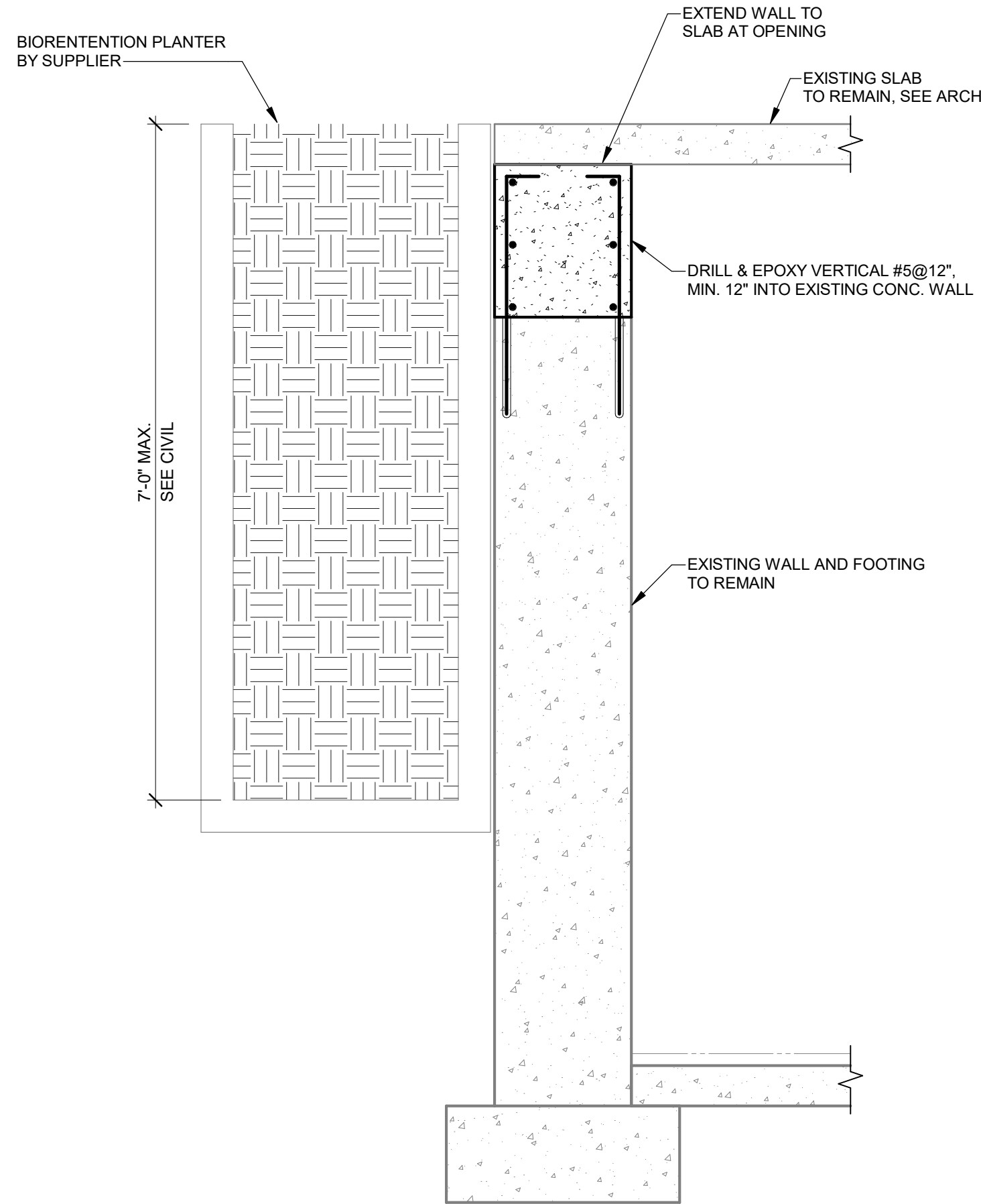
SECTION
SCALE: 3/4" = 1'-0"

3
S0210

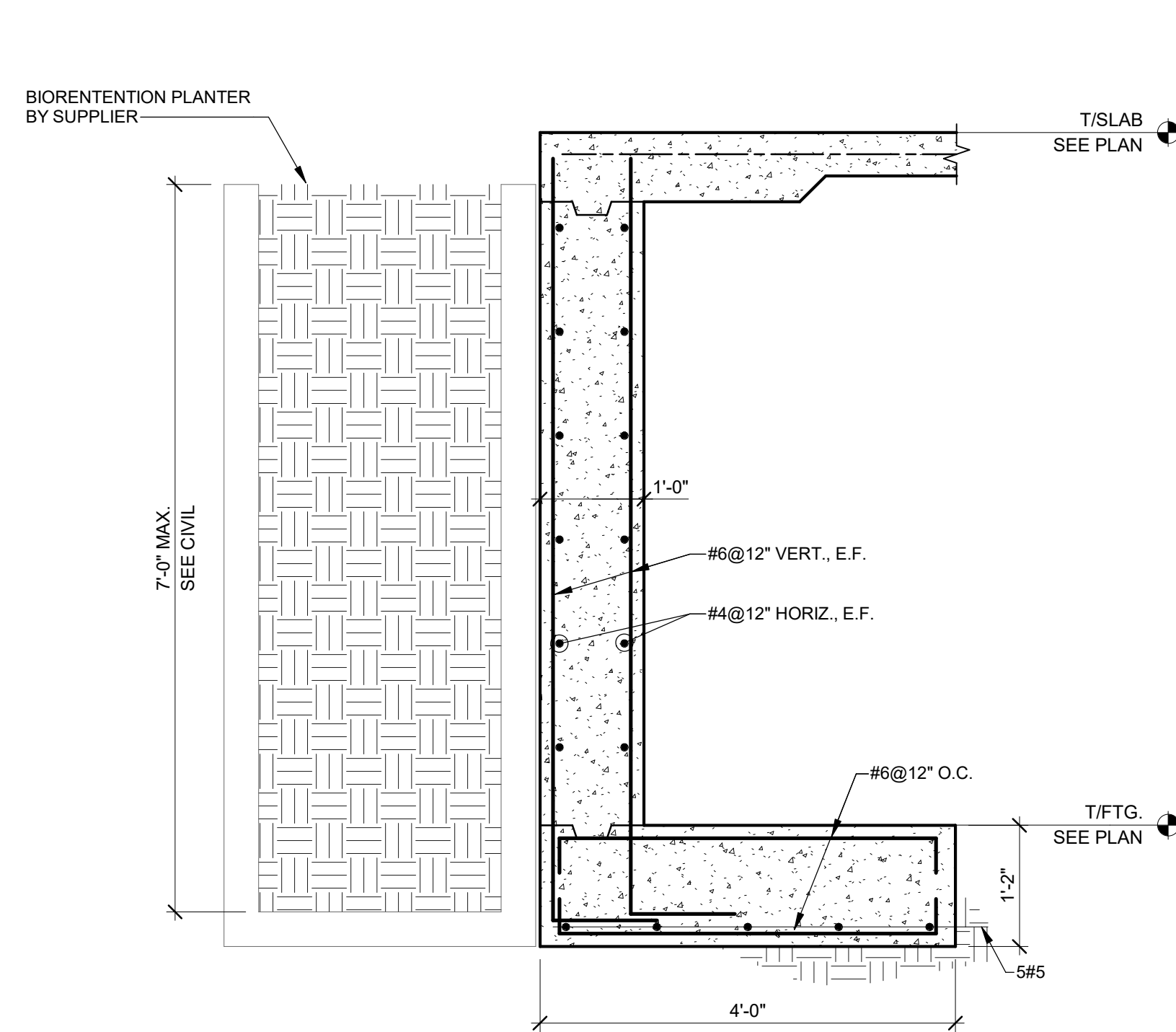
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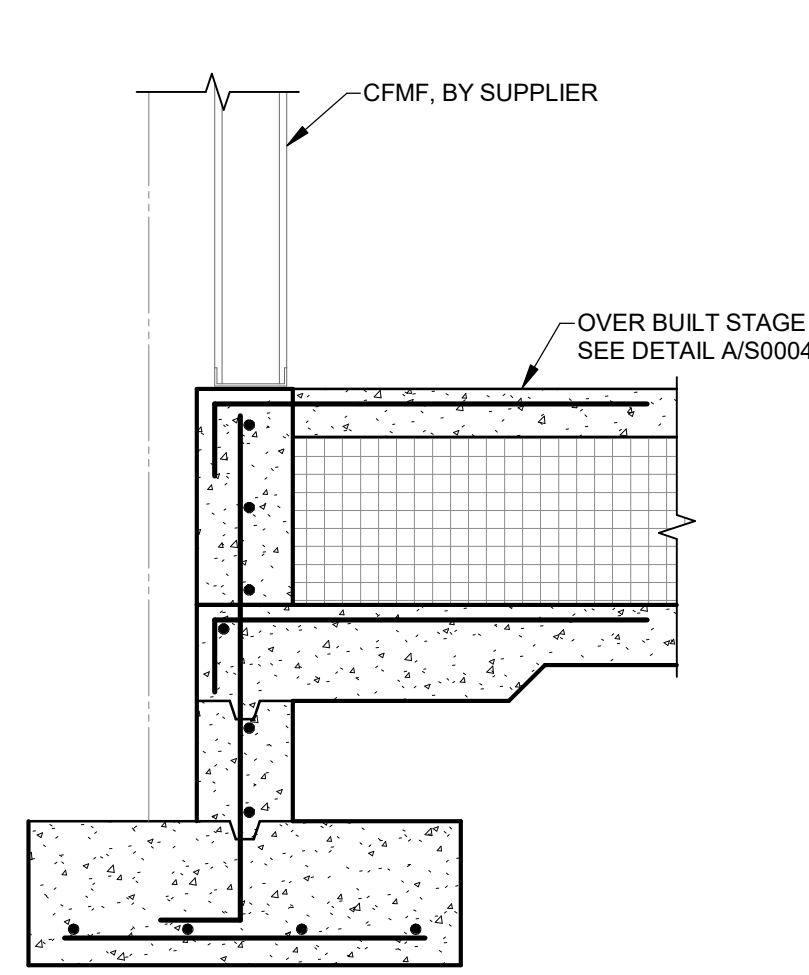
SECTION 1
SCALE: 3/4" = 1'-0"



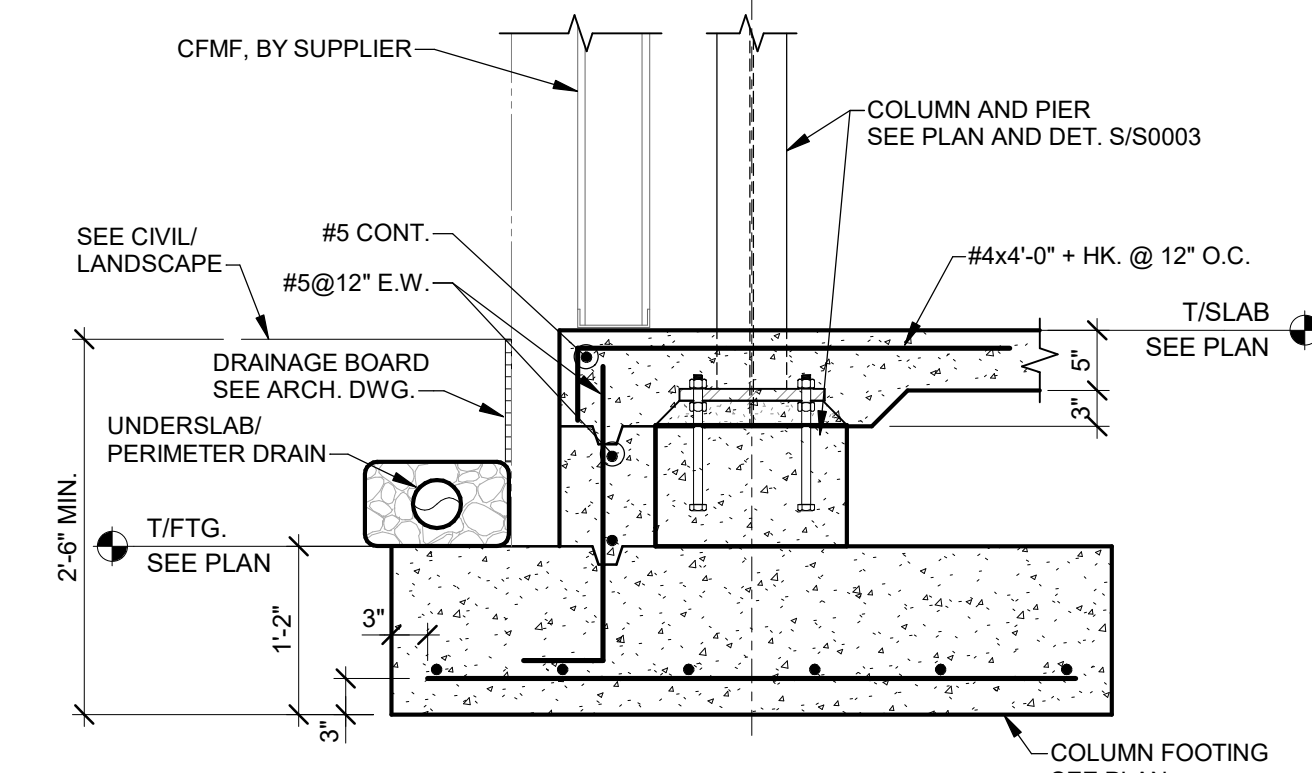
SECTION 2
SCALE: 3/4" = 1'-0"



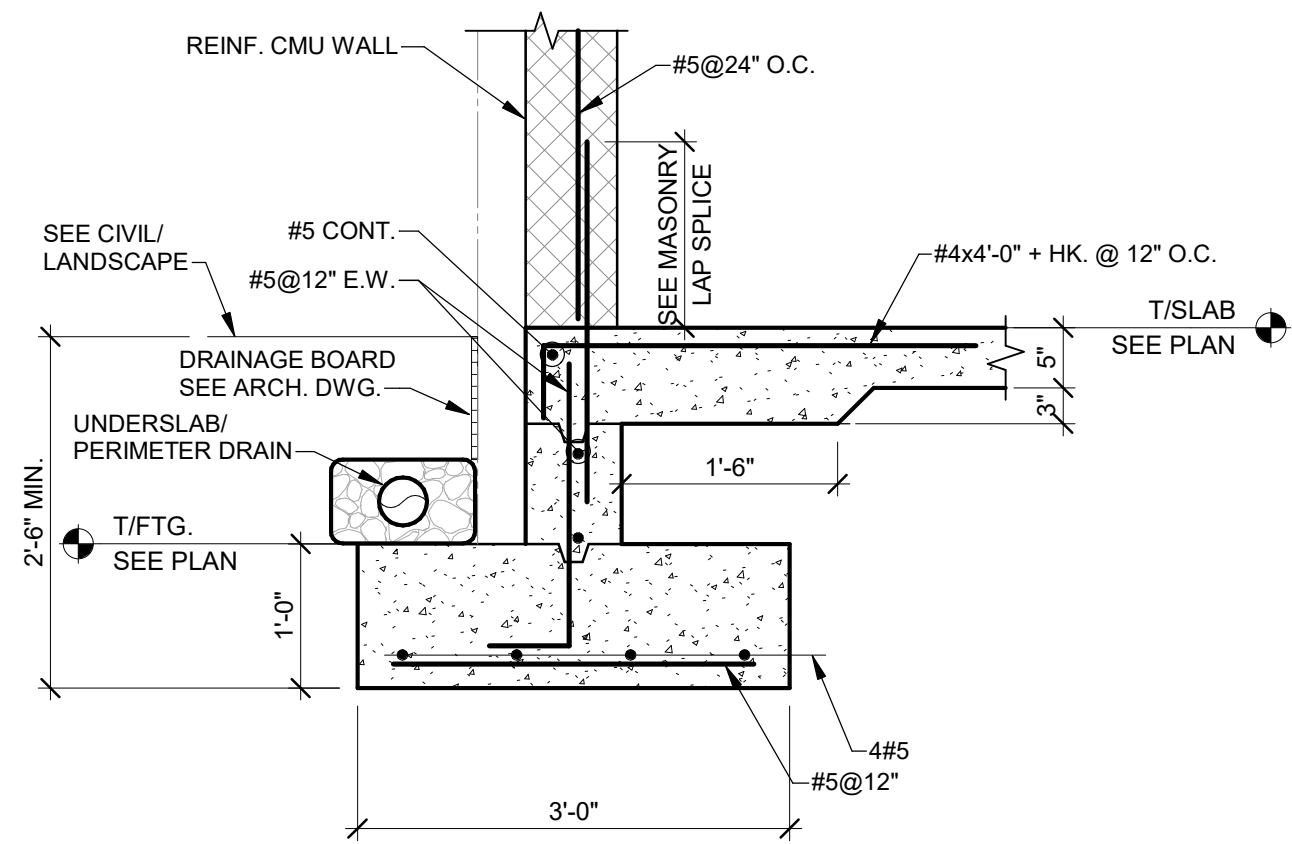
SECTION 3
SCALE: 3/4" = 1'-0"



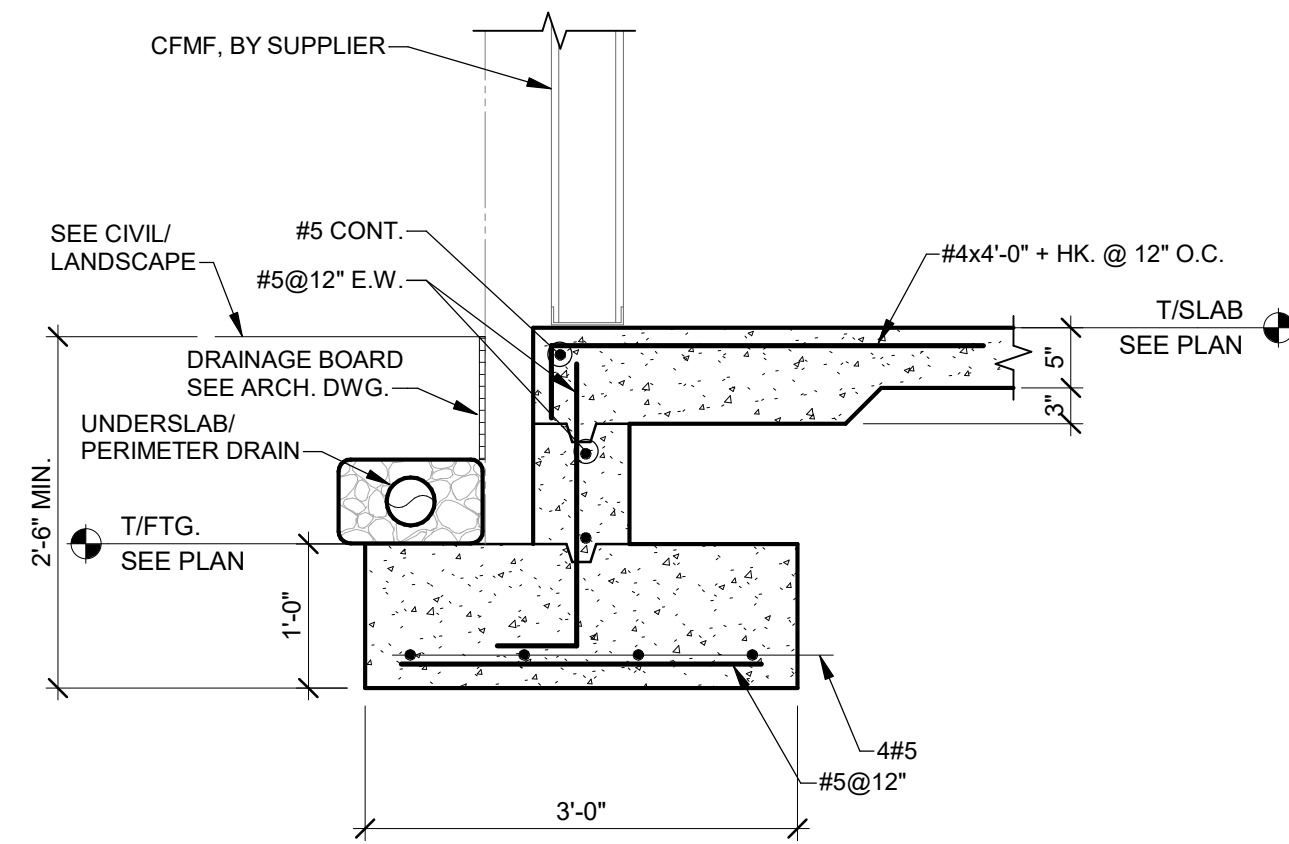
SECTION 4
SCALE: 3/4" = 1'-0"



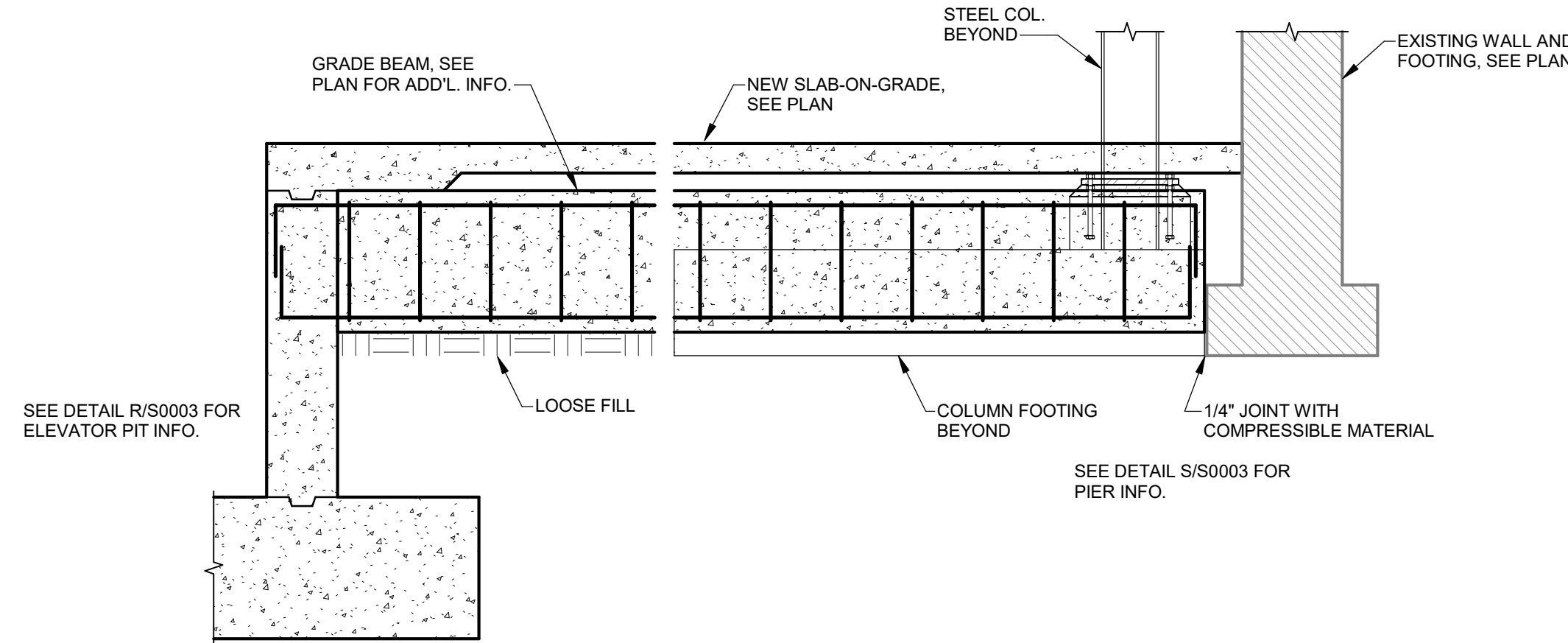
SECTION 5
SCALE: 3/4" = 1'-0"



SECTION 6
SCALE: 3/4" = 1'-0"



SECTION 7
SCALE: 3/4" = 1'-0"



SECTION 8
SCALE: 1/2" = 1'-0"